

GETTING A SEAT AT THE TABLE: PARTISAN POLL WORKERS AND ELECTORAL BIAS*

RAUL DUARTE[†]

ANDRES CARRIZOSA[‡]

September 2026

Does poll workers' partisanship affect electoral outcomes? Many countries have party representatives count the votes, expecting mutual control to ensure fairness. Yet parties differ in their capacity to staff every booth. Exploiting quasi-random assignment of voters to booths in Paraguay's 2018 general elections, we estimate that partisan poll workers decrease an opposing party's vote share by up to 1.1 percentage points and increase their own by up to 0.7. Effects are larger in proportional representation races, where established parties gain at the expense of smaller parties. Adversarial election administration thus translates unequal staffing capacity into biased official results.

JEL Codes: D72, D73, O17, K16

Keywords: Quasi-natural experiment, Election administration, Electoral irregularities

*We benefited from thoughtful comments by Francisco Cantú, William Clark, Claudio Ferraz, J. Andrew Harris, F. Daniel Hidalgo, Frances Hagopian, Alisha Holland, Torben Iversen, Jonathan N. Katz, Horacio Larreguy, Steven Levitsky, Andrew O'Donohue, Vincent Pons, Pia Raffler, Reed Rasband, Jonathan Weigel, panel participants at APSA 2022 and APSA 2023, and workshop participants at LACEA/RIDGE 2024, Instituto Desarrollo, and multiple Harvard workshops (Political Economy, Development Economics, Political Economy of Development, Comparative Politics, and the Latin American Discussion group). We also thank José Tomás Sánchez for earlier contributions to this project. We thank the Harvard Kennedy School and the Institute for Humane Studies for research funding towards this project. All errors are our own.

[†]Yale University (raul.duarte@yale.edu). Corresponding author.

[‡]Instituto Desarrollo (acarrizosa@gmail.com).

... *it does not matter how many votes a candidate gets, but how many the poll workers record on the vote tallies.*

– Carlos María Ljubetic, electoral justice adviser quoted in ABC Color (2018a)

To this day, ensuring the integrity of the vote count is an ongoing process in many countries. This concern is not unique to developing countries: Harris (1934) documented extensive fraud by partisan precinct officers across American cities in the early twentieth century, arguing that bipartisan monitoring failed when parties were of unequal strength. A growing economics literature documents how election administration affects whether recorded votes reflect voter preferences. Voting technology bears on who gets elected and downstream policy outcomes (Card and Moretti 2007; Djourelova and Tsankova 2025; Fujiwara 2015), the identity of election officials influences electoral integrity (Callen and Long 2015; Neggers 2018), and monitoring by citizens or technology can reduce irregularities (Garbiras-Díaz and Montenegro 2022; Gonzalez 2021).

A common institutional response to the problem of electoral integrity has been *adversarial controls*: systems in which political parties are given formal roles at polling stations, and mutual monitoring between competing parties is expected to ensure a fair count. Partisan poll watchers who observe the vote count are found in countries throughout the globe, including Argentina (Casas, Díaz and Trindade 2017), Mexico (Ascencio and Rueda 2019), Turkey (Gazete Duvar 2021), and the United States (Brennan Center 2021). In several countries, partisan representatives go further and serve as the poll workers who actually count the votes—including Nicaragua, Honduras, Paraguay, Armenia, Ukraine, Kosovo, El Salvador, Costa Rica, and Colombia (Pomares, Page, Zárate and Abdala 2015; Sjoberg 2016). These systems rest on an empirical premise: that whoever counts the votes, mutual counting at the booth neutralizes any effect of their partisanship on the recorded count. This paper asks whether that premise holds. How much does the partisan composition of the poll workers who count the votes change the recorded vote shares, and through what channels? Answering it requires comparing booths whose poll workers differ but whose voters do not, as parties place and replace poll workers where their supporters are.

This paper estimates the causal effect of partisan poll worker composition on electoral outcomes in Paraguay’s 2018 general elections. Two features of the Paraguayan electoral system provide identification. First, within polling stations, voters are assigned to voting booths alphabetically by surname, and we show that surnames are largely orthogonal to political preferences (following Casas, Díaz and Trindade 2017). This quasi-random assignment of voters to booths implies that on average, voters across booths within a given polling station should have similar political preferences, allowing us to isolate poll worker effects from voter preferences. Second, we observe both the *de jure* (pre-registered) and *de facto* (election-day) allocation of poll workers: we hand-coded the national ID numbers from approximately 120,000 scanned vote tallies signed by poll workers and matched these ID numbers to partisan affiliation records. This matching gives us

a detailed picture of all 21,211 voting booths in the country, including who was supposed to count votes and who actually did. We address two key identification challenges—the secret ballot making individual-level manipulation unobservable (Baland and Robinson 2008), and the endogeneity of party capacity to party popularity—with our within polling station design and by controlling for the proportion of each party’s registered supporters at each booth.

Paraguay is well-suited for this study for two additional reasons. First, parties differ sharply in operational capacity. The Colorado party, which backed the 35-year dictatorship of Alfredo Stroessner and has won every presidential election since democratization except one in 2008, is the country’s most resourced party. The Liberal party is the main opposition, while smaller parties lack the resources to staff all booths, despite collectively capturing 43.3% of votes in the 2018 Senate race. Through our data work we document that more than a third of pre-registered poll workers are replaced on election day, and these replacements overwhelmingly favor the Colorado and Liberal parties at the expense of smaller parties. Second, the 2018 elections used separate paper ballots for six races: the president and governors were elected by plurality in single-member districts (SMDP), while the Senate, Chamber of Deputies, departmental councils, and MERCOSUR (Southern Common Market) parliament were elected by closed-list proportional representation (PR).¹ As the same voters and the same poll workers count ballots under different electoral rules, we can examine how electoral systems shape incentives for partisan poll workers within the same booth.

Our results show that representation at a voting booth benefits political parties, meaning that parties with higher operational capacity (to send poll workers systematically) benefit. On average, having one additional copartisan poll worker relative to a *fair* allocation (one Colorado, one Liberal, one other, as the Electoral Code intends) increases the Colorado party’s vote share by approximately 0.5 percentage points (pp) and the Liberal party’s by approximately 0.2 pp per booth. As these figures pool all six races, they understate the largest race-specific effects: in departmental council elections a Colorado majority raises the Colorado vote share by 0.66 pp, and a Colorado trifecta lowers the Liberal vote share by 1.1 pp (SA Table G21). These effects are stronger in PR races than in SMDP races, which follows from how each rule turns votes into seats. Under plurality only the margin between the top two candidates decides the seat, so the votes worth moving are the rival established party’s. Under closed-list PR every list’s votes enter the seat allocation and the votes needed for a seat fall with district magnitude, so votes shaved from the many small lists that PR sustains (Cox 1997) convert into seats for whichever established party controls the booth.

We contribute to several strands of the literature. First, on election administration and fraud: Casas, Díaz and Trindade (2017) study party observers in Argentina who watch the count, while we study workers who actually count votes, leaving greater scope for affecting elections. Neg-

¹See Supplemental Appendix (SA) Figure C1 for an image of the ballot used for the Senate election.

gers (2018) shows that the religious and caste composition of election officer teams affects election outcomes in India, with officers exercising discretion in verifying voter eligibility in favor of co-ethnics. We show that the partisan composition of vote counters directly shapes recorded vote shares. Callen and Long (2015) document how weak institutions enable electoral fraud in Afghanistan, while we uncover a subtler institutional failure where the adversarial design itself enables manipulation through unequal party capacity. Second, on monitoring and transparency: Garbiras-Díaz and Montenegro (2022) show that citizen oversight reduces irregularities, and Gonzalez (2021) shows that information technology enables monitoring that deters fraud. Our paper emphasizes that when the “monitors” are themselves partisan agents with unequal representation, the monitoring mechanism breaks down.

Third, on clientelism and vote manipulation: Bobonis, Gertler, Gonzalez-Navarro and Nichter (2022) show how economic vulnerability enables clientelism, the exchange of contingent benefits for political support (Hicken 2011). Hicken, Leider, Ravanilla and Yang (2018) show that vote-selling works as a temptation good for voters in the Philippines, and that a simple promise not to accept payments can curb it in low-payment races. In addition, Ferlenga and Knight (2025) show how opportunities for repeated voting enable fraud. Our findings are in line with Duarte, Finan, Larreguy and Schechter (2025) and Finan and Schechter (2012), which show how historically dominant parties maintain leverage through vote buying even amid democratization. Yet we identify a distinct channel: not vote buying or turnout manipulation, but distortion at the counting stage by partisan agents. Finally, we show that the electoral system moderates these effects: PR creates more scope for established-party collusion, connecting to the literature on how electoral rules change incentives for corruption and electoral malpractice (Birch 2007; Chang and Golden 2007; Persson, Tabellini and Trebbi 2003).

A closely related literature studies partisan poll watchers, who oversee the vote count and may indirectly influence electoral results (Ascencio and Rueda 2019; Casas, Díaz and Trindade 2017). These papers highlight how parties spend resources to deploy more poll watchers, how this advantages parties with greater capacity, and how parties may allocate poll watchers strategically. In contrast, our work focuses on poll workers, those actually counting votes and thus more directly able to influence election results. A parallel literature shows that voting technology affects outcomes (Djourelouva and Tsankova 2025; Fujiwara 2015), suggesting the mode of counting matters, and we ask a complementary question about how the identity of who counts also matters. Up to this point, findings on the effects of poll workers on electoral outcomes have been mixed. Challú, Seira and Simpson (2020) show that Mexican poll workers with lower education and higher workload make nonpartisan benign errors. Similarly, Warner, Harris, Brown and Arnold (2021) find that vote tally irregularities in Kenya’s 2013 presidential election do not correlate with election outcomes. Our paper, through its quasi-experimental design, finds that the partisan composition of

the poll workers who count the votes affects recorded electoral outcomes independently of voter preferences. We also document that this composition is shaped by parties’ unequal capacity to fill election-day vacancies. This dynamic has been commented on intuitively in Paraguayan political circles and the media (ABC Color 2018a; Colmán Gutiérrez 2018; Ferrara 2016), yet our study provides the first rigorous empirical evidence that the composition of partisan poll workers counting votes significantly affects electoral outcomes. Our results suggest that adversarial vote-counting systems require roughly equal party capacity to function as intended—a condition that fails in many settings.

I. Institutional Background

A. Electoral Tables and Poll Workers

In Paraguay, voting takes place at polling stations (*locales de votación*), which are typically public schools. Each polling station contains multiple voting booths, called “voting tables” (*mesas de votación*).² Any political party can send poll watchers (*veedores de mesa*) to observe the count,³ but three partisan poll workers called “table members” (*miembros de mesa*) conduct the vote count. The Electoral Code reserves one poll worker seat for each of the three largest parties in Congress.⁴ Since the Colorado and Liberal parties have always held the most or second most seats in Congress since democratization, they are always entitled to a seat at every electoral table. In 2018, the third seat was reserved for the left-wing Frente Guasu (FG) alliance.⁵ All other parties are excluded from having poll workers unless the top three do not fill all their entitled seats.⁶

Weeks before the election, parties pre-register their poll workers. One of the three is randomly selected to be the “president” of the table, while the other two are called *vocal* 1 and 2.⁷ Despite this nominal hierarchy, all three are equal in attributions, and disputes are resolved by simple majority (two out of three). At the end of the election day, the three poll workers count the votes

²Throughout this paper we use the terms “voting booth,” “voting table,” and “electoral table” interchangeably.

³Poll watchers can observe and file complaints, but have no formal powers in the count. Unlike the poll worker data, available information on poll watchers is more limited. The election authority only has pre-registered poll watcher data and poll watchers do not sign vote tallies. Hence, we are not able to determine the *de facto* poll watchers at each voting booth. Still, SA Tables E1 and E2 show that our main results are unchanged after controlling for pre-registered poll watchers.

⁴The Colorado, Liberal, and PEN parties dominated the framing of the Electoral Code at the time, and the three-member electoral table rule has remained.

⁵Technically, the FG is an electoral alliance of smaller left-wing political parties which sometimes run together (PPS, PFA, PPC, PPT, PCP, PCPS, PMPP, and PUP). Party acronyms are described in the Supplemental Appendix (SA) Section C.1.

⁶Staffing each voting table with a small number of poll workers, typically three to five, is common in Latin America (Pomares, Page, Zárate and Abdala 2015).

⁷See SA Figure C2 for a picture of a typical electoral table setup. SA Table E3 tested whether the partisanship of the poll worker president has an independent effect on electoral results. The results do not show a clear pattern, which likely reflects that the official result within a table is the majority’s (a minimum of 2 out of 3 poll workers).

together and each writes a vote tally (*acta electoral*) for each race. All three are supposed to sign each tally, acknowledging agreement with the recorded results.⁸ A worker who disagrees may refuse to sign, but the tally stands as written by the majority, so the majority's count is the one transmitted. Each worker's copy of the tally also reaches the election authority: the quick count is a copy transmitted from the station, which carries the majority's count and is the one scanned and published (the source of our ID numbers), while contestations are adjudicated by comparing the three workers' copies under the same two-of-three rule (see SA Section C for details). Crucially, the 2018 Electoral Code stipulated that after the election the actual paper ballots are discarded, leaving the table-level tally as the lowest-level electoral document available for inspection (Última Hora 2018a). This means that the three poll workers have effective final authority over the recorded vote count at their table.

B. Poll Worker Replacements and Incentives

Despite the formal one-party-one-seat rule, in practice the allocation of poll workers frequently departs from this design. Vacancies on election day are common: when a pre-registered poll worker does not appear, any voter present at the polling station can fill the seat. These replacements are most often filled by Colorado and Liberal party supporters, through two channels. First, established parties actively pay and transport their poll workers to ensure they appear (Lachi 2009), a capacity that smaller parties lack. Focus groups with political brokers confirm that poll worker representation is considered “fundamental” for parties, and that some poll workers are bribed to skew results or simply not appear (Lachi 2009; Paraguay.com 2018).⁹ Second, parties encourage affiliated voters to arrive at polling stations early so that if pre-registered poll workers from other parties fail to appear, their own supporters can fill the vacancies. Since established parties have more affiliated voters, this practice further tilts the *de facto* allocation in their favor.

The result is that the *de facto* composition of poll workers often differs substantially from the pre-registered fair allocation of one Colorado, one Liberal, and one FG worker. Many tables end up with two or even three poll workers from the same party, what we call an “advantage” or “trifecta” respectively. Because the official result at each table is determined by majority agreement among the three poll workers, a party holding two or three seats effectively controls the recorded count. Moreover, booth-level results can only be challenged during the counting process by poll workers, poll watchers, or party officers present, which prevents smaller parties from contesting irregularities.

How can three poll workers alter a count that other parties are entitled to watch? Three features

⁸In a random sample of 300 booths from the tally images, every race tally carries all three signatures in 97.0% of booths (SA Table E11). ID numbers are omitted more often, on at least one race tally in 18.0% of booths, which is why the digitized data draw each worker's ID number from any of the booth's tallies.

⁹The article from Paraguay.com (2018) shows an anonymous political broker interviewed by journalists claiming that booths can be “bought” by bribing poll workers to distort the results.

of the process make it possible. First, the tally is the record. (ABC Color 2018a) describe the system as one where “the tally kills the vote”: what matters is not how many votes a candidate receives but how many the poll workers write on the tally, and a former electoral justice adviser has called the composition of the table “one of the great weaknesses of the Paraguayan electoral system” for this reason (ABC Color 2018a). As the ballots are destroyed once the tally is signed, a tally that misstates the count cannot be checked against the ballots afterward (U.S. Embassy in Asunción 2008). Second, the workers write the tally themselves and decide disputes by majority, so two workers from the same party, or from parties with a common interest, control what is recorded. Third, the parties most exposed to a biased count are rarely in the room. Poll watchers have no formal role in the count, and even on the pre-registered poll watcher rolls (the only ones we observe), the Colorado party had a watcher at 28 percent of booths, the Liberal party at 3 percent, FG at 2 percent, and all remaining parties combined at 12 percent. At the typical booth, therefore, no one outside the three workers is present on behalf of the parties whose votes are at stake. A mode of manipulation that participants describe follows from these features: votes cast for parties with no one at the table are recorded on the tally as votes for the parties that are present, leaving the booth’s total untouched (ABC Color 2018a; Lachi 2009). SA Section C documents one such case in a challenged Senate tally with visible corrections (SA Figures C3 and C4). The booth had been assigned one Colorado, one Liberal, and one other party poll worker who was replaced by a second Colorado worker. No party had pre-registered a poll watcher there, and the corrections add 15 votes each to the Colorado and Liberal lists and remove 20 from Patria Querida and 10 from Hagamos. Two further modes are possible in principle, annulling ballots cast for opponents and stuffing or suppressing turnout, and Section IV.B uses blank votes, null votes, and turnout to distinguish among the three.

C. Predictions

These institutional features generate two direct predictions about the effect of poll worker composition on electoral outcomes.

Prediction 1 *An established party having a majority of poll workers at a table (two or three out of three) should have higher vote shares at those tables and opposing parties should have lower vote shares.*

Prediction 2 *Parties that completely lack poll worker representation at a table should tend to receive lower vote shares at those tables.*

Because we group the parties other than the Colorado and Liberal parties, we test Prediction 2 on their combined vote share.

A third prediction concerns how the electoral system moderates these effects. Qualitative evidence from focus groups with political party activists (Lachi 2009) and the country's newspaper of record (ABC Color 2018a) suggests that Colorado and Liberal poll workers may collude to redistribute among themselves the votes of parties that lack representation at the table.¹⁰ After the 2018 elections, a Paraguayan columnist reported that his vote for a small party representing indigenous peoples was recorded as zero on the published table-level results (Colmán Gutiérrez 2018), and another voter made a similar allegation (ABC Color 2018b). Further reports can be found in the Supplemental Appendix.

We expect this type of collusion to be more prevalent in PR races than in SMDP races, because the two rules translate votes into seats differently (Lijphart 1994; Taagepera and Shugart 1989). The president and the governors are elected by plurality in a single-member district, so only the margin between the top two candidates matters for the outcome: moving a vote from a small party to one's own gains one vote of margin, moving a vote from the rival gains two. Manipulation under plurality therefore concentrates on the rival established party, and it pays only where the race is close, since votes moved in a race decided by a wide margin change nothing. The legislative races use closed and blocked party lists with seats allocated by the D'Hondt formula: the Senate and the MERCOSUR parliament in a single national district with magnitudes of 45 and 18 seats, the Chamber of Deputies in 18 departmental districts with magnitudes from 1 to 20 (a mean district magnitude of 4.4), and the departmental councils in 17 districts with magnitudes from 7 to 21.¹¹ Under closed-list PR every list's votes enter the seat allocation, and the share of votes that buys a seat falls with district magnitude (Taagepera and Shugart 1989), so votes shaved from any list improve the seat quotients of whichever established party controls the booth, and the lower the threshold, the more likely a given transfer is to move a seat. Because voters cannot alter the order of a closed list, the party organization rather than any individual candidate captures the gain (Carey and Shugart 1995), which is the setting in which electoral rules have been found most permissive of malfeasance (Persson, Tabellini and Trebbi 2003), although Chang and Golden (2007) find that this ranking of closed and open lists reverses at large district magnitudes. Larger magnitudes also sustain more lists (Cox 1997): Table 1 shows that the presidential race had ten candidates but an effective number of 2.3 parties, whereas the Senate race had 29 lists, an effective number of 5.3 parties, and 43 percent of valid votes cast for lists outside the top two. More competing lists means more small-list votes for established-party poll workers to redistribute among themselves, under a seat rule in which those votes count.

¹⁰See SA Figure C5 for a newspaper infographic from ABC Color (2018a) alluding to how colluding coalitions of poll workers could alter the count.

¹¹Lists were closed and blocked in 2018. Law 6318/2019 unblocked them from the 2021 municipal elections onward.

Prediction 3 *A majority of established party poll workers will have a stronger negative effect on the vote share of small, non-established parties under PR than under SMDP.*

Two corollaries follow within each rule. Within PR, the collusion effect should increase with district magnitude, while within plurality races, the gains of the party that controls the booth should be larger where the race is close. Section IV.A tests both.

II. Data

A. Data Sources

Our unit of observation is an electoral table by race combination. The 2018 general election had approximately 1,100 polling stations containing roughly 21,000 electoral tables, each producing vote tallies for six races.¹² We combine three data sources: booth-level electoral results, voter-level partisan affiliations, and poll worker identities.

Table 1 describes the six races. The two plurality races were closer contests among fewer effective parties: the presidential winner led by 3.9 percentage points with 6 percent of valid votes cast outside the top two candidates, and 6 of the 17 governor races were decided by less than 5 points. The closed-list PR races drew many more lists and spread the vote more widely, most of all the Senate, where 29 lists competed, and the two established parties together received 57 percent of valid votes. The Chamber of Deputies is the race whose district magnitude varies, from one seat in Alto Paraguay and Boquerón to twenty in Central, and the effective number of parties rises with it, from 1.8 in Alto Paraguay to 5.3 in Central (SA Table E5). Section IV.A uses this variation, and the governor margins, to test the seat-vote logic behind Prediction 3.

We obtained booth-level vote counts published online by the Superior Tribunal of Electoral Justice (Tribunal Superior de Justicia Electoral, TSJE), Paraguay’s election authority (TSJE 2018f), and were able to procure the 2018 electoral registry, which records each voter’s national ID number and table assignment (TSJE 2018d). Because the TSJE administers party primaries, it collects partisan affiliation data from political parties for each registered voter (TSJE 2018e). Matching the registry to the partisan affiliation data through voters’ ID numbers allows us to compute the proportion of partisan support for each party at each table.

We observe poll worker partisanship from two sources. Before each election, parties pre-register their volunteer poll workers with the TSJE, giving us the *de jure* allocation (TSJE 2018c). However, on election day many pre-registered poll workers do not appear, so the pre-registered data alone do not capture the *de facto* composition of poll workers at each table.

To measure the *de facto* allocation, we scraped the TSJE’s website for scans of quick count vote tallies from each table and race (TSJE 2018a), leading to approximately 120,000 tallies in total.¹³

¹²Residents of the capital and abroad do not vote for governors nor departmental legislators.

¹³SA Figure D1 shows all the quick count tallies from one electoral table.

Table 1: The six races of the 2018 general election

Race	Rule	Districts	District magnitude	Lists on the ballot	Effective number of parties	Winner's vote share	Top-two margin	Share outside the top two
President	Plurality	1	1	10	2.25	49.0	3.9	6.0
Governor	Plurality	17	1	7.2	2.41	49.6	8.7	9.6
				[4–14]	[2.05–4.12]	[35.7–62.7]	[0.3–34.0]	[1.4–40.9]
Senate	Closed-list PR	1	45	29	5.31	32.5	8.3	43.3
Chamber of Deputies	Closed-list PR	18	4.4	15.5	3.20	47.3	15.0	20.4
			[1–20]	[7–32]	[1.84–6.57]	[30.3–71.4]	[1.5–54.3]	[5.4–54.0]
Departmental council	Closed-list PR	17	14.5	16.7	3.33	44.5	12.4	23.5
			[7–21]	[7–31]	[2.47–6.00]	[31.1–51.5]	[0.1–25.5]	[11.3–48.7]
MERCOSUR parliament	Closed-list PR	1	18	29	3.65	42.1	11.9	27.8

Note: One row per race. Districts is the number of seat-allocating districts (the country for the president, the Senate, and the MERCOSUR parliament, the department otherwise) and district magnitude the number of seats per district. Lists on the ballot counts the candidates or lists that received votes in the district. The effective number of parties is the Laakso and Taagepera (1979) index $1/\sum_i s_i^2$ over the vote shares s_i of all lists in the district. The winner's vote share, the margin between the first- and second-placed lists, and the share outside the top two are in percent of valid votes. For races with several districts, cells report the unweighted mean across districts with the minimum and maximum in brackets. Votes are aggregated from the booth-level results used in the analysis, which exclude the booths whose tallies were annulled after a challenge and therefore record no valid votes, so they differ slightly from the official totals.

These tallies required poll workers to sign and record their national ID numbers.¹⁴ We digitized these ID numbers and matched them against the pre-registered roster to identify replacements. We then merged in the partisan affiliation data to obtain each replacement poll worker's party. Furthermore, we recovered the ID numbers of all three poll workers at 20,365 of the 21,211 booths (96.0 percent).¹⁵

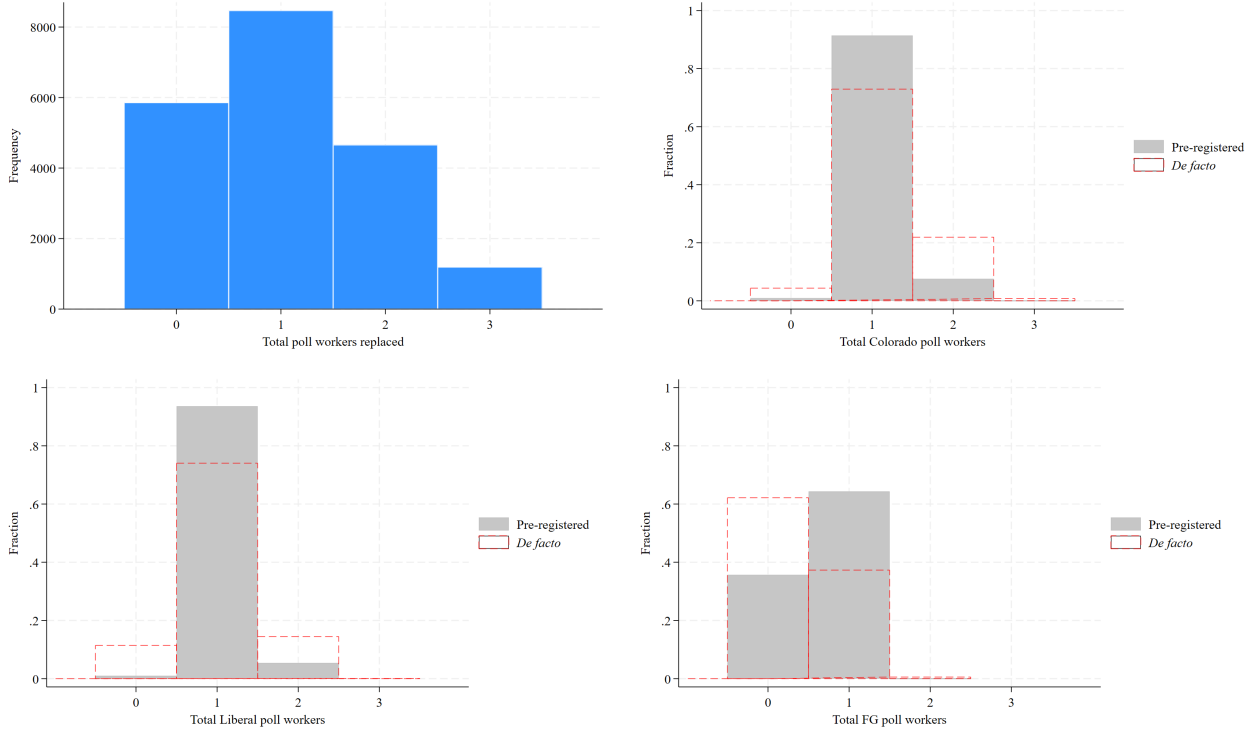
B. Descriptive Patterns

Figure 1 summarizes the gap between the *de jure* and *de facto* allocation of poll workers. Panel 1 shows that most tables experienced at least one replacement. More than 21,000 poll workers (over a third of those pre-registered) were replaced. Panels 2–4 show the consequences: the pre-registered allocation (gray histograms) is relatively balanced, with each major party typically holding one seat

¹⁴Although some poll workers could have forged the signature and ID number of pre-registered poll workers we find this unlikely for several reasons. The first panel of Figure 1 shows more than a third of all pre-registered poll workers get replaced, showing that a substantial number of replacement poll workers acknowledge they are not the pre-registered ones. It was also not expected that the quick count vote tallies would be made available to the wider public, because these elections were the first ones where this occurred. Third, working as a poll worker is anecdotally a task that ambitious members of the established parties (Colorado and Liberal) perform to climb their parties' political ladder. Recording their names and ID numbers honestly allows party officials to know who supported the party as poll workers during the elections.

¹⁵Most of the remaining 846 booths are remote booths (631 in total) whose results were transmitted by telephone, so their tallies carry the name of the election authority employee who took the call instead of the poll workers' ID numbers, the rest have one or more illegible or missing ID numbers. Our regressions use the booths with all three ID numbers.

Figure 1: Pre-registered and *de facto* allocation of partisan poll workers



per table, but the *de facto* allocation (dashed outlines) is skewed toward the Colorado and Liberal parties, with FG and smaller parties losing a substantial share of their poll workers. Further details on the data construction, descriptive statistics, and comparisons between pre-registered and *de facto* poll workers are in SA Section D.

III. Empirical Strategy

Our identification strategy exploits the quasi-random assignment of voters to booths within polling stations. As described in Section I, voters within each station are sorted alphabetically by surname and assigned to booths in groups of 200.¹⁶ If surnames are uncorrelated with political preferences, the assignment of voters to booths is as good as random conditional on the polling station, so booths within a station have comparable electorates. The assignment of poll workers to booths is not random, however: parties choose where to pre-register workers and which vacancies to fill. Our identifying assumption therefore conditions on the polling station by electoral race fixed effects, the affiliation shares of each booth's voters, and, in robustness checks, the pre-registered allocation. Given these controls, the election-day composition of the poll workers is assumed to be unrelated to the booth's potential recorded outcomes. Section III.A examines the two ways in which this assumption could fail and the checks that address them. Following Casas, Díaz and

¹⁶Given that most polling stations do not have an exact multiple of 200 voters, the last booth in each station often does not have exactly 200 voters.

Trindade (2017), we first test the comparability of electorates at each booth within polling stations directly.

To our knowledge, we conduct the most direct test of this identification strategy in the literature. Prior work using alphabetic assignment has relied on proxies for political preferences: Cantú (2014) tests correlations with TV viewership and conditional cash transfer receipt, while Casas, Díaz and Trindade (2017) test correlations with congressional candidates' parties and municipal employee wages. We instead test the association between surname initials and voters' actual partisan affiliation as recorded by the election authority. We confirm that surname initials are uncorrelated with partisan support, and that our results are robust to controlling for the distribution of surname initials. Detailed results are in Appendix A.

This design allows us to exploit variation within polling stations between booths. Polling station fixed effects absorb all station-level confounders, including local socioeconomic conditions, political competitiveness, state capacity, and urban-rural differences. One limitation is that we cannot capture variation in electoral malfeasance across polling stations, as station fixed effects net these out.

We operationalize the partisan composition of poll workers using indicator variables that together cover all booths. A “trifecta” indicates all three poll workers support the same party (Colorado or Liberal, entered separately). An “advantage” indicates two of three poll workers support the same party, with four subcategories: Colorado advantage + Liberal, Colorado advantage + Other, Liberal advantage + Colorado, and Liberal advantage + Other, where “other” denotes a poll worker not registered as supporting either the Colorado or Liberal party.¹⁷ The omitted category is a fair allocation: one Colorado, one Liberal, and one other poll worker. A residual “other poll worker allocation” category captures remaining configurations (e.g., two FG poll workers or two unaffiliated poll workers). These indicators are mutually exclusive and exhaustive.

If we treated our data as a natural experiment, we would estimate polling station fixed effects with booth-level vote shares as dependent variables for (1) the Colorado party, (2) the Liberal party

¹⁷FG, the third party entitled to a seat, is grouped with the other parties rather than treated separately because it is the party whose workers are most often gone on election day: FG had a pre-registered worker at 64 percent of booths but an actual one at 38 percent, held two of the three seats at 113 booths (0.6 percent), and never held all three. In addition, FG's own vote share is not a separate outcome in the presidential race or in the departments where it ran inside the GANAR alliance.

or GANAR alliance,¹⁸ and (3) all other parties:

$$Y_{ab} = \beta_0 + \beta_1 \text{Colorado_Trif}_{ab} + \beta_2 \text{Colorado_Adv_Oth}_{ab} + \beta_3 \text{Colorado_Adv_Liberal}_{ab} \\ + \beta_4 \text{Liberal_Adv_Colorado}_{ab} + \beta_5 \text{Liberal_Adv_Oth}_{ab} + \beta_6 \text{Liberal_Trif}_{ab} + \beta_7 \text{Other_pw_alloc}_{ab} \\ + \theta_a + \varepsilon_{ab},$$

where Y is an electoral outcome for voting booth b belonging to polling station a , and θ is a polling station fixed effect. The coefficients of interest are the β 's, which capture the effect of different partisan poll worker allocations on electoral results.

However, we must address two potential selection issues. First, parties may strategically pre-register poll workers at certain tables over others based on the local composition of voters. Second, parties with greater local operational capacity may be better able to fill election-day vacancies with copartisans. We explore these two threats below.

A. Exploring Selection

Given our use of polling station fixed effects, the selection concerns are within-station strategies parties can take. We distinguish two types: *central strategies* and *capacity-based network strategies*, both of which threaten identification if not controlled for.

Central strategies involve pre-election decisions by party organizations. Knowing each table's copartisan voter share, parties may place copartisan poll workers where their own voters concentrate (to protect votes) or where opposing voters concentrate (to dampen the opposition). Because this behavior is systematic, it can confound our estimates if uncontrolled.

Capacity-based network strategies, in contrast, operate on election day. Parties with more affiliated voters near a given table are better positioned to fill vacancies with copartisans. This creates a mechanical correlation: (1) tables with more copartisan voters are more likely to record favorable outcomes for that party, and (2) those same tables are more likely to receive copartisan replacement poll workers.

Table 2 tests for these selection mechanisms, regressing the proportion of affiliated voters at each table (Colorado, Liberal, and other parties) on both the pre-registered and *de facto* partisan allocation of poll workers, with polling station fixed effects.

Table 2 offers no evidence of central strategies and some evidence of capacity-based selection by the Liberal party. The pre-registered rows are informative only for the two allocations that occur at a meaningful number of booths, a Colorado advantage + Liberal (1,541 booths) and a Liberal advantage + Colorado (1,099 booths), and both are precisely estimated zeros. Relative to a fair pre-registered slate in the same station, a booth with a pre-registered Colorado advantage

¹⁸For several races, the Liberal party and several smaller parties banded together to present a single "GANAR" alliance (Liberal party, PRF, PDP, PEN, Avanza País, PMAS, and FG). Yet in practice, the Liberal party mainly ran its own candidates through the alliance, with FG running separately.

Table 2: Exploring party allocation strategies according to the proportion of affiliated voters

	% of Colorado voters (1)	% of Liberal voters (2)	% of other party voters (3)
Prereg. Colorado trifecta	1.5656 (1.9736)	-0.8783 (1.2282)	-1.3278 (0.6598)
Prereg. Colorado advantage + Other	-2.4710 (1.2812)	-0.3698 (1.0738)	-0.1382 (0.4615)
Prereg. Colorado advantage + Liberal	0.0903 (0.2092)	-0.1924 (0.1768)	0.0870 (0.0638)
Prereg. Liberal advantage + Colorado	0.0156 (0.2296)	0.1055 (0.2105)	0.0388 (0.0703)
Prereg. Liberal advantage + Other	-5.9477 (3.0004)	4.2494 (6.0294)	0.2792 (0.2751)
Prereg. other poll worker allocation	-5.8645 (2.6991)	3.4061 (2.5910)	-0.3407 (0.4298)
<i>De facto</i> Colorado trifecta	0.1796 (0.4106)	0.0973 (0.3027)	-0.0496 (0.1796)
<i>De facto</i> Colorado advantage + Other	-0.0645 (0.1945)	-0.0037 (0.1551)	0.0407 (0.0903)
<i>De facto</i> Colorado advantage + Liberal	-0.0497 (0.1155)	0.0784 (0.0989)	-0.0212 (0.0390)
<i>De facto</i> Liberal advantage + Colorado	-0.1689 (0.1318)	0.2472 (0.1209)	-0.0086 (0.0473)
<i>De facto</i> Liberal advantage + Other	-0.3169 (0.3499)	0.8677 (0.3291)	-0.0517 (0.1231)
<i>De facto</i> Liberal trifecta	1.9738 (1.4385)	0.2528 (1.2088)	-0.4122 (0.4611)
<i>De facto</i> other poll worker allocation	-0.1349 (0.1352)	0.1771 (0.1116)	0.0144 (0.0523)
Mean of Dependent Variable	34.1749	17.3445	4.0764
Polling station FE	✓	✓	✓
Observations	20,354	20,354	20,354
R^2	0.7523	0.7930	0.4503

Note: A unit of observation is a voting booth. Standard errors are robust. The regressors correspond to either pre-registered or election-day (*de facto*) partisan poll worker allocations. A trifecta indicates all three poll workers from the same party work at the voting booth. There is no pre-registered Liberal trifecta in our data so we do not include it as a regressor. Advantage indicates that two out three poll workers from the same party work at the voting booth. Other refers to a poll worker that is not registered as supporting the Colorado or Liberal party. The excluded categories are voting booths with a pre-registered “fair allocation” and, for the *de facto* regressors, voting booths with a *de facto* “fair allocation”, in each case one Colorado poll worker, one Liberal poll worker, and one “other” poll worker. The “other poll worker allocation” refers to any remaining poll worker allocation after accounting for “fair allocations” and the other allocations referenced with the independent variables. Of the 195 booths in the pre-registered “other” category, 183 have no pre-registration record and are absorbed by the polling station fixed effects, so the coefficient rests on the remaining 12 booths.

+ Liberal has 0.09 pp more Colorado voters (SE 0.21) and a booth with a pre-registered Liberal advantage + Colorado has 0.11 pp more Liberal voters (SE 0.21), against means of 34.2 and 17.3 percent. Neither party pre-registers its majorities where either its own or the other party's voters concentrate.¹⁹

The *de facto* rows tell a different story for the Liberal party. Relative to a booth that ended election day with a fair allocation, one that ended with two Liberal poll workers and one other has 0.87 pp more Liberal voters (SE 0.33), and one with two Liberal poll workers and one Colorado poll worker has 0.25 pp more (SE 0.12). No *de facto* Colorado allocation is associated with the share of Colorado voters: the estimates for the Colorado trifecta and the two Colorado advantages lie between -0.06 and 0.18 pp, with standard errors of 0.12 to 0.41 , against a mean of 34.2 percent. The *de facto* Liberal trifecta, which occurs at only 17 booths, is too rare to estimate precisely. Colorado-majority booths are indistinguishable from fair booths in the composition of their voters, both before and after election-day replacements.

This asymmetry is consistent with the parties' organizational structures. The Colorado party, which has dominated the central bureaucracy since its 1954–1989 dictatorship, runs coordinating groups of public employees inside state institutions that register civil servants and their relatives, monitor whether they voted, and commit each employee to mobilizing additional voters (Última Hora 2018b, 2022, 2023). This machinery can supply a copartisan replacement at any booth, whatever the number of Colorado voters standing in that booth's line, so the party's election-day allocations do not track booth-level voter composition. The Liberal party, whose strength is concentrated at the municipal and departmental level in certain strongholds (BTI 2024) and is fragmented across intra-party factions with different regional bases (Carrizosa 2018), fills vacancies through local networks, and a Liberal replacement is more likely to be at hand where more Liberal voters are present.

The Liberal-party correlations show that election-day selection is a potential confounder: booths with more Liberal voters are both more likely to receive Liberal replacements and more likely to record Liberal votes. We address this by controlling for the proportion of voters supporting the Colorado, Liberal, and other parties at each table.²⁰ In addition, because we pool all electoral races, we include fixed effects for every combination of polling station and electoral race. This

¹⁹The remaining pre-registered categories cannot speak to this question. A pre-registered Colorado trifecta occurs at 5 booths, a Colorado advantage + Other at 8, and a Liberal advantage + Other at 2, so their coefficients rest on a handful of within-station comparisons. The residual pre-registered category (195 booths) is no better: 183 of them, 134 abroad and 49 in Paraguarí, have no pre-registration record at all and make up every sampled booth of their eight polling stations, so the fixed effects absorb them and the coefficient rests on 12 booths.

²⁰These covariates are informative: 76.6 percent of registered voters have a party affiliation recorded in the 2018 electoral registry. The three controls count voters registered with a single party, as these are "pure" partisans (on average 34.2 percent Colorado, 17.3 percent Liberal, and 4.1 percent another party at a booth). The remaining 21 percent of voters are registered with more than one party, as affiliations are not regulated.

allows us to compare only across booths from the same station and race, accounting for differences in candidate characteristics and party options across races. Our main specification does not control for pre-registered allocations, since the actual poll worker composition on election day is what matters for the count.²¹ Our main specification is:

$$\begin{aligned}
Y_{abc} = & \beta_0 + \beta_1 \text{Colorado_Trif}_{ab} + \beta_2 \text{Colorado_Adv_Oth}_{ab} + \beta_3 \text{Colorado_Adv_Liberal}_{ab} \\
& + \beta_4 \text{Liberal_Adv_Colorado}_{ab} + \beta_5 \text{Liberal_Adv_Oth}_{ab} + \beta_6 \text{Liberal_Trif}_{ab} + \beta_7 \text{Other_pw_alloc}_{ab} \\
& + \delta_1 \text{voter_prop_Colorado}_{ab} + \delta_2 \text{voter_prop_Liberal}_{ab} + \delta_3 \text{voter_prop_Other}_{ab} + \eta_{ac} + \epsilon_{abc}.
\end{aligned} \tag{1}$$

where Y is an outcome for race c at voting booth b belonging to station a , and η is a polling station by electoral race fixed effect. The coefficients of interest are the β 's, which capture the effect of partisan poll worker allocations on electoral results after controlling for voters' preferences and parties' allocation strategies.²²

B. Electoral System Heterogeneity

We next test whether poll worker effects differ by the electoral system. As separate ballots exist for separate races, poll workers have independent opportunities to alter their behavior across races at the same table.

We generate an indicator variable for PR races (senatorial, MERCOSUR parliament, deputies, and departmental council) and interact it with each poll worker allocation indicator:

$$\begin{aligned}
Y_{acb} = & \beta_0 + \alpha_1 \text{Colorado_Trif}_{ab} \times PR_c + \alpha_2 \text{Colorado_Adv_Oth}_{ab} \times PR_c + \alpha_3 \text{Colorado_Adv_Liberal}_{ab} \times PR_c \\
& + \alpha_4 \text{Liberal_Adv_Colorado}_{ab} \times PR_c + \alpha_5 \text{Liberal_Adv_Oth}_{ab} \times PR_c + \alpha_6 \text{Liberal_Trif}_{ab} \times PR_c \\
& + \alpha_7 \text{Other_pw_alloc}_{ab} \times PR_c + \beta_1 \text{Colorado_Trif}_{ab} + \beta_2 \text{Colorado_Adv_Oth}_{ab} + \beta_3 \text{Colorado_Adv_Liberal}_{ab} \\
& + \beta_4 \text{Liberal_Adv_Colorado}_{ab} + \beta_5 \text{Liberal_Adv_Oth}_{ab} + \beta_6 \text{Liberal_Trif}_{ab} + \beta_7 \text{Other_pw_alloc}_{ab} \\
& + \delta_1 \text{voter_prop_Colorado}_{ab} + \delta_2 \text{voter_prop_Liberal}_{ab} + \delta_3 \text{voter_prop_Other}_{ab} + \eta_{ac} + \epsilon_{abc},
\end{aligned} \tag{2}$$

where Y is an electoral outcome for race c at voting booth b belonging to polling station a , and η is a polling station by electoral race fixed effect. The PR main effect is absorbed by the station-

²¹Table G5 shows that adding them leaves the estimates essentially unchanged. SA Table G6 goes further and keeps only the 86 percent of booths whose pre-registered allocation was fair, so that all variation in the *de facto* allocation comes from election-day replacements. The estimates are again similar. SA Table G9 splits the fair reference category by the party of the third poll worker: the coefficients of interest are unchanged, and relative to a fair allocation whose third worker is registered with FG, one whose third worker is registered with another party records a 0.35 pp lower Liberal/GANAR share and a 0.25 pp higher other-party share.

²²Throughout our analyses we use robust standard errors. Because the six races of a booth are counted by the same three workers, SA Tables G7 and G8 re-estimate Tables 3 and 4 with standard errors clustered by voting booth, which allows arbitrary correlation across the races counted at the same booth. In Table 3 the standard errors roughly double, while in Table 4 the main effects' standard errors rise by about 30 percent and the interaction terms become more precise. The two common Colorado-majority configurations, the Liberal advantage + Colorado effect, and the PR differential for the Colorado advantage + Liberal configuration all remain significant at the 5 percent level.

race fixed effects. The coefficients of interest are the α 's, which capture the differential effect of partisan poll worker allocations in PR races relative to SMDP races.

IV. Results

Our main dependent variables are the vote shares each political party receives at each table.²³ Overall, we find small effects with a consistent sign: booths where the Colorado party holds a poll worker majority record higher Colorado and lower Liberal vote shares, and the Liberal majority booths raise the Liberal vote share.

Table 3 estimates equation (1), pooling all races, to test Predictions 1 and 2 from Section I.C. The dependent variables are vote shares for the Colorado party, the Liberal/GANAR alliance, and other parties. Column 1 shows that the Colorado vote share increases by 0.43–0.51 pp when the Colorado party has two poll workers. A Liberal trifecta reduces the Colorado vote share by 1.95 pp, though we interpret this cautiously given that only 17 tables have a Liberal trifecta compared to 157 with a Colorado trifecta.

Column 2 shows that the Liberal/GANAR vote share increases by 0.21 pp when two Liberal poll workers and one Colorado poll worker are present. The Liberal/GANAR vote share decreases symmetrically when the Colorado party holds a majority: by 0.38 pp with two Colorado and one Liberal worker, 0.50 pp with two Colorado and one other worker, and 0.51 pp with a Colorado trifecta. “Other poll worker allocations” reduce the Liberal/GANAR vote share by 0.16 pp, these configurations (which consist of 12% of tables in our sample) include cases where other parties hold a majority.

Column 3 shows that other parties’ vote share decline when they lack representation: vote shares decrease by 0.13–0.17 pp when all three poll workers belong to the Colorado and Liberal parties, regardless of which holds the majority. Other poll worker allocations increase other parties’ vote share by 0.13 pp. In sum, holding a poll worker majority benefits a party’s vote share while lacking representation reduces it, consistent with Predictions 1 and 2.²⁴

In SA Section E we present additional specifications and robustness checks. Notably, replacing our composition indicators with simple presence indicators (such as whether at least one Colorado, Liberal, or other-party worker is present) yields weaker results, consistent with the institutional requirement that a majority of poll workers approve the official tally. The estimates are also essentially unchanged when we control for each booth’s pre-registered allocation (Table G5), which indicates that they are driven by election-day deviations from the TSJE pre-registered allocations.

²³The vote shares have valid votes (turnout minus null and blank votes) as the denominator as only valid votes are taken into account when calculating the congressional seat shares in PR races and determining the winner in plurality races. We focus on vote shares instead of absolute votes as vote shares do not confound the results with differences in turnout. In Appendix B we estimate the total number of votes affected by partisan poll workers through a simulation exercise.

²⁴SA Tables G16 to G21 show that our results hold for each race separately.

Table 3: Effect of partisan poll worker composition on election results

	Colorado vote share (1)	Liberal/GANAR vote share (2)	Other party vote share (3)
Colorado trifecta	0.2820 (0.2029)	-0.5121 (0.1847)	0.2301 (0.1752)
Colorado advantage + Other	0.4318 (0.0954)	-0.4969 (0.0790)	0.0651 (0.0820)
Colorado advantage + Liberal	0.5125 (0.0493)	-0.3800 (0.0455)	-0.1325 (0.0433)
Liberal advantage + Colorado	-0.0328 (0.0525)	0.2063 (0.0505)	-0.1735 (0.0483)
Liberal advantage + Other	-0.2126 (0.1611)	-0.0035 (0.1406)	0.2161 (0.1442)
Liberal trifecta	-1.9481 (0.5951)	0.4756 (0.5947)	1.4724 (0.4649)
Other poll worker allocation	0.0246 (0.0646)	-0.1593 (0.0597)	0.1347 (0.0578)
Mean of Dependent Variable	41.5808	34.4445	23.9746
Polling station-elect. race FE	✓	✓	✓
Colorado voter prop. control	✓	✓	✓
Liberal voter prop. control	✓	✓	✓
Other party voter prop. control	✓	✓	✓
Observations	117,229	117,229	117,229
R^2	0.8425	0.8869	0.9308

Note: A unit of observation is a voting booth-electoral race combination. Standard errors are robust. The Liberal always ran separately or through the GANAR alliance, hence we have an outcome variable for Liberal/GANAR vote shares. Vote shares are calculated relative to valid votes. The excluded category corresponds to voting booths with “fair allocations” having one Colorado party poll worker and one Liberal poll worker.

A. Heterogeneity by Electoral System

As discussed in Section I, PR races provide greater scope for established-party collusion because more small parties compete. Table 4 estimates equation (2) to test Prediction 3. Column 1 shows no differential effect of poll worker allocations on Colorado vote shares by electoral system. Column 2 shows that the interaction of PR race with a Colorado majority plus one Liberal worker is positive and significant at the 10 percent level (0.17 pp), indicating that Liberal/GANAR vote shares increase more under this configuration in PR races. Column 3 shows that the interaction of a PR election with a Colorado majority plus one Liberal worker reduces other parties' vote shares (0.30 pp, significant at the 1 percent level), and that the interaction with a Liberal majority plus one Colorado worker does so as well (0.14 pp, significant at the 10 percent level).

Two patterns support the collusion interpretation consistent with Prediction 3. First, having three established-party poll workers, regardless of which party holds the majority, is differentially detrimental to other parties in PR races, with the caveat that the differential for a Liberal majority is estimated less precisely. Second, a Colorado majority with one Liberal worker differentially increases the Liberal vote share in PR races, suggesting that the minority Liberal worker is less harmed by the Colorado party majority compared to SMDP. Two caveats apply. The electoral rule is bundled with the office, since the plurality races are executive contests and the PR races legislative ones, so the interaction also absorbs differences in salience, ballot complexity, and the number of competing lists. And other parties hold 43 percent of valid votes in the Senate race but 6 percent in the presidential race (Table 1), so there are more small-party votes to lose under PR. The within-rule tests further below seek to address this.

Because the coefficients in Table 4 are interactions, the effect of an allocation within a given electoral system is the sum of its main effect and its interaction with the PR indicator. SA Figures E1 to E3 plot these marginal effects separately for SMDP and PR races, along with their difference, for each outcome. They make the collusion pattern concrete. In SMDP races, a booth with two Colorado poll workers and one Liberal poll worker leaves other parties' vote share unchanged (0.07 pp, not significant), whereas in PR races the same booth lowers it by 0.23 pp ($p < 0.01$). A booth with two Liberal poll workers and one Colorado poll worker likewise moves from no effect in SMDP races (-0.08 pp) to -0.22 pp in PR races ($p < 0.01$). On the Liberal/GANAR vote share, a Colorado majority with one Liberal worker costs the Liberals 0.50 pp in SMDP races but 0.32 pp in PR races. Colorado vote shares, by contrast, respond to the same allocation by similar amounts under both systems (0.43 pp in SMDP and 0.55 pp in PR races), which is why Column 1 shows no significant interactions.

Why does the electoral system matter? Under PR, every list's votes count toward the allocation of seats, so votes shaved from small lists translate directly into seats for whichever established party controls the booth. Under plurality, a vote taken from a small party adds one vote to the winner's

margin and a vote taken from the rival adds two, so the votes worth moving are the rival established party's, and only where the race is close. The marginal effects above follow this logic: established-party majorities harm other parties only in PR races, and a Colorado majority harms the Liberal list less in PR races, where its target shifts to the small lists. An alternative reading would run through coalition membership rather than the counting rule. In the presidential race, and in some departments' governor, deputies, and departmental council races, the Liberal party ran jointly with FG and other small parties under the GANAR label, so a Liberal poll worker there had reason to protect rather than harm those parties' votes, whereas in the Senate and MERCOSUR races, and elsewhere, it ran on its own list. SA Table E4 tests this channel directly by interacting the allocation indicators with whether the Liberal party ran under GANAR rather than with the electoral system. Alliance status makes no difference: a Liberal majority with one Colorado worker reduces other parties' vote share by 0.18 pp where the Liberals ran on their own list, and the interaction with running under GANAR is 0.01 pp and far from significant. Within the deputies and departmental council races, where alliance status varies across departments but the electoral system does not, the corresponding figures are 0.25 pp and 0.004 pp. The differential is therefore about how votes are counted, not about who shares the Liberal list. These findings suggest that the irregularity documented in SA Figure C3 may be small but as systematic as anecdotal media reports suggest (ABC Color 2018a; Colmán Gutiérrez 2018).²⁵

The two corollaries stated in Section I.C test this logic within each rule, using the variation described in Table 1. SA Table E6 re-estimates the deputies race with each allocation interacted with the log of the district's magnitude, which runs from one seat to twenty. The harm that two established-party workers inflict on other parties grows with magnitude: the interaction is -0.28 pp per log unit for a Liberal majority with one Colorado worker (wild-cluster-bootstrap $p = 0.06$ with 18 department clusters) and -0.07 pp for a Colorado majority with one Liberal worker. In the eight districts electing one or two deputies, where closed-list PR comes closest to plurality, neither allocation reduces other parties' vote share (0.14 and 0.20 pp), whereas in the larger districts both do (-0.25 and -0.34 pp). SA Table E7 pools the deputies and departmental council races and adds allocation-by-department and allocation-by-race dummies, so that magnitude is identified only from the gap between the two races' magnitudes within a department (together with other differences between the two offices within a department), the interactions are -0.40 and -0.41 pp per log unit (bootstrap p of 0.07 and 0.08). SA Table E8 turns to the governor race, the plurality race whose margin varies across the 17 departments, and interacts each allocation with the department's 2018 top-two margin. A Colorado majority with one Liberal worker raises the Colorado share by

²⁵In SA Table E9, we pool Colorado and Liberal poll workers as "traditional party" poll workers (the term used in Paraguayan political discourse for these established parties) and contrast them with other party poll workers. The results reinforce the collusion interpretation: a traditional party trifecta significantly reduces other parties' vote shares, and this effect is compounded in PR races.

0.53 pp and lowers the Liberal share by 0.65 pp in a race decided by a hair, and each ten points of margin shrink these effects by 0.25 and 0.31 pp (bootstrap p of 0.20 and 0.06), so that they vanish in the lopsided races. For a Liberal majority with one Colorado worker the gradient runs the other way. With 17 departments this is a low-powered test, and the margin is the realized 2018 margin, which the allocations may themselves have shifted, so the closeness interaction describes heterogeneity by ex-post margin rather than a test of anticipated closeness. We read the results as consistent with the seat-vote logic for the magnitude channel and inconclusive for closeness.

B. Mechanisms

Section I.A described three ways in which poll workers could alter a booth’s recorded result: recording votes cast for absent parties as votes for the parties present, annulling ballots cast for opponents, and adding or suppressing ballots. Each leaves a different trace. Reassignment on the tally changes the division of votes but not their number, so it should leave turnout, blank votes, and null votes unaffected. Annulment raises the null vote share at booths with a partisan majority. Ballot stuffing or suppression moves turnout. We therefore use each of these process outcomes as a dependent variable in SA Table E10, which lets the data discriminate among the modes even though we cannot observe the count itself.²⁶ An alternative mechanism is the irregularity mentioned by ABC Color (2018a), the newspaper of record of Paraguay, and also documented in SA Figure C3. This latter residual category (among others we are not able to examine with specific dependent variables) consists of retaining the same total number of votes (thus not affecting turnout, null votes, or blank votes), but redistributing votes from parties lacking poll workers among those that have workers present.

If partisan poll workers monitor turnout-buying agreements or suppress opposition turnout, we would expect turnout to vary with poll worker composition. We find limited evidence: tables with a Liberal advantage + Other composition have slightly higher turnout (approximately 0.6 additional votes) and Colorado trifectas slightly lower turnout (approximately 0.7 fewer votes), but these results are ambiguous—they could reflect genuine vote-counting or turnout-buying enforcement. The remaining configurations show no significant effects, possibly because these strategies may offset each other (e.g., stuffing ballots vs. discarding opposition votes). Next, if poll workers mark blank ballots for their party, we would expect fewer blank votes at tables with a partisan majority. We find no evidence of this in SA Table E10. Finally, if poll workers annul opposition votes, we would expect more null votes at tables with a partisan majority. Instead, we find significantly fewer null votes at Colorado trifectas and at tables where both established parties are present and one of them holds the majority, suggesting that established-party poll workers “protect” borderline

²⁶A null vote is a ballot invalidated during the count, typically because the voter marked more than one candidate, made extraneous marks, or otherwise defaced the ballot. Null votes are distinct from blank votes, in which the voter deposits an unmarked ballot.

Table 4: Effect of partisan poll worker composition on election results by electoral system

	Colorado vote share (1)	Liberal/GANAR vote share (2)	Other party vote share (3)
Colorado trifecta $\times$ PR race	-0.5625 (0.4166)	0.1364 (0.3948)	0.4261 (0.3124)
Colorado advantage + Other $\times$ PR race	0.0317 (0.1976)	0.0117 (0.1750)	-0.0434 (0.1440)
Colorado advantage + Liberal $\times$ PR race	0.1232 (0.1032)	0.1718 (0.0982)	-0.2949 (0.0754)
Liberal advantage + Colorado $\times$ PR race	0.0514 (0.1097)	0.0887 (0.1072)	-0.1401 (0.0848)
Liberal advantage + Other $\times$ PR race	0.1243 (0.3316)	-0.1741 (0.3116)	0.0498 (0.2596)
Liberal trifecta $\times$ PR race	0.0301 (1.1852)	-0.6823 (1.2005)	0.6522 (0.8066)
Other poll worker allocation $\times$ PR race	0.1203 (0.1356)	-0.1856 (0.1292)	0.0653 (0.1024)
Colorado trifecta	0.6614 (0.3292)	-0.6040 (0.3248)	-0.0574 (0.1983)
Colorado advantage + Other	0.4104 (0.1575)	-0.5047 (0.1485)	0.0943 (0.0877)
Colorado advantage + Liberal	0.4295 (0.0831)	-0.4957 (0.0816)	0.0663 (0.0449)
Liberal advantage + Colorado	-0.0673 (0.0883)	0.1469 (0.0874)	-0.0796 (0.0511)
Liberal advantage + Other	-0.2959 (0.2627)	0.1130 (0.2648)	0.1830 (0.1667)
Liberal trifecta	-1.9683 (0.9019)	0.9328 (0.9306)	1.0355 (0.4775)
Other poll worker allocation	-0.0564 (0.1096)	-0.0345 (0.1075)	0.0909 (0.0635)
Mean of Dependent Variable	41.5808	34.4445	23.9746
Polling station-elect. race FE	✓	✓	✓
Colorado voter prop. control	✓	✓	✓
Liberal voter prop. control	✓	✓	✓
Other party voter prop. control	✓	✓	✓
Observations	117,229	117,229	117,229
R^2	0.8425	0.8869	0.9308

Note: A unit of observation is a voting booth-electoral race combination. Standard errors are robust. The Liberal party always ran separately or through the GANAR alliance (while mainly proposing its own candidates), hence we have an outcome variable for Liberal/GANAR vote shares. Vote shares are calculated by dividing by the number of valid votes (i.e., all votes excluding null and blank votes) and multiplying by 100. The excluded category corresponds to voting booths with “fair allocations” having one Colorado party poll worker and one Liberal poll worker.

ballots from being discarded. However, these coefficients have the opposite sign to what annulment of opposition ballots would produce, and they are a fraction of the corresponding vote share effects in Table 3 for the same configuration, so this channel cannot account for our main findings.

To summarize, the process outcomes are most consistent with reassignment on the tally as the main channel. Annulment and ballot stuffing would leave traces in null votes and turnout, and those traces are small: the null-vote effects are negative rather than positive, and a fraction of the vote share effects, and the turnout effects are a fraction of a vote. Recording votes cast for absent parties as votes for the parties present, while preserving the booth’s total, as the modified tally in SA Figure C3 illustrates, is the mode that best fits this pattern: sizable effects on the division of votes and almost none on their total number. As the ballots themselves were destroyed, this evidence is consistent with reassignment but still not a direct verification of it. For example, other manipulations that preserve the booth’s total, such as enforcing vote-buying agreements (Duarte et al. 2025), would leave the same traces.

V. Conclusion

This paper shows that the partisan composition of poll workers who count the votes affects recorded electoral outcomes. On average, having one additional copartisan poll worker relative to a fair allocation yields approximately a 0.5 pp vote share increase for the Colorado party and a 0.2 pp increase for the Liberal party at each voting booth, independent of voter preferences.²⁷ Our within-station design identifies effects only from variation in poll worker composition across booths within the same station, so it does not capture what happens at stations where every booth has the same partisan composition. If parties deploy uniform strategies across the booths of a station, part of their aggregate effect is absorbed by the fixed effects.

Despite these modest magnitudes, a statistical simulation exercise in Appendix B shows that under a counterfactual fair allocation (one Colorado, one Liberal, and one other poll worker at every table), one departmental council seat would have changed hands from the Colorado to the Liberal party. In close races, small margins determine representation. We also find that these effects are heterogeneous across electoral systems: established-party poll workers are differentially harmful to smaller parties in PR races, where every list’s votes count toward seats, more small lists compete, and the scope for collusion is greater. Parties that lack poll worker representation at a booth are systematically disadvantaged regardless of which established party holds the majority.

These findings are unlikely to be specific to Paraguay. In Honduras’s 2017 elections, media reports revealed recordings of training sessions for the incumbent party’s poll workers that included instructions for electoral irregularities. Among these strategies was obtaining credentials

²⁷While our statistical findings point to an average effect from partisan poll worker majorities, this does not mean all voting booths with this configuration had biased vote counts. Our results do not allow us to uncover the individual voting booths driving our results.

from smaller parties’ poll workers (The Economist 2017), an effort similar to the replacements we document. Those elections were narrowly won by the incumbent amid numerous reported irregularities (New York Times 2017). More broadly, using data on 111 Latin American presidential elections from 1990 to 2018, we document a positive cross-country correlation between the use of partisan poll workers and established-party vote shares (SA Section F).²⁸ These dynamics also have historical roots in the United States. Harris (1934) cataloged how partisan precinct officers in Philadelphia, Chicago, and other cities exploited their positions to manipulate vote counts, and argued that the bipartisan monitoring system failed whenever one party locally dominated, a finding we can corroborate in the Paraguayan context.

Our results speak to the design of adversarial electoral control systems. Their premise is that mutual counting by partisan poll workers makes the count accurate regardless of who sits at the booth. That premise requires that no party control the booth, and in Paraguay it failed at the margin that matters: more than a third of pre-registered poll workers were replaced on election day, the replacements favored the two established parties, and the booths where the balance tipped recorded systematically different results. Three features of this evidence travel beyond the case. The bias arises from election-day replacements, not from the official roster, so rules that balance the roster on paper do not close the channel. The differential harm to small parties is larger in the proportional representation races, where every list’s votes count toward seats, which suggests that the counting rule shapes what poll workers have to gain. And the asymmetry in capacity that produces the bias is the normal condition in the countries that use partisan poll workers, most of which pair an entrenched established party with weaker challengers. Lehoucq (2002) argues that non-partisan election administration produces higher electoral integrity than partisan administration. Our findings supply booth-level evidence for why, and identify the condition under which the partisan alternative breaks down.

One alternative is the random selection of non-partisan poll workers from the general citizenry, as practiced in Mexico, where a double lottery process selects and trains citizens in a jury-duty-style system (Challú, Seira and Simpser 2020). Our findings suggest that moving from partisan to non-partisan poll workers—or at minimum, ensuring that no single party holds a majority of poll workers at any given table—could reduce the biases we document.

The methodology we develop of comparing pre-registered (*de jure*) and actual (*de facto*) poll worker allocations can be applied to other elections where similar data exist.²⁹ More broadly, our findings suggest that the institutional details of election administration — who counts the votes

²⁸Countries that allow partisan poll workers differ in other respects, so we read it only as descriptive evidence that the pattern we identify in Paraguay is not idiosyncratic, not as a causal estimate.

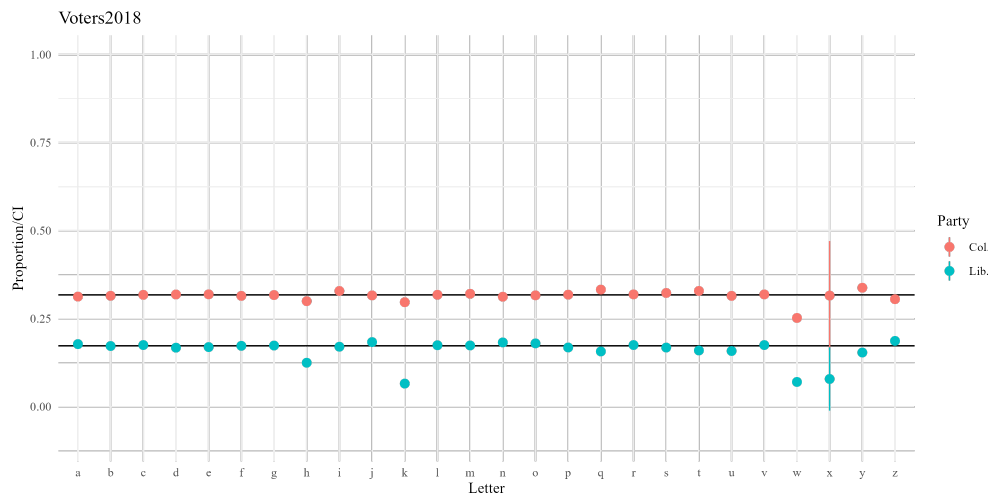
²⁹Although Paraguay has had general elections in 2023, we are unable to replicate our methodology since the election authority no longer required poll workers to include their ID numbers on the quick-count vote tallies, reducing transparency.

and how those counters are selected — deserve the same scrutiny that economists have devoted to electoral rules, voting technology, and election monitoring.

A. Tests for Correlation Between Voter Preferences and Names

Figure A1 plots the distribution of Colorado and Liberal party support among registered voters of the 2018 election for each of the indicated initials of voters' surnames. The horizontal line plots the overall average of Colorado and Liberal party support. If surnames are independent of party identification, we expect a flat distribution of partisan support among the surname initials. We see that voters support these two parties at around the same overall rate across almost all surname initials, with an overall flat distribution of Colorado and Liberal party support. Although there are a few outliers (h, k, w, x, and y), they are spread around different letters and these are letters which have comparatively fewer voters. Figure A2 shows that the few surname initials farther from the overall means all have substantially lower frequencies. This can explain why the proportions for these letters do not approach the overall mean as much as other surname initials. Also, neither Colorado nor Liberal party supporters have surnames concentrated towards the end of the alphabet or vice versa. Hence, voters' first letter of their last name does not correlate with their political preferences.

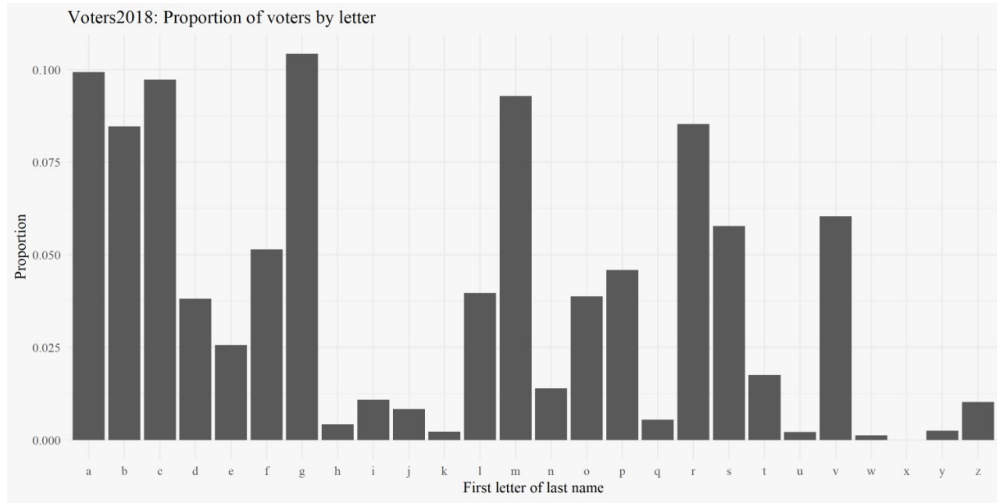
Figure A1: Distribution of party support by voter initials



Note: This figure plots the distribution of Colorado and Liberal party support among registered voters of the 2018 election for each of the indicated initials of voters' surnames. The horizontal line plots the overall average of Colorado and Liberal party support. Figure A2 shows that the surname initials that are farther away from the overall means (h, k, w, x, and y) have substantially lower frequencies. This can explain why the proportions for these letters do not approach the overall mean as much as other surname initials.

Additionally, because we observe every voter's name and booth assignment, we can control for each booth's share of voters by surname initial as a robustness check. Table A1 includes these alphabetic proportion control variables—with "A" being the excluded category. We find that our

Figure A2: Proportion of voters' surname initials by letter



Note: This figure plots the proportion of voters' surname initials by letter. This figure shows how the surname initials that are farthest away from the overall means (h, k, w, x, and y) in Figure A1 have substantially lower frequencies.

main estimates are robust to including these control variables.

We ran additional robustness checks including controlling for the proportion of the 100 most common surnames in the country at each voting booth, and the proportion of voters whose surnames match those of candidates. The main pattern is robust to these controls: the coefficients for the common Colorado-majority configurations and for the Liberal advantage + Colorado configuration keep their sign, size, and precision, while the sparse trifecta categories remain imprecisely estimated. We also test whether the alphabetic ranking of voters' surnames (which determines the booth where they vote) is uncorrelated with electoral results. SA Tables G3, G11, and G12 report the surname controls, and SA Tables G29 to G31 report placebo tests using the surname initial rankings of voting booths.

Table A1: Effect of partisan poll worker composition on election results controlling for surname initial proportions at voting booths

	Colorado vote share (1)	Liberal/GANAR vote share (2)	Other party vote share (3)
Colorado trifecta	0.3642 (0.2025)	-0.5501 (0.1843)	0.1858 (0.1754)
Colorado advantage + Other	0.4606 (0.0951)	-0.4853 (0.0788)	0.0246 (0.0823)
Colorado advantage + Liberal	0.5284 (0.0497)	-0.3680 (0.0457)	-0.1604 (0.0437)
Liberal advantage + Colorado	-0.0349 (0.0524)	0.2253 (0.0505)	-0.1904 (0.0484)
Liberal advantage + Other	-0.2381 (0.1588)	0.0314 (0.1397)	0.2067 (0.1444)
Liberal trifecta	-2.0625 (0.5968)	0.5901 (0.5918)	1.4724 (0.4577)
Other poll worker allocation	0.0456 (0.0644)	-0.1691 (0.0597)	0.1236 (0.0579)
Mean of Dependent Variable	41.5808	34.4445	23.9746
Polling station-elect. race FE	✓	✓	✓
Colorado voter prop. control	✓	✓	✓
Liberal voter prop. control	✓	✓	✓
Other party voter prop. control	✓	✓	✓
Surname initials prop.	✓	✓	✓
Observations	117,229	117,229	117,229
R^2	0.8435	0.8876	0.9309

Note: A unit of observation is a voting booth-electoral race combination. Standard errors are robust. The Liberal party always ran separately or through the GANAR alliance (while mainly proposing its own candidates), hence we have an outcome variable for Liberal/GANAR vote shares. Vote shares are calculated by dividing by the number of valid votes (i.e., all votes excluding null and blank votes) and multiplying by 100. A trifecta indicates all three poll workers from the same party work at the voting booth. Advantage indicates that two out three poll workers from the same party work at the voting booth. Other refers to a poll worker that is not registered as supporting the Colorado or Liberal party. The excluded category corresponds to voting booths with “fair allocations” having one Colorado party poll worker, one Liberal poll worker, and one “other” poll worker. The “other poll worker allocation” refers to any remaining poll worker allocation after accounting for “fair allocations” and the other allocations referenced with the independent variables. “Surname initials prop.” denotes the inclusion of separate indicator variables for the proportion of voters having a surname initial at each booth, with one indicator variable per initial.

B. Simulation Exercise

To estimate the substantive effects of poll workers' partisanship on electoral outcomes, we ran simulations that sought to answer the following question:

If all electoral tables had a fair allocation of partisan poll workers in the 2018 Paraguayan general elections, how much would electoral results have changed?

To answer this question we compared two different data sets. First, we have the actual electoral outcomes recorded in the 2018 general elections. Second, using our regression models for separate elections (SA Tables G16 to G21) we generated a data set of predicted turnout for each electoral table assuming a fair allocation. Hence, for example, if a given table had a Colorado party trifecta, we would replace that table's turnout with what our models predict would have been the turnout for that table if it had a fair allocation instead, and we do this replacement for all tables that had any poll worker allocation other than the fair allocation. Third, we repeated this exercise with vote outcomes. Using our regression models for separate elections we predicted what the electoral outcomes would have been for each voting booth assuming a fair allocation. Fourth, we take these vote share predictions, and we multiply them by the turnout predictions to arrive at our predicted number of votes that each party should have received if there was a fair allocation in that table. For this exercise each party's vote share is defined over all votes cast at the booth, so that the predicted shares multiplied by the predicted turnout sum to the booth's predicted total.³⁰ Specifically, we took all tables with non-fair allocations recorded in our data, and we replaced these cases with predicted outcomes of our models had a given table instead had a fair allocation. Then, with our true vote outcomes and our predicted vote outcomes, we take the difference between (1) true and predicted vote shares, and (2) true and predicted seat distributions for different parties. These exercises give us a sense of how important partisan poll workers are for electoral outcomes in Paraguay. The exercise uses the point predictions of the regression models and does not propagate the sampling uncertainty of the estimated coefficients through the seat allocation, so it illustrates the magnitude of the effects rather than providing confidence intervals for the number of votes or seats affected.

Table B1 shows the simulation results for the presidential election. All else equal, if all tables had a fair allocation of partisan poll workers the ANR (Colorados) would have had 3,052 fewer votes and the Liberal/GANAR alliance would have received 931 more votes. The presidential winner holds, though the simulation still moves thousands of votes.

Table B2 shows the simulation results for the senatorial election. Although some vote differences are sizable—the ANR would have received 2,575 fewer votes under fair allocations—none is large enough to shift a seat.

³⁰However, rather than joining smaller parties into a single “other” category, for the exercises shown in this section our models treated each party separately.

Table B1: Comparison between true and predicted presidential race results

Party	Pred. Vote	True Vote	Vote Diff.
ANR	1,159,013	1,162,065	-3,052
GANAR	1,068,792	1,067,861	931
PVP	80,365	80,554	-189
MNAP	15,064	15,064	-0
PMPP	9,464	9,478	-14
MRP	9,017	9,097	-80
PFA	8,408	8,293	115
SN	6,972	7,018	-46

Note: Party acronyms are defined in SA Section C.1.

Table B2: Comparison between true and predicted Senate race results

Party	Pred. Vote	True Vote	Vote Diff.	Pred. Seats	True Seats	Seat Diff.
ANR	733,089	735,664	-2,575	17	17	0
PLRA	549,968	549,032	936	13	13	0
FG	265,723	265,392	331	6	6	0
PPQ	155,693	155,671	22	3	3	0
PPH	103,475	103,526	-51	2	2	0
PDP	84,216	84,192	24	2	2	0
UNACE	48,438	48,131	307	1	1	0
CN	57,302	56,927	375	1	1	0

Note: Party acronyms are defined in SA Section C.1.

Table B3: Comparison between true and predicted MERCOSUR parliament race results

Party	Pred. Vote	True Vote	Vote Diff.	Pred. Seats	True Seats	Seat Diff.
ANR	914,662	917,471	-2,809	10	10	0
PLRA	661,207	660,090	1,117	7	7	0
FG	100,764	100,051	713	1	1	0
PPQ	84,366	84,478	-112	0	0	0
UNACE	45,873	45,577	296	0	0	0
PPH	39,324	39,584	-260	0	0	0
PVP	38,052	38,444	-392	0	0	0
MRP	23,839	23,851	-12	0	0	0

Note: Party acronyms are defined in SA Section C.1.

Table B4: Comparison between true and predicted deputy race results

Party	Pred. Vote	True Vote	Vote Diff.	Pred. Seats	True Seats	Seat Diff.
ANR	887,847	890,412	-2,565	42	42	0
PLRA	410,254	409,632	622	17	17	0
GANAR	274,395	273,892	503	13	13	0
PPQ	103,588	103,676	-88	3	3	0
PEN	74,335	74,118	217	2	2	0
PPH	74,403	74,415	-12	2	2	0
CN	32,968	32,675	293	1	1	0
PFA	20,350	20,213	137	0	0	0

Note: Party acronyms are defined in SA Section C.1.

Given that the MERCOSUR parliament is elected using a national district we may expect simulations to behave similarly, which Table B3 confirms. Although predicted differences are generally larger in the MERCOSUR races than in the Senate races in absolute terms, every seat allocation holds.

Table B4 contains the results of the simulation exercise for the Chamber of Deputies in Paraguay. Again, we can see that there are comparable vote changes to what we have seen previously, but no race flips. Similarly, in gubernatorial elections the true results showed 13 Colorado winners 1 GANAR winner and 3 Liberal winners, and predicted results show no differences in allocations. Therefore, no governor races were impacted by the effects of poll worker allocations either.

Finally, Table B5 shows one seat changes among Colorado (ANR) and Liberal (PLRA) outcomes in these elections. Again, the change in votes received by each party seems comparable to what we have seen in other models, but in this case, because of the lower-level of departmental council elections, these vote differences were sufficient to change one seat allocation.

Table B5: Comparison between true and predicted departmental council race results

Party	Pred. Vote	True Vote	Vote Diff.	Pred. Seats	True Seats	Seat Diff
ANR	802,041	804,736	-2,695	130	131	-1
PLRA	444,336	442,825	1,511	56	55	1
GANAR	241,506	241,173	333	48	48	0
PVP	41,614	42,000	-386	3	3	0
PPQ	46,879	46,950	-71	2	2	0
UNACE	51,243	50,837	406	1	1	0
PEN	27,541	27,407	134	1	1	0
AP	15,374	15,514	-140	1	1	0

Note: Party acronyms are defined in SA Section C.1.

References

- ABC Color**, “Robo de votos es ‘común’,” 2018.
- , “Votó por primera vez y le ‘robaron’, aseguró,” 2018.
- Albertus, Michael and Victor Menaldo**, *Authoritarianism and the Elite Origins of Democracy*, Cambridge University Press, 2018.
- Ascencio, Sergio J. and Miguel R. Rueda**, “Partisan Poll Watchers and Electoral Manipulation,” *American Political Science Review*, 2019, 113 (3), 727–742.
- Baker, Andy and Kenneth F. Greene**, “Latin American Elections and Ideology Database,” Data set, University of Colorado Boulder 2011. Retrieved from <https://web.archive.org/web/20230330055805/https://www.colorado.edu/faculty/baker/latin-american-elections-and-ideology>.
- **and Kenneth Greene**, “The Latin American left’s mandate: free-market policies and issue voting in new democracies,” *World Politics*, 2011, 63 (1), 43–77.
- Baland, Jean-Marie and James Robinson**, “Land and power: Theory and evidence from Chile,” *American Economic Review*, 2008, 98 (5), 1737–65.
- Birch, Sarah**, “Electoral systems and electoral misconduct,” *Comparative Political Studies*, 2007, 40 (12), 1533–1556.
- Bobonis, Gustavo, Paul Gertler, Marco Gonzalez-Navarro, and Simeon Nichter**, “Vulnerability and clientelism,” *American Economic Review*, 2022, 112 (11), 3627–3659.
- Brennan Center**, “Who Watches the Poll Watchers?,” *Brennan Center for Justice*, 2021.
- BTI**, “Paraguay Country Report 2024,” bti-project.org/en/reports/country-report/PRY, 2024.
- Callen, Michael and James D. Long**, “Institutional corruption and election fraud: Evidence from a field experiment in Afghanistan,” *American Economic Review*, 2015, 105 (1), 354–81.
- Cantú, Francisco**, “Identifying irregularities in Mexican local elections,” *American Journal of Political Science*, 2014, 58 (4), 936–951.
- Card, David and Enrico Moretti**, “Does voting technology affect election outcomes? Touch-screen voting and the 2004 presidential election,” *The Review of Economics and Statistics*, 2007, 89 (4), 660–673.
- Carey, John M. and Matthew Soberg Shugart**, “Incentives to Cultivate a Personal Vote: A Rank Ordering of Electoral Formulas,” *Electoral Studies*, 1995, 14 (4), 417–439.
- Carrizosa, Andres**, “Paraguay 2017: Competencia Política en las cámaras, en las calles y en las urnas,” *Revista de Ciencia Política*, 2018, 38 (2), 335–360.
- Casas, Agustín, Guillermo Díaz, and Andre Trindade**, “Who monitors the monitor? Effect of party observers on electoral outcomes,” *Journal of Public Economics*, 2017, 145, 136–149.
- Challú, Cristian, Enrique Seira, and Alberto Simpser**, “The quality of vote tallies: Causes and

- consequences,” *American Political Science Review*, 2020, 114 (4), 1071–1085.
- Chang, Eric C. C. and Miriam A. Golden**, “Electoral Systems, District Magnitude and Corruption,” *British Journal of Political Science*, 2007, 37 (1), 115–137.
- Colmán Gutiérrez, Andrés**, “El país de los votos perdidos,” *Última Hora*, 2018.
- Cox, Gary**, *Making Votes Count: Strategic Coordination in the World’s Electoral Systems*, Cambridge University Press, 1997.
- Djourelova, Milena and Teodora Tsankova**, “The Electoral Effects of Voting Technology: Evidence from Bulgaria,” *Working Paper*, 2025.
- Duarte, Raul, Frederico Finan, Horacio Larreguy, and Laura Schechter**, “Brokering Votes With Information Spread Via Social Networks,” *The Review of Economic Studies*, 2025.
- Economist Intelligence Unit**, “Democracy Index 2021: The China Challenge,” 2022.
- Ferlenga, Francesco and Brian Knight**, “Vote early and vote often? Detecting electoral fraud from the timing of 19th century elections,” *Journal of Public Economics*, 2025, 243, 105317.
- Ferrara, Marta**, “Mapa de Debilidades del Sistema Electoral, Paraguay 2016,” *Semillas para la Democracia* 2016.
- Finan, Frederico and Laura Schechter**, “Vote-buying and reciprocity,” *Econometrica*, 2012, 80 (2), 863–881.
- Fujiwara, Thomas**, “Voting technology, political responsiveness, and infant health: Evidence from Brazil,” *Econometrica*, 2015, 83 (2), 423–464.
- Garbiras-Díaz, Natalia and Mateo Montenegro**, “All eyes on them: A field experiment on citizen oversight and electoral integrity,” *American Economic Review*, 2022, 112 (8), 2631–2668.
- Gazete Duvar**, “Turkey’s opposition parties say ready to ensure election safety,” 2021.
- Gonzalez, Robert**, “Cell phone access and election fraud: evidence from a spatial regression discontinuity design in Afghanistan,” *American Economic Journal: Applied Economics*, 2021, 13 (2), 1–51.
- Harris, Joseph P.**, *Election Administration in the United States*, Brookings Institution Press, 1934.
- Hicken, Allen**, “Clientelism,” *Annual Review of Political Science*, 2011, 14, 289–310.
- , **Stephen Leider, Nico Ravanilla, and Dean Yang**, “Temptation in vote-selling: Evidence from a field experiment in the Philippines,” *Journal of Development Economics*, 2018, 131, 1–14.
- King, Gary, Michael Tomz, and Jason Wittenberg**, “Making the most of statistical analyses: Improving interpretation and presentation,” *American Journal of Political Science*, 2000, pp. 347–361.
- Laakso, Markku and Rein Taagepera**, ““Effective” Number of Parties: A Measure with Application to West Europe,” *Comparative Political Studies*, 1979, 12 (1), 3–27.
- Lachi, Marcello**, “Recolección de Fondos y Gastos Electorales en las Elecciones Municipales,” 2009. Transparencia Paraguay.

- Lehoucq, Fabrice**, “Can parties police themselves? Electoral governance and democratization,” *International Political Science Review*, 2002, 23 (1), 29–46.
- Lijphart, Arend**, *Electoral Systems and Party Systems: A Study of Twenty-Seven Democracies, 1945–1990*, Oxford: Oxford University Press, 1994.
- Lupu, Noam**, *Party brands in crisis: Partisanship, brand dilution, and the breakdown of political parties in Latin America*, Cambridge University Press, 2016.
- Neggers, Yusuf**, “Enfranchising your own? Experimental evidence on bureaucrat diversity and election bias in India,” *American Economic Review*, 2018, 108 (6), 1288–1321.
- New York Times**, “Honduran President Declared Winner, but O.A.S. Calls for New Election,” 2017.
- Paraguay.com**, “Detallan el sistema de la compra y venta de votos,” 2018.
- Persson, Torsten, Guido Tabellini, and Francesco Trebbi**, “Electoral Rules and Corruption,” *Journal of the European Economic Association*, 2003, 1 (4), 958–989.
- Pomares, Julia, María Page, Soledad Zárate, and M. Belén Abdala**, “¿Quién cuida los votos?,” CIPPEC 2015.
- Roodman, David, Morten Ørregaard Nielsen, James G. MacKinnon, and Matthew D. Webb**, “Fast and Wild: Bootstrap Inference in Stata Using boottest,” *Stata Journal*, 2019, 19 (1), 4–60.
- Rueda, Miguel**, “Small aggregates, big manipulation: Vote buying enforcement and collective monitoring,” *American Journal of Political Science*, 2017, 61 (1), 163–177.
- Sjoberg, Fredrik**, “Bring the party back in: Institutional design for ‘smart election fraud’,” *Electoral Studies*, 2016, 44, 307–318.
- Taagepera, Rein and Matthew Soberg Shugart**, *Seats and Votes: The Effects and Determinants of Electoral Systems*, New Haven: Yale University Press, 1989.
- The Economist**, “Is Honduras’s ruling party planning to rig an election?,” 2017.
- TSJE**, “Actas de Transmisión de Resultados Electorales Preliminares (TREP), Elecciones Generales 2018,” Scanned tally images, Tribunal Superior de Justicia Electoral (TSJE), Asunción, Paraguay 2018.
- , “Impugnaciones electorales, Elecciones Generales 2018,” Legal records obtained from the Tribunal Superior de Justicia Electoral (TSJE) and coded by the authors, Asunción, Paraguay 2018.
- , “Nómina de agentes electorales preinscritos, Elecciones Generales 2018,” Data set obtained from the Tribunal Superior de Justicia Electoral (TSJE), Asunción, Paraguay 2018.
- , “Registro Cívico Permanente (padrón electoral), 2011–2018,” Data set obtained from the Tribunal Superior de Justicia Electoral (TSJE), Asunción, Paraguay 2018.
- , “Registros de afiliación partidaria,” Data set obtained from the Tribunal Superior de Justicia Electoral (TSJE) and from individual political parties, Asunción, Paraguay 2018.

— , “Resultados Electorales, Elecciones Generales 2018: recuento definitivo por mesa,” Data set, Tribunal Superior de Justicia Electoral (TSJE), Asunción, Paraguay 2018. Retrieved from <https://tsje.gov.py>.

Última Hora, “Cartes había vetado proyecto que permitía guardar papeletas de voto,” 2018.

— , “Con planillas quieren controlar voto de estatales y familiares,” 2018.

— , “Reflotan las coordinadoras coloradas para captar los votos de funcionarios,” 2022.

— , “La ANR prepara su maquinaria electoral para seguir en el poder,” 2023.

U.S. Embassy in Asunción, “Three Weeks ’til Elections,” Confidential diplomatic cable 08ASUNCION195, WikiLeaks Public Library of US Diplomacy, March 2008. https://wikileaks.org/plusd/cables/08ASUNCION195_a.html.

Warner, Zach, J. Andrew Harris, Michelle Brown, and Christian Arnold, “Hidden in plain sight? Irregularities on statutory forms and electoral fraud,” *Electoral Studies*, 2021, 74, 102411.

Getting a seat at the table: Partisan poll workers and electoral bias

Supplemental Appendix

Sections are lettered continuing from Appendices A and B of the paper.

C	Additional Background Information and Context	1
D	Data Construction and Description	6
E	Additional Results	11
F	Establishment party support and partisan poll workers	27
G	Additional Robustness Checks	30

C. Additional Background Information and Context

Figure C1: Image of the paper ballot for the senatorial election

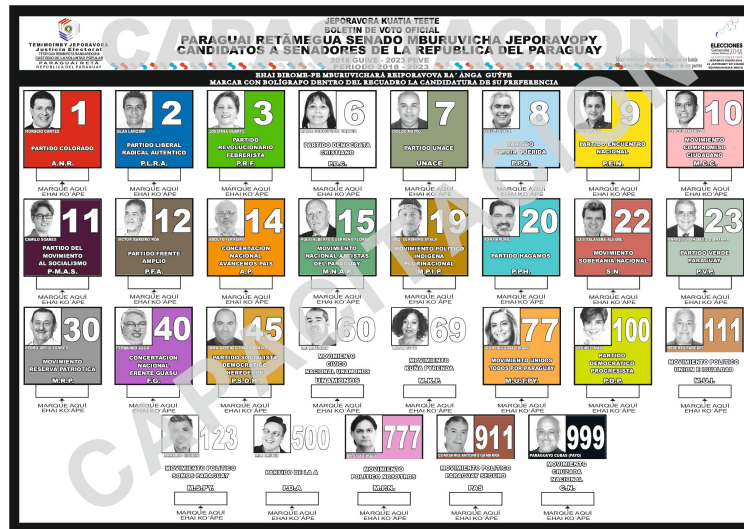


Figure C2: Image of an electoral table with voting booth and poll workers



Note: this figure shows a common setup for electoral tables in Paraguayan elections. Polling stations tend to be in public schools and voting booths tend to be installed in classrooms. The three poll workers set up a table for checking voters' ID cards outside a classroom before allowing them to vote. The picture is from: <https://es-us.finanzas.yahoo.com/noticias/paraguay-celebrar%C3%A1-elecciones-municipales-10-204613200.html>.

As a motivating example of what is mentioned in Section I, we describe an example of a table where the official results were challenged because the quick count tally had visible modifications to

its counts. Figure C3 shows the senatorial race quick count tally for an electoral table with visible modifications. After reviewing the final outcomes for this table, we were able to verify that these changes were incorporated into the final count, which implies the following changes to the original tally: Colorado party +15, Liberal party +15, Patria Querida (PPQ) -20, and Hagamos (PPH) -10.

This electoral table that had its official results challenged had two poll workers from the Colorado party and one from the Liberal party (SA Figure C4). Neither PPQ nor PPH had poll worker representation at this electoral table. Thus, with this electoral table we observe a presumptive case of collusive irregularities benefiting the established parties present, at the expense of smaller parties.¹ We expect this type of collusive irregularities to be more common with the PR races for legislative positions compared to the executive elections (president and governor) which are determined by SMDP.

The main irregularities that colluding coalitions can commit include miscounting and tampering with the paper ballots (see Figure C5 for a newspaper infographic from ABC Color (2018a) alluding to how colluding coalitions of poll workers could alter the count). ABC Color (2018a) also indicates how vote monitoring could occur with paper ballots.² Carlos María Ljubetic, a previous director of the TSJE, acknowledged this can occur as follows. Political brokers buying votes tell voters to write in particular marks on their paper ballots that the partisan poll workers could tally to confirm the vote buying agreements were followed through on. At a lecture one of us attended, Ljubetic described a second monitoring method: poll workers can mark the ballots of particular voters to learn how they voted.³ Because poll workers must sign each paper ballot before the voter casts it, they can vary their signatures to make a given ballot recognizable during the count. Finally, poll workers could become more or less careful when counting votes depending on the presence of poll workers from other parties.

These sorts of irregularities have been an open secret even among the diplomatic community in Paraguay. A leaked diplomatic cable due to WikiLeaks coming from the U.S. Embassy in Asunción (2008) remarked the following a few weeks before the 2008 Paraguayan general elections:

In coming weeks before the elections, all sides will engage in vote-buying, as well as buying (or attempted buying) of the political party representatives who will sit at each polling station on election day. No one does this better than the Colorados. Castiglioni supporters predict the Colorados will steal around 100,000 votes at roughly 700 polling stations (where they already dominate) by altering the electoral tally sheets (“Actas”).

¹This electoral table is particularly flagrant given the visible modifications to the tally. There are other cases where the counts were presumably modified from the start, so there are no visible modifications. For example, the cases mentioned by Colmán Gutiérrez (2018) and ABC Color (2018b) show that the parties they voted for received 0 votes according to the quick count tallies of their voting booths.

²Even under a secret ballot, brokers can enforce vote-buying agreements by monitoring the published tallies of small polling stations (Rueda 2017).

³Ljubetic gave the lecture to DENDE, a Paraguayan civil society organization.

Figure C3: Quick count tally of voting booth whose official results were challenged, with visible modifications

Elecciones Generales y Departamentales - 2018

Justicia Electoral
CUSTODIO DE LA VOLUNTAD POPULAR

A 085884

TRANSMISION DE RESULTADOS ELECTORALES PRELIMINARES - TREP

SENADORES

Departamento : (11) CENTRAL
Distrito : (3) FERNANDO DE LA MORA
Zona : (6) FERNANDO DE LA MORA
Local : (1) COLANC, FERNANDO DE LA MORA

CTX : 851
Mesa N°: 066
Seguridad: 206403

N°	Partido	Votos en blanco	Votos en letras
1	PARTIDO COLORADO	150	Arriba y abajo Cuarenta
2	PARTIDO LIBERAL RADICAL AUTENTICO	43	(deiz sen) Treinta dos
3	PARTIDO REVOLUCIONARIO FERRERISTA	1	una
6	PARTIDO DEMOCRATA CRISTIANO	0	cero
7	PARTIDO UNICE	2	dos
8	PARTIDO PATRIA QUERIDA	25	(treinta) Cero
9	PARTIDO ENCUENTRO NACIONAL	1	una
10	MOVIMIENTO COMPROMISO CIUDADANO	2	dos
11	PARTIDO DEL MOVIMIENTO AL SOCIALISMO	2	dos
12	PARTIDO FRENTE AMPLIO	0	cero
14	CONCERTACION NACIONAL AVANCEMOS PAIS	0	cero
15	MOVIMIENTO NACIONAL ARTISTAS DEL PARAGUAY	0	cero
19	MOVIMIENTO POLITICO INDIENIA PLURINACIONAL	1	una
20	PARTIDO HAGAMOS	10	diez
22	MOVIMIENTO POLITICO SOBERANIA NACIONAL	1	una
23	PARTIDO VERDE PARAGUAY	2	dos
30	MOVIMIENTO RESERVA PATRIOTICA	3	tres
40	CONCERTACION NACIONAL FRENTE GUARU	6	seis
45	PARTIDO SOCIALISTA DEMOCRATICO HEREDEROS	0	cero
60	MOVIMIENTO CIVICO NACIONAL UNIMOROS	0	cero
69	MOVIMIENTO KUÑA PYRENDI	2	dos
77	MOVIMIENTO UNIDOS TODOS POR PARAGUAY	0	cero
100	PARTIDO DEMOCRATICO PROGRESISTA	7	siete
111	MOVIMIENTO POLITICO UNION E IGUALDAD	0	cero
123	MOVIMIENTO POLITICO SONOS PARAGUAY	3	tres
900	PARTIDO DE LA A	1	una
777	MOVIMIENTO POLITICO NOSOTROS	0	cero
911	MOVIMIENTO POLITICO PARAGUAY SEGURO	1	una
999	MOVIMIENTO CRUZADA NACIONAL	3	siete
	VOTOS NULOS	6	seis
	VOTOS EN BLANCO	2	dos
	SUMA TOTAL DE VOTOS	138	cento treinta y ocho

Firma del vocal de mesa
CIN: 7-677-145

Firma del presidente de mesa
CIN: 7-876-645

Firma del vocal de mesa
CIN: 8-11145

Note: the picture above shows the Senate race quick count tally for a voting booth whose official results were challenged with visible modifications. The modifications to the results are as follows: Colorado party +15, Liberal party +15, PPQ -20, PPH (Hagamos) party -10.

Figure C4: Stata output showing the partisan support and ID numbers of poll workers from a voting booth that had its official results challenged

mesaid	f_v1_part	f_p_part	f_v2_part	f_v1_cedula	f_p_cedula	f_v2_cedula
11_3_0_1_66	PLRA	ANR	ANR	3653742	1540605	4177951

Note: the picture above shows Stata output of the partisan support and the ID numbers of the poll workers from an electoral table that had its official results challenged. The data on partisan support and registration come from the election authority (the TSJE) and the political parties themselves. Two of the poll workers belong to the Colorado party and one belongs to the Liberal party. Neither the PPQ nor PPH had poll worker representation at this electoral table.

Figure C5: Newspaper infographic describing collusion at electoral tables for tabulation irregularities



Note: the text above states: “Vote tallies ‘kill’ votes: Carlos María Ljubetic, former electoral justice adviser considers that one of the great weaknesses of the Paraguayan electoral system is in the composition of electoral tables. This mechanism allows the vote tallies to be modified, which are ultimately the ones that define the candidacies. Small parties can receive a certain number of votes. However, the poll workers of the table, who generally represent the two major parties, can agree and share the votes of those who do not have their own poll watchers or empowered party officers. In this way, regardless of what is expressed at the polls, their votes can go to the traditional parties.”

Most agree (and history proves) that this is easy to do and difficult to prove, because the ballots are destroyed once voting ceases and the Actas are signed.

The previous quote highlights how diplomats from the United States Embassy in Paraguay were convinced that political parties would bribe poll workers sitting at the electoral tables. It also highlights their belief that the hegemonic Colorado party does this better than other political parties. Luis Castiglioni was Paraguay's vice president from 2003 to 2008 and a Colorado presidential candidate who lost the party's 2008 primaries. Finally, the cable also claims that vote counting irregularities are "easy to do and difficult to prove" given that the paper ballots are destroyed.

C.1. Political party name acronyms

In this subsection we list out all the political party name acronyms we mentioned in the main text:

- **ANR:** Asociación Nacional Republicana – Partido Colorado
- **AP:** Avancemos País
- **CN:** Movimiento Cruzada Nacional
- **MNAP:** Movimiento Nacional Artistas del Paraguay
- **MRP:** Movimiento Reserva Patriótica
- **PCP:** Partido Comunista Paraguay
- **PCPS:** Partido Convergencia Popular Socialista
- **PDP:** Partido Democrático Progresista
- **PEN:** Partido Encuentro Nacional
- **PFA:** Partido Frente Amplio
- **PLRA:** Partido Liberal Radical Auténtico
- **PMAS:** Partido del Movimiento al Socialismo
- **PMPP:** Partido del Movimiento Patriótico Popular
- **PPC:** Partido de Participación Ciudadana
- **PPH:** Partido Hagamos
- **PPQ:** Partido Patria Querida
- **PPS:** Partido País Solidario
- **PPT:** Partido Popular Tekojoja

- **PRF:** Partido Revolucionario Febrerista
- **PUP:** Partido de la Unidad Popular
- **PVP:** Partido Verde Paraguay
- **SN:** Movimiento Soberanía Nacional
- **UNACE:** Partido UNACE (Unión Nacional de Ciudadanos Éticos)

D. Data Construction and Description

The pre-registered poll worker allocation data comes from a spreadsheet obtained from the TSJE, which indicates the partisan poll worker representation that was officially approved two weeks in advance of the elections. It notes all the pre-registered electoral agents, that is, poll workers, poll watchers (*veedores*), and empowered party officers (*apoderados*). Most importantly, this data has the political party these pre-registered electoral agents will work for during the election and their national ID numbers.

In addition, the TSJE uploaded scans of all single quick count tallies for all electoral races at each table. We compiled the data on *de facto* poll workers from scans of these approximately 120,000 handwritten election-day quick count tallies written by the poll workers themselves. The election authority briefly published these on its website, which we were able to scrape from before it took these results down. As the TSJE already shared the electoral results at the voting booth-level, the key additional piece of information that the tally scans provide are the ID numbers of the poll workers at the bottom of each tally (see Figure D1). Using data from the election authority and political parties, we can match the IDs of the poll workers to the parties they work for (if they are a pre-registered electoral agent) and the parties they are registered to. For example, Figure D2 shows the pre-registered poll workers for the electoral table from Figure D1. In this example, comparing the pre-registered poll workers with those on the tallies shown, we can observe that the Colorado and Liberal pre-registered poll workers appeared on election day, while the FG one was replaced. This shows an example of the difference between pre-registered and *de facto* allocations which we describe in the main paper.

As mentioned above, to find the *de facto* poll workers we rely on the quick count vote tallies that the TSJE scanned and uploaded to its website after the elections. Remote polling stations are an exception: they transmitted their results by telephone, and those tallies record the name of the election authority employee who took the call in place of the poll workers' ID numbers, so they yield no *de facto* composition (SA Figure D3). From the tallies we digitized the ID numbers of those poll workers actually present at the electoral table on election day. Digitized numbers that did not match any registered voter were re-checked against the tally image and corrected where the handwriting had been misread. We also matched numbers by string distance to the pre-registered

Figure D1: Example quick count vote tallies from a single voting booth

Elecciones Generales y
Departamentales - 2018

Justicia Electoral
CURSOS DE LA VIGILANCIA POPULAR

A 096191

TRANSMISION DE RESULTADOS ELECTORALES PRELIMINARES - TRIP

PRESIDENCIA DE LA REPUBLICA

CTX : 147

Departamento: (1) CONCEPCION
Distrito: (6) CONCEPCION
Zona: (6) CONCEPCION
Local: (1) C.R.E.C. JANN E. OLEARY

Mesa N°: 001

Seguridad: 858850

Boleta	Partido	Mostrador	Mostrador	Votos en blanco	Votos en letras
1	PARTIDO COLORADO	63	Sesenta y cinco		
2	ALIANZA OHHO	46	University 46s		
12	PARTIDO INDEPENDI AMPLIO	1	Uno		
15	MOVIMIENTO NACIONAL ARTISTAS DEL PARAGUAY	0	Cero		
22	MOVIMIENTO SOBERANIA NACIONAL	0	Cero		
23	PARTIDO VERDE PARAGUAY	2	Dos		
30	MOVIMIENTO RESERVA PATRIOTICA	0	Cero		
45	PARTIDO SOCIALISTA DEMOCRATICO REPERCUROS	1	Uno		
50	PARTIDO DEL MOVIMIENTO PATRIOTICO POPULAR	2	Dos		
60	MOVIMIENTO CIVICO NACIONAL UNIMACHOS	2	Dos		
	VOTOS NEGROS	2	Dos		
	VOTOS EN BLANCO	3	Tres		
	SUMA TOTAL DE VOTOS	122	Ciento veintidos		

Firma del Jefe de Mesa

FIRMA DEL VOCAL DE MESAS
CALLE: 30 de Mayo

FIRMA DEL PRESIDENTE DE MESA
CALLE: 30 de Mayo

FIRMA DEL VOCAL DE MESAS
CALLE: 30 de Mayo

5063.089

Elecciones Generales y
Departamentales - 2015

Justicia Electoral
CURTIIDO DE LA VOLENTAD POPULAR

A 096189

TRANSMISOR DE RESULTADOS ELECTORALES EN PRESENCIA DE TESTIGOS

SEÑALADORES

CTX. 147

Departamento: (1) CONCEPCION
Municipalidad: (8) CONCEPCION
Zona: (1) CONCEPCION
Local: (1) C.R.E.C. JUAN E. O'LEARY

Mesa No: 001
Secundaria: 080848

Partido		Votos en sección	Votos en tablero
1	PARTIDO ORAZCO	47	Cuarenta y Siete
2	PARTIDO LIBERAL, RADICAL, AUTOTITULO	29	Veintinueve
3	PARTIDO REVOLUCIONARIO FRENTERISTA	4	Uno
6	PARTIDO DEMOCRATICO CRISTIANO	0	Cero
7	PARTIDO UNICO	0	Cero
8	PARTIDO PATRIA GUERRA	10	Diez
9	PARTIDO MOVIMIENTO NACIONAL	0	Cero
10	MOVIMIENTO COMPROMISO CIOCIANO	0	Cero
11	MOVIMIENTO DEL MOVIMIENTO DEL MOVIMIENTO	0	Cero
12	PARTIDO FRENTE AMPLIO	0	Cero
14	CONCERTACION NACIONAL AVANCEMOS PAIS	0	Cero
15	MOVIMIENTO NACIONAL ARTISTAS DEL PARAGUAY	0	Cero
19	MOVIMIENTO POLITICO INDIGENA PLURINACIONAL	0	Cero
20	PARTIDO HAGAMOS	4	Cuatro
22	MOVIMIENTO POLITICO SOBERANIA NACIONAL	0	Cero
23	PARTIDO VERDE PARAGUAY	8	Ocho
30	MOVIMIENTO RESERVA PATRIOTICA	0	Cero
40	CONCERTACION NACIONAL FRENTE GUAY	9	Nueve
45	PARTIDO SOCIALISTA MOVIMIENTO RENESADOR	0	Cero
60	MOVIMIENTO CIVICO NACIONAL UNIONES	0	Cero
69	MOVIMIENTO KURA PYRENDA	1	Uno
77	MOVIMIENTO UNIDOS TODOS POR PARAGUAY	0	Cero
100	PARTIDO DEMOCRATICO PROGRESISTA	3	Tres
111	MOVIMIENTO POLITICO UNO EN IGUALDAD	0	Cero
123	MOVIMIENTO POLITICO SOROS PARAGUAY	0	Cero
130	MOVIMIENTO KURA PYRENDA	0	Cero
800	MOVIMIENTO POLITICO INDIGENA	0	Cero
777	MOVIMIENTO POLITICO INDIGENAS	2	Doce
911	MOVIMIENTO POLITICO PARAGUAY RESISTO	2	Doce
999	MOVIMIENTO CRUZADA NACIONAL	1	Uno
VOTOS MALOS		3	Tres
VOTOS EN BLANCO		6	Seis
SUJETO TOTAL DE VOTOS		139	Ciento Veintinueve y Nueve

Intendente, Jefe de Mesa
C.R.E.C. JUAN E. O'LEARY

PRIMA DEL C.R.E.C. DE MESA
C.R.E.C. JUAN E. O'LEARY

Intendente, Jefe de Mesa
C.R.E.C. JUAN E. O'LEARY

[illegible]

Elecciones Generales y
Departamentales - 2011

Justicia Electoral
CURSOS DE LA VOLUNTAD POPULAR

A 096130

TRANSMISION DE RESULTADOS ELECTORALES PRELIMINARES - TRIP

GOBERNADOR

CTX : 147

Departamento : (1) CONCEPCION
Circuito : (6) CONCEPCION
Zona : (6) CONCEPCION
Local : (1) C.R.E.C. JUAN E. O'LEARY

Mesa N°: 001

Seguridad: 126772

Boleta	Partido	Nombre del Candidato	Votos en Boletines	Votos en Letras
1	PARTIDO COLOMBIO		55	Cinuenta y tres
2	PARTIDO LIBERAL RADICAL AUTENTICO		59	Cinuenta y nueve
3	PARTIDO REVOLUCIONARIO FERRERISTA		1	Uno
7	PARTIDO UNICO		0	Cero
16	MOVIMIENTO NACIONAL ARTISTAS DEL PARAGUAY		0	Cero
23	PARTIDO VERDE PARAGUAY		2	Dos
46	PARTIDO SOCIALISTA DEMOCRATICO HEREDEROS		0	Cero
VOTOS MALOS			2	Dos
VOTOS EN BLANCO			5	Cinco
BUENA TOTAL DE VOTOS			122	Ciento veintidos
Firma del Presidente de Mesa				
Firma del Vocal de Mesa				
Firma del Vocal Secundario				

Firma del Presidente de Mesa
Firma del Vocal de Mesa
Firma del Vocal Secundario

Firma del Presidente de Mesa
Firma del Vocal de Mesa
Firma del Vocal Secundario

Firma del Presidente de Mesa
Firma del Vocal de Mesa
Firma del Vocal Secundario

356707

Elecciones Generales y
Departamentales - 2016

Justicia Electoral
CORTES DE LA VOLUNTAD POPULAR

A 095186

TRANSMISION DE RESULTADOS ELECTORALES PRELIMINARES - TRIP

JUNTA DEPARTAMENTAL

CTX : 147

Departamento : (1) CONCEPCION

Circuito : (8) CONCEPCION

Zona : (4) CONCEPCION

Local : (1) C.A.E.C. JUAN E. O'LEARY

Mesa N°: 001

Seguridad: 809092

N° de Lista	Partido/ Movimiento/ Círculo/ Organización	Votos en sufragio	Votos en lista
1	PARTIDO COLOMBIO	45	Cuarenta y Cinco
2	PARTIDO LIBERAL, PATRIAL, AUTENTISTO	28	Veintiocho y Diecho
3	PARTIDO REVOLUCIONARIO FRENTE AMPLIO	4	Cuatro
7	PARTIDO UNICE	1	Uno
9	PARTIDO ENCUENTRO NACIONAL	1	Uno
12	PARTIDO FRENTE AMPLIO	1	Uno
14	CONCERTACION NACIONAL, AVANCEMOS PAIS	1	Uno
16	MOVIMIENTO NACIONAL, ARTISTAS DEL PARAGUAY	0	Cero
23	PARTIDO VERDE PARAGUAY	1	Uno
40	CONCERTACION NACIONAL, FRENTE GUARU	4	Cuatro
46	PARTIDO SOCIALISTA, DEMOCRATICO, HEREDEROS	0	Cero
80	PARTIDO DEL MOVIMIENTO PATRISTICO POPULAR	1	Uno
911	MOVIMIENTO PARAGUAY SEGURO	9	Nueve
VOTOS NULOS		3	Tres
VOTOS EN BLANCO		19	Diece
SUMA TOTAL DE VOTOS		122	Ciento Veintidos

Verifica Censura, Entrega, etc. -
 FIRMA DEL Jefe de Mesa
 C.I.M. 99.585.0124

FIRMA DEL REPRESENTANTE DE MESA
 C.I.M. 99.585.0124

Firma del Jefe de Mesa
 FIRMA DEL Jefe de Mesa
 C.I.M. 99.585.0124

Elecciones Generales y
Departamentales - 2018

Justicia Electoral
Oficina Nacional de Elecciones

A 096188

TRANSMISION DE RESULTADOS ELECTORALES POR MUNICIPIOS - TRIP

PARLAMENTARIOS DEL MERCOSUR CTX : 147

Departamento: (1) CONCEPCION
Sección: (4) CONCEPCION
Zona: (1) CONCEPCION
Local: (1) C.R.E.C. JUAN E. CULIARY

Mesa N°: 001

Secuencia: 405450

N°	Partido / Movimiento / Candidato	Votos en blanco	
		Votos en blanco	Votos en blanco
1	PARTIDO COLORADO	49	49
2	PARTIDO LIBERAL RADICAL APATERA	40	40
3	PARTIDO REVOLUCIONARIO FERRERISTA	1	Uno
6	PARTIDO DEMOCRATA CRISTIANO	0	Cero
7	PARTIDO UNICE	1	Uno
8	PARTIDO PATRIA QUERIDA	0	Cero
9	PARTIDO ENCUENTRO NACIONAL	1	Uno
10	MOVIMIENTO COMPROMISO CRUCIAGUANO	0	Cero
11	PARTIDO DEL MOVIMIENTO AL FOCALISMO	1	Uno
12	PARTIDO FRENTE AMPLIO	0	Cero
14	CONCENTRACION NACIONAL AVANZADOS PAS	1	Uno
15	MOVIMIENTO NACIONAL ARTISTAS DEL PARAGUAY	0	Cero
19	MOVIMIENTO POLITICO INDIGENA PURPURACIONAL	1	Uno
20	PARTIDO IGANAGNI	3	Tres
21	PARTIDO LA JUVENTUD	1	Uno
22	MOVIMIENTO SOBERANIA NACIONAL	0	Cero
23	PARTIDO VERDE PARAGUAY	2	Uno
26	MOVIMIENTO RESERVA PATRIOTICA	0	Cero
40	CONCENTRACION NACIONAL FRENTE GASU	4	Cuatro
46	PARTIDO SOCIALISTA DEMOCRATICO HEMERENDERO	0	Cero
50	PARTELO DEL MOVIMIENTO POLITICO POPULAR	0	Cero
60	MOVIMIENTO CIVICO NACIONAL UNAMOROS	0	Cero
77	MOVIMIENTO UNIDOS TODOS POR PARAGUAY	0	Cero
100	PARTIDO DEMOCRATICO PROGRESISTA	1	Uno
119	MOVIMIENTO POLITICO UNION E IGUALDAD	0	Cero
123	MOVIMIENTO POLITICO GRUPO PARAGUAY	0	Cero
508	PARTIDO DE LA A	0	Cero
911	MOVIMIENTO POLITICO PARAGUAY SEGURO	1	Uno
999	MOVIMIENTO CRUZADA NACIONAL	0	Cero
VOTOS NULOS		3	Tres
VOTOS EN BLANCO		10	Diez
SUMA TOTAL DE VOTOS		129	Ciento Veintidos

Juan E. Culinary
C.R.E.C. JUAN E. CULIARY

PRIMA DEL VOTO
C.R.E.C. JUAN E. CULIARY

PRIMA DEL VOTO
C.R.E.C. JUAN E. CULIARY

Note: This figure shows scans of all the quick count vote tallies from a single electoral table.

Figure D2: Pre-registered poll workers from example quick count tally shown

0	CONCEPCION	0	CONCEPCION	1	C.R.E.C JUAN E. O'LEARY	1	3561873	F	9/16/1980	PRESIDENTE	TITULAR	1	ANR
0	CONCEPCION	0	CONCEPCION	1	C.R.E.C JUAN E. O'LEARY	1	3563089	F	9/13/1980	VOCAL	TITULAR	2	PLRA
0	CONCEPCION	0	CONCEPCION	1	C.R.E.C JUAN E. O'LEARY	1	5587364	F	12/24/1989	VOCAL	TITULAR	40	FG

Note: This figure shows the pre-registered poll workers from the quick count vote tally shown on Figure D1. Comparing the pre-registered poll workers with those on the tally, the Colorado and Liberal pre-registered poll workers appeared on election day, while the FG one was replaced.

Figure D3: Example vote tallies transmitted via voice for remote polling stations

Elecciones Generales y Departamentales - 2018

TRANSMISION DE RESULTADOS ELECTORALES PRELIMINARES - TREP - VIA VOZ

GOBERNADOR CTX : 1220

Departamento : (16) ALTO PARAGUAY
Distrito : (2) CARMELO PERALTA
Zona : (11) PUERTO BASTREX (LA ESPERANZA)
Local : (801) LIC. MAC. LA ESPERANZA

Mesa N°: 002
Seguridad: 734211

Nro. Lista	Partido/ Movimiento/ Alianza/ Concertación	Votos en blanco	Votos en letra
1	PARTIDO COLORADO	56	Cinuenta y siete
5	ALIANZA GANAR	72	Sesenta y dos
23	PARTIDO VERDE PARAGUAY	0	Cero
40	CONCERTACION NACIONAL FRENTE GUASU	0	Cero
77	MOVIMIENTO UNIDOS TODOS POR PARAGUAY	0	Cero
VOTOS NULOS		2	Doce
VOTOS EN BLANCO		4	Cuatro
SUMA TOTAL DE VOTOS		135	Ciento treinta y cinco

NOMBRE DEL FUNCIONARIO TREP RECEPTOR: Miguel Jacinto
NRO. DEL RECEPTOR: 35 NRO. LINEA RECEPTOR: 6180154

Elecciones Generales y Departamentales - 2018

TRANSMISION DE RESULTADOS ELECTORALES PRELIMINARES - TREP - VIA VOZ

PRESIDENTE DE LA REPUBLICA CTX : 1220

Departamento : (16) ALTO PARAGUAY
Distrito : (2) CARMELO PERALTA
Zona : (11) PUERTO BASTREX (LA ESPERANZA)
Local : (801) LIC. MAC. LA ESPERANZA

Mesa N°: 002
Seguridad: 379176

Nro. Lista	Partido/ Movimiento/ Alianza/ Concertación	Votos en blanco	Votos en letra
1	PARTIDO COLORADO	74	Sesenta y cuatro
5	ALIANZA GANAR	53	Cinuenta y tres
12	PARTIDO FRENTE AMPLIO	0	Cero
16	MOVIMIENTO NACIONAL ARTISTAS DEL PARAGUAY	0	Cero
22	MOVIMIENTO SOBERANIA NACIONAL	0	Cero
23	PARTIDO VERDE PARAGUAY	0	Cero
30	MOVIMIENTO RESERVA PATRIOTICA	0	Cero
45	PARTIDO SOCIALISTA DEMOCRATICO HEREDEROS	0	Cero
60	PARTIDO DEL MOVIMIENTO PATRIOTICO POPULAR	0	Cero
80	MOVIMIENTO CIVICO NACIONAL UNANIMOS	0	Cero
VOTOS NULOS		0	Cero
VOTOS EN BLANCO		8	Ocho
SUMA TOTAL DE VOTOS		135	Ciento treinta y cinco

NOMBRE DEL FUNCIONARIO TREP RECEPTOR: Miguel Jacinto
NRO. DEL RECEPTOR: 35 NRO. LINEA RECEPTOR: 6180154

Elecciones Generales y Departamentales - 2018

TRANSMISION DE RESULTADOS ELECTORALES PRELIMINARES - TREP - VIA VOZ

SENADORES CTX : 1220

Departamento : (16) ALTO PARAGUAY
Distrito : (2) CARMELO PERALTA
Zona : (11) PUERTO BASTREX (LA ESPERANZA)
Local : (801) LIC. MAC. LA ESPERANZA

Mesa N°: 002
Seguridad: 828567

Nro. Lista	Partido/ Movimiento/ Alianza/ Concertación	Votos en blanco	Votos en letra
1	PARTIDO COLORADO	45	Cuarenta y siete
2	PARTIDO LIBERAL RADICAL AUTENTICO	9	Nueve
3	PARTIDO REVOLUCIONARIO FERRERISTA	0	Cero
6	PARTIDO DEMOCRATA CRISTIANO	0	Cero
7	PARTIDO UNACE	0	Cero
8	PARTIDO PATRIA GUERRA	12	Doce
9	PARTIDO ENCUENTRO NACIONAL	0	Cero
10	MOVIMIENTO COMPROMISO CIUDADANO	0	Cero
11	PARTIDO DEL MOVIMIENTO AL SOCIALISMO	0	Cero
12	PARTIDO FRENTE AMPLIO	0	Cero
14	CONCERTACION NACIONAL AVANCEMOS PAIS	6	Seis
15	MOVIMIENTO NACIONAL ARTISTAS DEL PARAGUAY	0	Cero
18	MOVIMIENTO POLITICO INDIGENA PLURINACIONAL	0	Cero
20	PARTIDO JAGUARI	0	Cero
22	MOVIMIENTO POLITICO SOBERANIA NACIONAL	0	Cero
23	PARTIDO VERDE PARAGUAY	0	Cero
30	MOVIMIENTO RESERVA PATRIOTICA	0	Cero
40	CONCERTACION NACIONAL FRENTE GUASU	4	Cuatro
45	PARTIDO SOCIALISTA DEMOCRATICO HEREDEROS	0	Cero
60	MOVIMIENTO CIVICO NACIONAL UNANIMOS	0	Cero
69	MOVIMIENTO KURA PYRENDA	0	Cero
77	MOVIMIENTO UNIDOS TODOS POR PARAGUAY	0	Cero
100	PARTIDO DEMOCRATICO PROGRESISTA	0	Cero
111	MOVIMIENTO POLITICO UNION E IGUALDAD	0	Cero
123	MOVIMIENTO POLITICO SOMOS PARAGUAY	0	Cero
500	PARTIDO DE LA A	0	Cero
777	MOVIMIENTO POLITICO NOSOTROS	0	Cero
911	MOVIMIENTO POLITICO PARAGUAY SEGURO	0	Cero
999	MOVIMIENTO CRUZADA NACIONAL	0	Cero
VOTOS NULOS		4	Cuatro
VOTOS EN BLANCO		53	Cinuenta y tres
SUMA TOTAL DE VOTOS		135	Ciento treinta y cinco

NOMBRE DEL FUNCIONARIO TREP RECEPTOR: Miguel Jacinto
NRO. DEL RECEPTOR: 35 NRO. LINEA RECEPTOR: 6180154

Note: This figure shows a picture of a quick count vote tally for remote polling stations where electoral results were transmitted to the election authority through a call (via voice) instead of a scan of the vote tally. The vote tally is for a booth in a polling station in the Carmelo Peralta district from the Alto Paraguay department in the Chaco region. Instead of showing the ID numbers of the poll workers at the bottom, it has the name of the election authority employee who received the results via a call.

roster and to the electoral registries to catch single-digit transcription errors, and numbers that remained ambiguous were flagged and resolved by hand from the image. Then, we merge these ID numbers with nation-wide party registration data sets we obtained from the TSJE (because the election authority collects this data from individual parties) and from individual parties themselves. By merging these data sets we can uncover who the poll workers support even if they are not on our pre-registered electoral agents data. When the sources disagree we give priority to the party for which a person pre-registered as an electoral agent in 2018, then to the 2013 poll worker roster (as these also indicate who these individuals have supported as poll workers), and then to the party registrations in the electoral registries of 2017, 2015, 2013, and 2011, in that order. A person pre-registered by the GANAR alliance is assigned the member party in which he or she is registered, and the GANAR label is kept only when no party registration is found. By merging these data, we find some cases of poll workers registered to multiple parties. Among all *de facto* poll workers, less than 4% of them (2,514 out of 63,633) are multi-party poll workers and we consider them as a separate category because they are not “pure” partisans.⁴ We also find 5,166 poll workers (around 8% of all poll workers) not registered to any political party. Although these poll workers may be working for some party, and we cannot determine this, they may also be truly independent voters who replaced pre-registered poll workers that did not appear on election day. Figure 1 shows how the allocation of partisan poll workers changes for the main parties due to replacements of pre-registered poll workers.

We next discuss poll workers’ partisanship by role. For the pre-registered poll workers, Table D1 shows that the “president” is the most equally distributed among parties (as presidents are randomly assigned among the three members of each table). Yet we can notice how the FG and other parties have less booth presidents than the Colorado and Liberal parties (although the Colorado party still has approximately 500 more booth presidents than the Liberal party). With the *de facto* poll workers the Colorado and Liberal party gain 758 and 147 table presidents respectively. On the other hand, the FG and other parties lose 2,210 and 644 table presidents respectively. Given that the *vocal* 1 and 2 share the same role and are interchangeable, we can just consider the sum of the two. The Colorado and Liberal party have roughly the same number of *vocal* poll workers at around 15,050. On the other hand, the FG and other parties have 8,743 and 3,084 respectively. With the replacements we can observe that the Colorado party gains 1,046 more *vocal* poll workers, whereas the other parties lose poll workers. The Liberal party loses 1,137, the FG loses 3,555, and the other parties lose 1,350. Finally, we can observe that the replacements vis-à-vis the pre-registered poll worker allocations only fully benefit the Colorado party, as only the Colorado party obtains more poll workers than its pre-registered amount.

⁴In Paraguay registering to multiple parties is not regulated.

Table D1: Partisan representation of pre-registered and *de facto* poll workers by poll worker function

	President	Vocal 1 + Vocal 2	Vocal 1	Vocal 2
Pre-registered poll workers				
Colorado	7,523	15,048	13,901	1,147
Liberal	7,060	15,053	7,066	7,987
FG	4,877	8,743	1	8,742
Other party	1,508	3,086	2	3,084
<i>De facto</i> poll workers				
Colorado	8,281	16,094	9,550	6,544
Liberal	7,207	13,916	6,800	7,116
FG	2,667	5,188	1,939	3,249
Other party	864	1,736	629	1,107
Multi-party	732	1,782	754	1,028
Unaffiliated	1,460	3,706	1,539	2,167
Difference between number of <i>de facto</i> and pre-registered poll workers				
Colorado	758	1,046	-4,351	5,397
Liberal	147	-1,137	-266	-871
FG	-2,210	-3,555	1,938	-5,493
Other party	-644	-1,350	627	-1,977

Note: This table indicates the partisan support of the pre-registered and *de facto* poll workers and their difference by poll worker function.

E. Additional Results

Tables E1 and E2 below show that our main estimates are unchanged after taking into account pre-registered poll watchers. As we observe only pre-registered watchers, this does not rule out that actual attendance is correlated with replacements and outcomes. It shows that the pre-registered presence of watchers does not account for the estimates. The coefficients for the Colorado and other party poll watchers align with what we would have expected. While the direction of the coefficients for Liberal poll watchers might not be intuitive, this could stem from two main reasons we can think of: (1) measurement error, as we only observe the pre-registered poll watchers and not the *de facto* ones; (2) perhaps the Liberal party strategically sends poll watchers to voting booths where it expects the Colorados to perform particularly well.

On Table E3 we tested whether the partisanship of the poll worker president independently impacts electoral results. The results do not show a clear pattern, likely reflecting that the official result within a table is the majority's (a minimum of 2 out of 3 poll workers). Hence, even if poll worker president tallies show results a certain way, if the two vocals indicate something else (and the vocals' results agree with each other), the official result for that table is what the vocals indicated. The specification does not hold the parties of the other two workers fixed, so the coefficients compare booths by the president's party.

Figures E1 to E3 plot the marginal effects implied by Table 4 under each electoral system, discussed in Section IV.A. For every allocation and outcome, the left panel shows the difference between PR and SMDP races (the interaction coefficient), and the middle and right panels show the effect within PR races and within SMDP races, respectively.

Table E4 tests the coalition channel discussed in Section IV.A: it replaces the PR indicator of Table 4 with an indicator for whether the Liberal party ran under the GANAR list in that race and department, pooling all races in Columns (1) to (3) and keeping only the deputies and departmental council races, where alliance status varies across departments within the same electoral system, in Columns (4) to (6). The interactions with the GANAR indicator are insignificant throughout for the collusion-relevant allocations, so the test is consistent with the electoral-rule interpretation.

Tables E5 to E8 support the discussion of the seat-vote logic in Section IV.A. Table E5 lists, for each department, the magnitudes of the deputies and departmental council districts, the number of lists and the effective number of parties in the deputies race, and the governor race's number of candidates, winner's share, and top-two margin. Tables E6 and E7 interact the allocation indicators with district magnitude within the PR races, and Table E8 interacts them with the closeness of the governor race. Because the interacted variable takes one value per department, the robust standard errors understate the uncertainty of the interaction coefficients. Each table therefore also reports wild-cluster-bootstrap p -values clustered on department, with Webb weights and 9,999 replications

Table E1: Effect of partisan poll worker composition on election results, controlling for pre-registered poll watcher presence

	Colorado vote share (1)	Liberal/GANAR vote share (2)	Other party vote share (3)
Colorado trifecta	0.2865 (0.2029)	-0.5233 (0.1847)	0.2368 (0.1754)
Colorado advantage + Other	0.4358 (0.0954)	-0.5043 (0.0790)	0.0685 (0.0820)
Colorado advantage + Liberal	0.5175 (0.0494)	-0.3865 (0.0455)	-0.1310 (0.0433)
Liberal advantage + Colorado	-0.0269 (0.0525)	0.1988 (0.0506)	-0.1719 (0.0483)
Liberal advantage + Other	-0.2032 (0.1610)	-0.0099 (0.1406)	0.2131 (0.1442)
Liberal trifecta	-1.9268 (0.5972)	0.4565 (0.5953)	1.4702 (0.4658)
Other poll worker allocation	0.0279 (0.0646)	-0.1633 (0.0597)	0.1354 (0.0579)
Colorado poll watcher	0.1551 (0.0929)	-0.2489 (0.0877)	0.0938 (0.0866)
Liberal poll watcher	0.8295 (0.1547)	-0.5486 (0.1520)	-0.2809 (0.1234)
Other poll watcher	-0.0880 (0.0786)	-0.0714 (0.0737)	0.1594 (0.0730)
Mean of Dependent Variable	41.5808	34.4445	23.9746
Polling station-elect. race FE	✓	✓	✓
Colorado voter prop. control	✓	✓	✓
Liberal voter prop. control	✓	✓	✓
Other party voter prop. control	✓	✓	✓
Observations	117,229	117,229	117,229
R^2	0.8425	0.8869	0.9308

Note: A unit of observation is a voting booth-electoral race combination. Standard errors are robust. The excluded category corresponds to voting booths with “fair allocations” having one Colorado party poll worker and one Liberal poll worker.

Table E2: Effect of partisan poll worker composition on election results by electoral system, controlling for pre-registered poll watcher presence

	Colorado vote share (1)	Liberal/GANAR vote share (2)	Other party vote share (3)
Colorado trifecta $\times$ PR race	-0.5626 (0.4164)	0.1363 (0.3948)	0.4263 (0.3126)
Colorado advantage + Other $\times$ PR race	0.0319 (0.1975)	0.0112 (0.1749)	-0.0431 (0.1440)
Colorado advantage + Liberal $\times$ PR race	0.1232 (0.1032)	0.1715 (0.0982)	-0.2948 (0.0754)
Liberal advantage + Colorado $\times$ PR race	0.0514 (0.1098)	0.0886 (0.1071)	-0.1400 (0.0848)
Liberal advantage + Other $\times$ PR race	0.1244 (0.3315)	-0.1742 (0.3116)	0.0498 (0.2596)
Liberal trifecta $\times$ PR race	0.0288 (1.1897)	-0.6830 (1.2019)	0.6541 (0.8081)
Other poll worker allocation $\times$ PR race	0.1204 (0.1356)	-0.1859 (0.1292)	0.0655 (0.1024)
Colorado trifecta	0.6659 (0.3291)	-0.6151 (0.3248)	-0.0509 (0.1986)
Colorado advantage + Other	0.4143 (0.1574)	-0.5118 (0.1484)	0.0975 (0.0877)
Colorado advantage + Liberal	0.4345 (0.0831)	-0.5021 (0.0816)	0.0676 (0.0449)
Liberal advantage + Colorado	-0.0614 (0.0884)	0.1395 (0.0875)	-0.0780 (0.0511)
Liberal advantage + Other	-0.2866 (0.2626)	0.1067 (0.2648)	0.1799 (0.1668)
Liberal trifecta	-1.9461 (0.9057)	0.9141 (0.9319)	1.0320 (0.4785)
Other poll worker allocation	-0.0531 (0.1096)	-0.0383 (0.1075)	0.0914 (0.0636)
Colorado poll watcher	0.1553 (0.0929)	-0.2488 (0.0877)	0.0935 (0.0866)
Liberal poll watcher	0.8294 (0.1548)	-0.5487 (0.1520)	-0.2807 (0.1234)
Other poll watcher	-0.0880 (0.0786)	-0.0715 (0.0737)	0.1595 (0.0730)
Mean of Dependent Variable	41.5808	34.4445	23.9746
Polling station-elect. race FE	✓	✓	✓
Colorado voter prop. control	✓	✓	✓
Liberal voter prop. control	✓	✓	✓
Other party voter prop. control	✓	✓	✓
Observations	117,229	117,229	117,229
R^2	0.8425	0.8869	0.9308

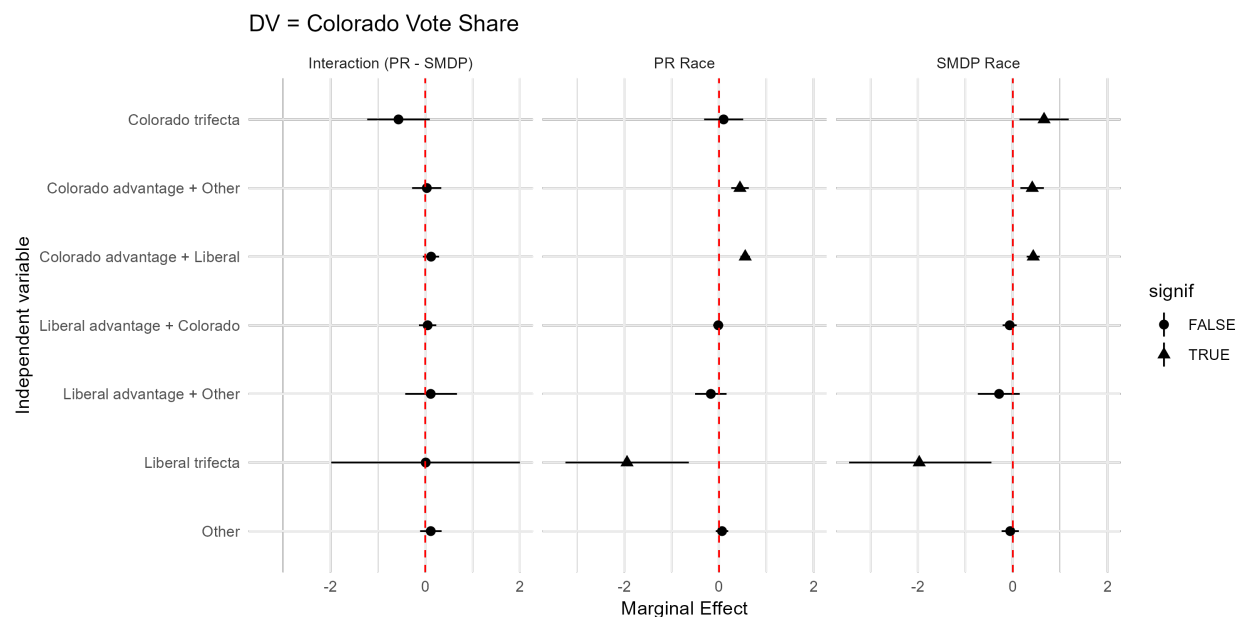
Note: A unit of observation is a voting booth-electoral race combination. Standard errors are robust. The excluded category corresponds to voting booths with “fair allocations” having one Colorado party poll worker and one Liberal poll worker.

Table E3: Effect of poll worker president's party on election results

	Colorado vote share (1)	Liberal/GANAR vote share (2)	Other vote share (3)	Blank vote share (4)	Null vote share (5)	Turnout (6)
Colorado poll worker president	0.1320 (0.0977)	-0.3191 (0.0897)	0.1871 (0.0863)	-0.0048 (0.0444)	-0.0733 (0.0383)	-1.1173 (0.1849)
Liberal poll worker president	0.0910 (0.0983)	-0.2154 (0.0903)	0.1243 (0.0868)	-0.0190 (0.0446)	-0.0891 (0.0385)	-1.1555 (0.1853)
Other party poll worker president	0.0024 (0.1027)	-0.3195 (0.0948)	0.3171 (0.0906)	-0.0253 (0.0470)	-0.0080 (0.0403)	-0.7478 (0.1923)
Multi-party poll worker president	-0.1127 (0.1275)	-0.1843 (0.1202)	0.2970 (0.1179)	-0.0972 (0.0591)	-0.0893 (0.0518)	-1.7160 (0.2640)
Mean of Dependent Variable	41.5808	34.4445	23.9746	5.2896	3.2264	122.1063
Polling station-elect. race FE	✓	✓	✓	✓	✓	✓
Colorado voter prop. control	✓	✓	✓	✓	✓	✓
Liberal voter prop. control	✓	✓	✓	✓	✓	✓
Other party voter prop. control	✓	✓	✓	✓	✓	✓
Observations	117,229	117,229	117,229	117,229	117,229	111,890
R^2	0.8423	0.8868	0.9308	0.6147	0.2120	0.5782

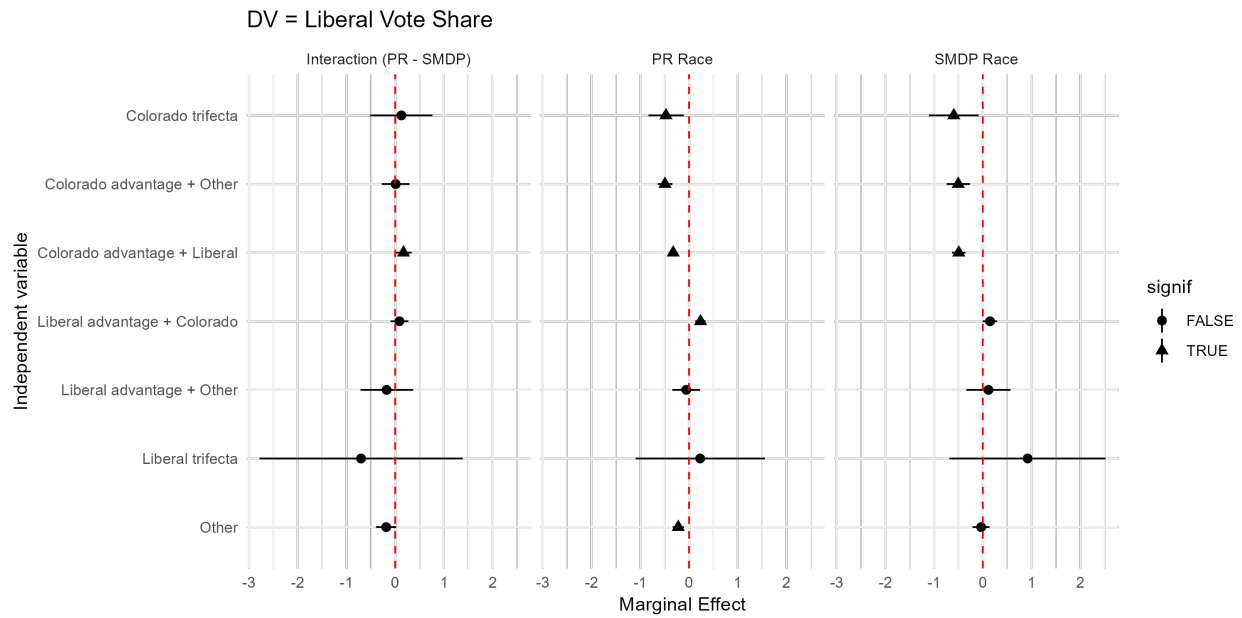
Note: A unit of observation is a voting booth-electoral race combination. Standard errors are robust. Vote shares for the first three columns are calculated by dividing by the number of valid votes (i.e., all votes excluding null and blank votes) and multiplying by 100. Vote shares for columns (4) and (5) are calculated by dividing by turnout (total votes including null and blank votes) and multiplying by 100. “X poll worker president” is an indicator variable for whether the poll worker president at the voting booth supports the X (where X can be Colorado, Liberal, Other party, or multiple parties). The excluded category is a president who is not registered with any party. The sample is restricted to voting booths for which we have the ID numbers of all three poll workers (which is around 96% of all voting booths). The number of observations on Column (6) is smaller as we restrict our sample to booths with exactly 200 registered voters.

Figure E1: Marginal effects on specification from Table 4. Effects under PR and SMDP calculated separately, and then interacted for Colorado vote share



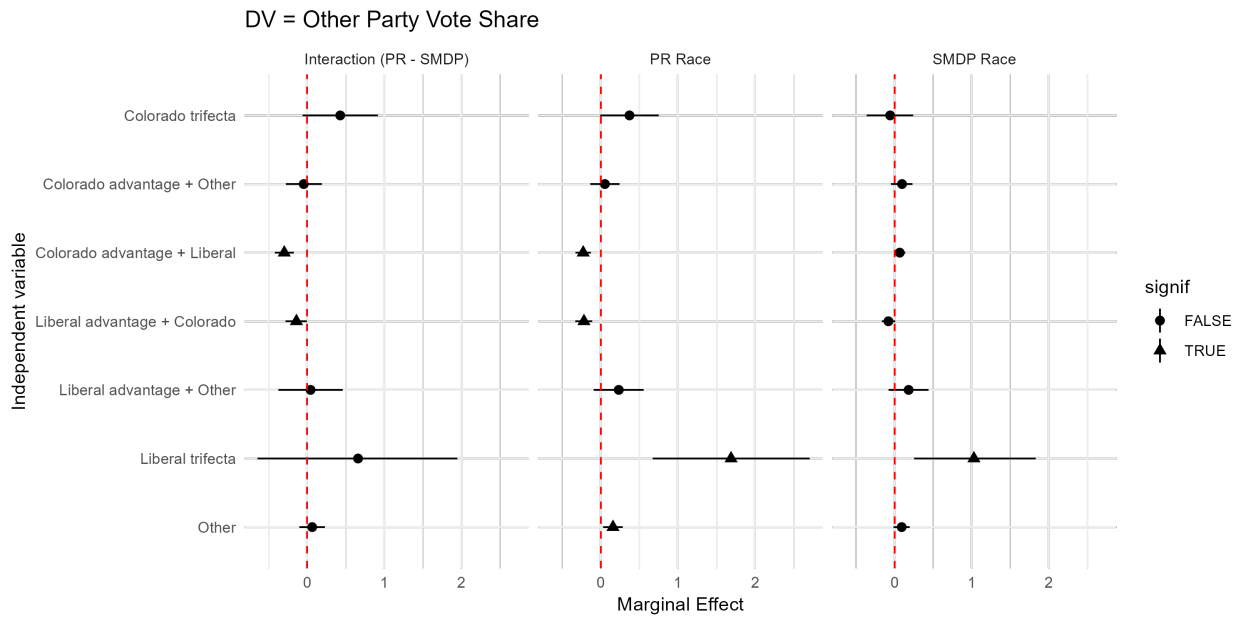
Note: Marginal effects were estimated using the Clarify package in R (King, Tomz and Wittenberg 2000). First, we held the race constant at PR=1, and estimated 10,000 iterations of the dependent variable holding each independent-variable dummy at 0 (PLRA_trifecta=0 for example), and then repeated the process holding PR=1 and setting each indicator to 1 (PLRA_trifecta=1 for example). We then differenced the estimates at 0 and 1, giving the first-difference marginal effect of each independent variable (e.g., PLRA_trifecta) on the outcome when PR=1. We then repeated this process for each independent variable, but holding PR=0. This gave us the marginal effect of each independent variable (PLRA_trifecta for example) on the dependent variable when PR=0. We then carried out the second difference estimate, taking the first difference for each indicator when PR=0 and subtracting that from the first difference for each indicator when PR=1. This provided us with our marginal effects for the interaction terms.

Figure E2: Marginal effects on specification from Table 4. Effects under PR and SMDP calculated separately, and then interacted for Liberal vote share



Note: Marginal effects were estimated using the Clarify package in R (King, Tomz and Wittenberg 2000). First, we held the race constant at PR=1, and estimated 10,000 iterations of the dependent variable holding each independent-variable dummy at 0 (PLRA_trifecta=0 for example), and then repeated the process holding PR=1 and setting each indicator to 1 (PLRA_trifecta=1 for example). We then differenced the estimates at 0 and 1, giving the first-difference marginal effect of each independent variable (e.g., PLRA_trifecta) on the outcome when PR=1. We then repeated this process for each independent variable, but holding PR=0. This gave us the marginal effect of each independent variable (PLRA_trifecta for example) on the dependent variable when PR=0. We then carried out the second difference estimate, taking the first difference for each indicator when PR=0 and subtracting that from the first difference for each indicator when PR=1. This provided us with our marginal effects for the interaction terms.

Figure E3: Marginal effects on specification from Table 4. Effects under PR and SMDP calculated separately, and then interacted for Other party vote share



Note: Marginal effects were estimated using the Clarify package in R (King, Tomz and Wittenberg 2000). First, we held the race constant at PR=1, and estimated 10,000 iterations of the dependent variable holding each independent-variable dummy at 0 (PLRA_trifecta=0 for example), and then repeated the process holding PR=1 and setting each indicator to 1 (PLRA_trifecta=1 for example). We then differenced the estimates at 0 and 1, giving the first-difference marginal effect of each independent variable (e.g., PLRA_trifecta) on the outcome when PR=1. We then repeated this process for each independent variable, but holding PR=0. This gave us the marginal effect of each independent variable (PLRA_trifecta for example) on the dependent variable when PR=0. We then carried out the second difference estimate, taking the first difference for each indicator when PR=0 and subtracting that from the first difference for each indicator when PR=1. This provided us with our marginal effects for the interaction terms.

Table E4: Effect of partisan poll worker composition on election results by whether the Liberal party ran under the GANAR list

	All races			Deputies and dept. council races		
	Colorado vote share (1)	Liberal/GANAR vote share (2)	Other party vote share (3)	Colorado vote share (4)	Liberal/GANAR vote share (5)	Other party vote share (6)
Colorado trifecta × GANAR list	-0.3148 (0.4080)	0.5602 (0.3849)	-0.2454 (0.3375)	-1.3512 (0.6850)	1.2227 (0.6559)	0.1286 (0.6987)
Colorado advantage + Other × GANAR list	0.0305 (0.1956)	-0.0026 (0.1656)	-0.0279 (0.1573)	0.2786 (0.3544)	-0.0634 (0.2799)	-0.2152 (0.3140)
Colorado advantage + Liberal × GANAR list	-0.0316 (0.1032)	-0.0574 (0.0969)	0.0890 (0.0825)	0.0709 (0.1893)	0.1779 (0.1749)	-0.2488 (0.1744)
Liberal advantage + Colorado × GANAR list	0.0736 (0.1081)	-0.0882 (0.1055)	0.0146 (0.0922)	0.2041 (0.1907)	-0.2001 (0.1888)	-0.0040 (0.1882)
Liberal advantage + Other × GANAR list	0.1239 (0.3449)	0.1775 (0.3137)	-0.3013 (0.2770)	0.3879 (0.6399)	0.5438 (0.5377)	-0.9317 (0.5220)
Liberal trifecta × GANAR list	-3.1038 (1.1471)	2.8948 (1.1562)	0.2090 (0.8814)	-6.8867 (1.7970)	6.2248 (1.7834)	0.6619 (1.7239)
Other poll worker allocation × GANAR list	-0.1294 (0.1336)	0.0609 (0.1251)	0.0685 (0.1126)	-0.1896 (0.2380)	-0.1755 (0.2202)	0.3651 (0.2286)
Colorado trifecta	0.4054 (0.2683)	-0.7314 (0.2285)	0.3260 (0.2465)	0.9429 (0.4255)	-0.9664 (0.3759)	0.0235 (0.4490)
Colorado advantage + Other	0.4191 (0.1222)	-0.4964 (0.0963)	0.0772 (0.1176)	0.3459 (0.2099)	-0.5256 (0.1667)	0.1797 (0.2028)
Colorado advantage + Liberal	0.5240 (0.0614)	-0.3595 (0.0551)	-0.1645 (0.0596)	0.5258 (0.1036)	-0.4352 (0.0953)	-0.0906 (0.1026)
Liberal advantage + Colorado	-0.0614 (0.0666)	0.2401 (0.0628)	-0.1787 (0.0680)	-0.0992 (0.1143)	0.3526 (0.1110)	-0.2535 (0.1219)
Liberal advantage + Other	-0.2583 (0.1940)	-0.0657 (0.1588)	0.3240 (0.1962)	-0.3729 (0.2913)	-0.2484 (0.2519)	0.6213 (0.3127)
Liberal trifecta	-0.6692 (0.7834)	-0.7170 (0.7821)	1.3862 (0.6889)	1.1254 (1.0863)	-2.8331 (1.0783)	1.7077 (1.1505)
Other poll worker allocation	0.0749 (0.0813)	-0.1831 (0.0734)	0.1082 (0.0801)	0.1060 (0.1359)	-0.1900 (0.1283)	0.0839 (0.1343)
Mean of Dependent Variable	41.5808	34.4445	23.9746	39.7285	32.2255	28.0461
Polling station-elect. race FE	✓	✓	✓	✓	✓	✓
Colorado voter prop. control	✓	✓	✓	✓	✓	✓
Liberal voter prop. control	✓	✓	✓	✓	✓	✓
Other party voter prop. control	✓	✓	✓	✓	✓	✓
Observations	117,229	117,229	117,229	38,268	38,268	38,268
R ²	0.8425	0.8869	0.9308	0.8317	0.8648	0.8986

Note: A unit of observation is a voting booth-electoral race combination. Standard errors are robust. “GANAR list” indicates that in the given race and department the Liberal party ran jointly with FG and other small parties under the GANAR label (in every department for the presidential race, in some departments for the governor, deputies, and departmental council races, and never for the Senate and MERCOSUR races). The excluded case is the Liberal party running on its own list. Columns (1) to (3) pool all races. Columns (4) to (6) keep the deputies and departmental council races, the two PR races in which alliance status varies across departments, so that the electoral system is held fixed. Where GANAR ran, the Liberal/GANAR vote share is the alliance’s share and the other-party share excludes the alliance’s members. Vote shares are calculated by dividing by the number of valid votes (i.e., all votes excluding null and blank votes) and multiplying by 100. A trifecta indicates all three poll workers from the same party work at the voting booth. Advantage indicates that two out of three poll workers from the same party work at the voting booth. Other refers to a poll worker that is not registered as supporting the Colorado or Liberal party. The excluded allocation category corresponds to voting booths with “fair allocations” having one Colorado poll worker and one Liberal poll worker. The Liberal trifecta rows rest on the 17 booths with a *de facto* Liberal trifecta.

(Roodman, Nielsen, MacKinnon and Webb 2019), for the two allocations in which both established parties hold seats and one of them a majority.

Table E5: District magnitudes and race descriptives by department

Department	District magnitude		Deputies race		Governor race		
	Deputies	Dept. council	Lists	ENEP	Lists	Winner's share	Top-two margin
Capital	7	–	30	6.57	–	–	–
Concepción	3	14	11	2.49	7	52.3	9.1
San Pedro	5	21	13	3.04	6	43.0	5.9
Cordillera	4	18	17	2.84	7	47.7	0.3
Guairá	3	14	11	2.79	8	62.7	34.0
Caaguazú	6	21	21	2.65	8	49.4	1.7
Caazapá	2	12	12	2.31	6	53.0	8.9
Itapúa	6	21	21	3.36	7	48.1	6.3
Misiones	2	11	12	2.22	6	50.7	4.3
Paraguarí	4	16	13	2.74	7	49.8	9.6
Alto Paraná	8	21	30	5.70	14	40.0	6.1
Central	20	21	32	5.31	8	46.4	2.0
Ñeembucú	2	9	7	2.20	4	50.0	1.9
Amambay	2	11	10	2.32	6	51.0	3.9
Canindeyú	2	12	9	2.24	7	57.8	20.5
Pdte. Hayes	2	10	12	4.60	10	35.7	12.4
Alto Paraguay	1	7	8	1.84	5	52.2	5.8
Boquerón	1	7	10	2.42	6	52.6	15.6

Note: One row per department. District magnitude is the number of seats the department elects to the Chamber of Deputies and to its departmental council (the capital elects deputies but has no governor or departmental council). Lists is the number of lists or candidates that received votes in the department. ENEP is the effective number of electoral parties of Laakso and Taagepera (1979). The governor race's winner share and top-two margin are in percent of valid votes. Votes are aggregated from the booth-level results used in the analysis, which exclude the booths whose tallies were annulled after a challenge and therefore record no valid votes.

Table E6: Effect of partisan poll worker composition on Chamber of Deputies results by district magnitude

	$Z = \ln(\text{district magnitude})$			$Z = 1(\text{district magnitude} \leq 2)$		
	Colorado vote share (1)	Liberal/GANAR vote share (2)	Other party vote share (3)	Colorado vote share (4)	Liberal/GANAR vote share (5)	Other party vote share (6)
Colorado trifecta $\times Z$	0.3377 (0.6548)	-0.0932 (0.5703)	-0.2445 (0.6244)	-3.6796 (2.4209)	0.4767 (2.2659)	3.2029 (1.4412)
Colorado advantage + Other $\times Z$	0.0500 (0.3536)	0.4674 (0.2501)	-0.5175 (0.2802)	0.7661 (0.8596)	-0.7427 (0.6082)	-0.0234 (0.5679)
Colorado advantage + Liberal $\times Z$	0.0022 (0.1567)	0.0725 (0.1403)	-0.0747 (0.1438)	-0.2153 (0.4177)	-0.1767 (0.3566)	0.3921 (0.2806)
Liberal advantage + Colorado $\times Z$	0.0260 (0.1657)	0.2527 (0.1590)	-0.2788 (0.1651)	0.4905 (0.4551)	-1.0348 (0.4260)	0.5442 (0.3006)
Liberal advantage + Other $\times Z$	-0.0902 (0.5813)	0.0106 (0.4609)	0.0797 (0.3986)	1.4369 (2.0085)	-1.2247 (1.5700)	-0.2122 (0.9886)
Liberal trifecta $\times Z$	3.2115 (1.2937)	-0.8998 (1.2109)	-2.3117 (0.8492)	-1.6014 (3.5753)	-1.4021 (2.1598)	3.0035 (2.1115)
Other poll worker allocation $\times Z$	0.0752 (0.2340)	0.0066 (0.2018)	-0.0818 (0.1891)	-0.4728 (0.6051)	0.0649 (0.5126)	0.4079 (0.4020)
Colorado trifecta	-0.3820 (1.4650)	0.2414 (1.2292)	0.1406 (1.2581)	0.4479 (0.4717)	0.0580 (0.4045)	-0.5059 (0.4800)
Colorado advantage + Other	0.3895 (0.7694)	-1.4069 (0.5415)	1.0174 (0.5524)	0.4016 (0.2390)	-0.4048 (0.1919)	0.0032 (0.2318)
Colorado advantage + Liberal	0.4509 (0.3420)	-0.3970 (0.3038)	-0.0539 (0.2773)	0.4767 (0.1206)	-0.2280 (0.1126)	-0.2487 (0.1225)
Liberal advantage + Colorado	-0.1648 (0.3648)	-0.1098 (0.3472)	0.2746 (0.3122)	-0.1595 (0.1330)	0.5007 (0.1333)	-0.3412 (0.1460)
Liberal advantage + Other	-0.1395 (1.3858)	0.0605 (1.1058)	0.0790 (0.8531)	-0.4551 (0.3657)	0.1843 (0.3092)	0.2708 (0.3743)
Liberal trifecta	-8.2230 (2.9418)	2.3887 (2.5812)	5.8343 (1.6995)	-1.9972 (1.6839)	0.8884 (1.4915)	1.1088 (1.1518)
Other poll worker allocation	-0.1289 (0.5280)	-0.2265 (0.4480)	0.3555 (0.3885)	0.0721 (0.1560)	-0.2189 (0.1451)	0.1468 (0.1598)
Mean of Dependent Variable	39.1551	30.2855	30.5593	39.1551	30.2855	30.5593
Polling station FE	✓	✓	✓	✓	✓	✓
Colorado voter prop. control	✓	✓	✓	✓	✓	✓
Liberal voter prop. control	✓	✓	✓	✓	✓	✓
Other party voter prop. control	✓	✓	✓	✓	✓	✓
WCB p : Colorado adv. + Liberal $\times Z$	0.987	0.513	0.485	0.704	0.652	0.068
WCB p : Liberal adv. + Colorado $\times Z$	0.820	0.134	0.056	0.185	0.067	0.188
Clusters (departments)	18	18	18	18	18	18
Observations	20,138	20,138	20,138	20,138	20,138	20,138
R^2	0.8487	0.8812	0.9232	0.8487	0.8812	0.9232

Note: A unit of observation is a voting booth. Standard errors are robust. The Liberal party always ran separately or through the GANAR alliance (while mainly proposing its own candidates), hence we have an outcome variable for Liberal/GANAR vote shares. Vote shares are calculated by dividing by the number of valid votes (i.e., all votes excluding null and blank votes) and multiplying by 100. A trifecta indicates all three poll workers from the same party work at the voting booth. Advantage indicates that two out three poll workers from the same party work at the voting booth. Other refers to a poll worker that is not registered as supporting the Colorado or Liberal party. The excluded category corresponds to voting booths with “fair allocations” having one Colorado party poll worker and one Liberal poll worker. The “other poll worker allocation” refers to any remaining poll worker allocation after accounting for “fair allocations” and the other allocations referenced with the independent variables. The sample is the Chamber of Deputies race. In Columns (1) to (3), Z is the natural log of the department’s district magnitude (1 to 20 seats), in Columns (4) to (6), Z is an indicator for the eight districts electing one or two deputies. Because Z takes one value per department, the last rows report wild-cluster-bootstrap p -values for the two interactions named, clustered on department with Webb weights and 9,999 replications (Roodman, Nielsen, MacKinnon and Webb 2019). The main effect of Z is absorbed by the polling station fixed effects.

Table E7: Effect of partisan poll worker composition on Chamber of Deputies and departmental council results by district magnitude, within departments

	Colorado vote share (1)	Liberal/GANAR vote share (2)	Other party vote share (3)
Colorado trifecta $\times \ln(\text{district magnitude})$	0.6814 (1.0622)	0.3237 (0.9911)	-1.0051 (1.2045)
Colorado advantage + Other $\times \ln(\text{district magnitude})$	0.3999 (0.5742)	0.2879 (0.4304)	-0.6877 (0.5053)
Colorado advantage + Liberal $\times \ln(\text{district magnitude})$	0.0579 (0.2726)	0.3422 (0.2511)	-0.4001 (0.2625)
Liberal advantage + Colorado $\times \ln(\text{district magnitude})$	0.0409 (0.2806)	0.3708 (0.2767)	-0.4117 (0.2942)
Liberal advantage + Other $\times \ln(\text{district magnitude})$	0.1943 (0.8836)	0.0439 (0.7277)	-0.2382 (0.7543)
Liberal trifecta $\times \ln(\text{district magnitude})$	2.6722 (1.7784)	0.0268 (2.0852)	-2.6990 (2.1236)
Other poll worker allocation $\times \ln(\text{district magnitude})$	0.1897 (0.3794)	-0.3464 (0.3481)	0.1567 (0.3494)
Colorado trifecta	-0.3379 (2.3819)	-0.0008 (2.0404)	0.3387 (2.6946)
Colorado advantage + Other	-1.0611 (1.2488)	-0.1366 (0.9119)	1.1978 (1.1579)
Colorado advantage + Liberal	-0.1376 (0.5954)	0.1950 (0.5309)	-0.0573 (0.5915)
Liberal advantage + Colorado	-1.1721 (0.9500)	0.2073 (0.7057)	0.9648 (1.0095)
Liberal advantage + Other	-0.8443 (2.3027)	0.4613 (1.6460)	0.3830 (2.1920)
Liberal trifecta	-11.8919 (3.6703)	-0.7791 (4.2477)	12.6710 (4.3218)
Other poll worker allocation	-0.3409 (0.8394)	0.9527 (0.7358)	-0.6117 (0.7933)
Mean of Dependent Variable	39.7285	32.2255	28.0461
Polling station-elect. race FE	✓	✓	✓
Allocation $\times$ department	✓	✓	✓
Allocation $\times$ council race	✓	✓	✓
Colorado voter prop. control	✓	✓	✓
Liberal voter prop. control	✓	✓	✓
Other party voter prop. control	✓	✓	✓
WCB p : Colorado adv. + Liberal $\times \ln(\text{district magnitude})$	0.594	0.054	0.071
WCB p : Liberal adv. + Colorado $\times \ln(\text{district magnitude})$	0.669	0.100	0.082
Clusters (departments)	18	18	18
Observations	38,268	38,268	38,268
R^2	0.8325	0.8655	0.8990

Note: A unit of observation is a voting booth-electoral race combination. Standard errors are robust. The Liberal party always ran separately or through the GANAR alliance (while mainly proposing its own candidates), hence we have an outcome variable for Liberal/GANAR vote shares. Vote shares are calculated by dividing by the number of valid votes (i.e., all votes excluding null and blank votes) and multiplying by 100. A trifecta indicates all three poll workers from the same party work at the voting booth. Advantage indicates that two out three poll workers from the same party work at the voting booth. Other refers to a poll worker that is not registered as supporting the Colorado or Liberal party. The excluded category corresponds to voting booths with “fair allocations” having one Colorado party poll worker and one Liberal poll worker. The “other poll worker allocation” refers to any remaining poll worker allocation after accounting for “fair allocations” and the other allocations referenced with the independent variables. The sample pools the Chamber of Deputies and departmental council races. District magnitude is the number of seats the department elects in the race (1 to 20 for deputies, 7 to 21 for councils). The regressions add the interaction of every allocation indicator with department dummies and with a departmental council indicator, so the interactions with $\ln(\text{district magnitude})$ are identified from the within-department difference in magnitude between the two races. Because district magnitude takes one value per department and race, the last rows report wild-cluster-bootstrap p -values for the two interactions named, clustered on department with Webb weights and 9,999 replications (Roodman, Nielsen, MacKinnon and Webb 2019). The main effect of district magnitude is absorbed by the polling station-electoral race fixed effects.

Table E8: Effect of partisan poll worker composition on governor results by the closeness of the race

	Colorado vote share (1)	Liberal/GANAR vote share (2)	Other party vote share (3)
Colorado trifecta $\times$ top-two margin	0.0494 (0.0719)	0.0031 (0.0465)	-0.0525 (0.0475)
Colorado advantage + Other $\times$ top-two margin	0.0158 (0.0390)	-0.0048 (0.0345)	-0.0110 (0.0236)
Colorado advantage + Liberal $\times$ top-two margin	-0.0248 (0.0156)	0.0314 (0.0148)	-0.0066 (0.0094)
Liberal advantage + Colorado $\times$ top-two margin	-0.0397 (0.0199)	0.0434 (0.0177)	-0.0037 (0.0123)
Liberal advantage + Other $\times$ top-two margin	0.0335 (0.0590)	-0.0688 (0.0794)	0.0353 (0.0612)
Liberal trifecta $\times$ top-two margin	-0.0767 (0.1897)	-0.0090 (0.1421)	0.0857 (0.1186)
Other poll worker allocation $\times$ top-two margin	0.0040 (0.0334)	0.0112 (0.0285)	-0.0152 (0.0185)
Colorado trifecta	0.4340 (0.6380)	-0.4385 (0.5654)	0.0045 (0.4009)
Colorado advantage + Other	0.3271 (0.3136)	-0.4562 (0.3003)	0.1291 (0.1743)
Colorado advantage + Liberal	0.5333 (0.1514)	-0.6549 (0.1481)	0.1216 (0.0859)
Liberal advantage + Colorado	0.1204 (0.1618)	-0.0426 (0.1573)	-0.0779 (0.0972)
Liberal advantage + Other	-0.5300 (0.4508)	0.3625 (0.4454)	0.1675 (0.2781)
Liberal trifecta	-1.2302 (1.7312)	0.6784 (1.7319)	0.5518 (0.9122)
Other poll worker allocation	-0.1507 (0.2238)	-0.1614 (0.2102)	0.3121 (0.1303)
Mean of Dependent Variable	46.9013	42.3609	10.7379
Polling station FE	✓	✓	✓
Colorado voter prop. control	✓	✓	✓
Liberal voter prop. control	✓	✓	✓
Other party voter prop. control	✓	✓	✓
WCB p : Colorado adv. + Liberal $\times$ top-two margin	0.197	0.059	0.437
WCB p : Liberal adv. + Colorado $\times$ top-two margin	0.057	0.009	0.649
Clusters (departments)	17	17	17
Observations	18,144	18,144	18,144
R^2	0.8143	0.8564	0.9083

Note: A unit of observation is a voting booth. Standard errors are robust. The Liberal party always ran separately or through the GANAR alliance (while mainly proposing its own candidates), hence we have an outcome variable for Liberal/GANAR vote shares. Vote shares are calculated by dividing by the number of valid votes (i.e., all votes excluding null and blank votes) and multiplying by 100. A trifecta indicates all three poll workers from the same party work at the voting booth. Advantage indicates that two out three poll workers from the same party work at the voting booth. Other refers to a poll worker that is not registered as supporting the Colorado or Liberal party. The excluded category corresponds to voting booths with “fair allocations” having one Colorado party poll worker and one Liberal poll worker. The “other poll worker allocation” refers to any remaining poll worker allocation after accounting for “fair allocations” and the other allocations referenced with the independent variables. The sample is the governor race. Each allocation is interacted with the department’s 2018 top-two margin in percentage points of valid votes, so the main effects are the effects in a race decided by a zero margin. Because the margin takes one value per department, the last rows report wild-cluster-bootstrap p -values for the two interactions named, clustered on departments with Webb weights and 9,999 replications (Roodman, Nielsen, MacKinnon and Webb 2019). The main effect of the margin is absorbed by the polling station fixed effects.

For a related follow-up test to the electoral system heterogeneity analysis, we generate new partisan allocation indicator variables by pooling together the Colorado and Liberal poll workers as “traditional party” poll workers, which we contrast with the remaining other party poll workers. We do this because we acknowledge that collusion between traditional party poll workers could even occur with a fair allocation having just one Colorado party and one Liberal poll worker, as they would constitute a majority of the poll worker team. We thus generate the following indicator variables: “traditional party trifecta,” “other party trifecta,” “traditional party advantage + other party,” and “other party advantage + traditional party.” Our excluded category then corresponds to voting booths with fair allocations having only one traditional party poll worker and one poll worker from another party. Thus, the indicator variable for “other poll worker allocations” here refers to any remaining poll worker allocation after accounting for fair allocations and the other allocations referenced above. These indicator variables are all mutually exclusive and together cover all electoral tables. We once again include the interactions between PR race and the indicator variables mentioned previously. This leads us to the following regression specification:

$$\begin{aligned}
Y_{abc} = & \beta_0 + \alpha_1 \text{Tradpart_Trif}_{ab} \times PR_c + \alpha_2 \text{Tradpart_Adv_Othpart}_{ab} \times PR_c \\
& + \alpha_3 \text{Othpart_Adv_Tradpart}_{ab} \times PR_c + \alpha_4 \text{Othpart_Trif}_{ab} \times PR_c + \alpha_5 \text{Other_pw_alloc}_{ab} \times PR_c \\
& + \beta_1 \text{Tradpart_Trif}_{ab} + \beta_2 \text{Tradpart_Adv_Othpart}_{ab} + \beta_3 \text{Othpart_Adv_Tradpart}_{ab} \\
& + \beta_4 \text{Othpart_Trif}_{ab} + \beta_5 \text{Other_pw_alloc}_{ab} \\
& + \delta_1 \text{voter_prop_Colorado}_{ab} + \delta_2 \text{voter_prop_Liberal}_{ab} + \delta_3 \text{voter_prop_Other}_{ab} + \eta_{ac} + \varepsilon_{abc}.
\end{aligned} \tag{3}$$

As an additional test of Prediction 3, Table E9 explores the heterogeneous effects of partisan poll workers by electoral system and uses regression specification (3). The difference with Table 4 is that we pool the two traditional parties together and contrast them with poll workers from other parties. In SMDP races, a traditional party trifecta of poll workers raises the Colorado vote share by 0.36 pp, and the interaction with the PR indicator is close to zero, so the estimate for PR races is similar (0.33 pp, the sum of the main effect and the interaction). A traditional party trifecta tends to favor the Colorado party, likely because the Colorado party holds the majority within these teams more often than the Liberal party does. For the Liberal/GANAR vote share, the main effect in SMDP races is negative and imprecise (−0.15 pp), while the interaction with the PR indicator is 0.35 pp, so a traditional party trifecta in PR races is associated with a 0.20 pp higher Liberal/GANAR vote share (the sum of the two coefficients). The PR–SMDP difference has the sign our collusion prediction implies. As the pooled categories combine mixed Colorado–Liberal teams with same-party majorities and trifectas, they capture established-party domination broadly.

Table E9: Effect of partisan poll worker composition on all election results by electoral system distinguishing traditional and other parties

	Colorado vote share (1)	Liberal/GANAR vote share (2)	Other party vote share (3)
Trad. party trifecta × PR race	-0.0343 (0.1837)	0.3534 (0.1726)	-0.3190 (0.1386)
Trad. party adv. + Other party × PR race	-0.1125 (0.1805)	0.2052 (0.1680)	-0.0927 (0.1353)
Other party adv. + Trad. party × PR race	-0.1342 (0.3366)	-0.3163 (0.3264)	0.4505 (0.2563)
Other party trifecta × PR race	-0.5722 (1.6076)	1.7118 (1.6334)	-1.1396 (1.6029)
Other poll worker allocation × PR race	-0.0745 (0.1892)	0.2489 (0.1776)	-0.1744 (0.1432)
Trad. party trifecta	0.3636 (0.1485)	-0.1537 (0.1428)	-0.2099 (0.0870)
Trad. party adv. + Other party	0.2296 (0.1459)	-0.0456 (0.1394)	-0.1840 (0.0854)
Other party adv. + Trad. party	0.1328 (0.2716)	0.2013 (0.2715)	-0.3341 (0.1596)
Other party trifecta	-0.9501 (1.2369)	0.4189 (1.4207)	0.5312 (0.9772)
Other poll worker allocation	0.1047 (0.1531)	0.1197 (0.1474)	-0.2245 (0.0901)
Mean of Dependent Variable	41.5808	34.4445	23.9746
Polling station-elect. race FE	✓	✓	✓
Colorado voter prop. control	✓	✓	✓
Liberal voter prop. control	✓	✓	✓
Other party voter prop. control	✓	✓	✓
Observations	117,229	117,229	117,229
R^2	0.8423	0.8868	0.9308

Note: A unit of observation is a voting booth-electoral race combination. Standard errors are robust. The parties belonging to the GANAR alliance vary across Paraguayan departments for the governor, deputies, and departmental council elections, with FG running independently in some departments. however, the Liberal party always runs separately or through GANAR (and mainly proposing its own candidates), hence we have an outcome variable for Liberal/GANAR vote shares for these races. Vote shares are calculated by dividing by the number of valid votes (i.e., all votes excluding null and blank votes) and multiplying by 100. A trifecta indicates all three poll workers at the voting booth either support the traditional parties (Colorado and Liberal) or some other party. Advantage indicates that two out three poll workers either support the traditional parties (Colorado and Liberal) or some other party. The excluded category corresponds to voting booths with “fair allocations” having one traditional party poll worker and one poll worker from another party. The “other poll worker allocation” refers to any remaining poll worker allocation after accounting for “fair allocations” and the other allocations referenced with the independent variables. “PR race” is an indicator variable for electoral races with a proportional representation (PR) system. The coefficient for PR race is omitted because it is collinear once we control for polling station-electoral race fixed effects. The sample is restricted to voting booths for which we have the ID numbers of all three poll workers (which is around 96% of all voting booths).

In addition, we can observe how the vote share of other parties varies symmetrically with the number of traditional party and other party poll workers. Whenever the traditional parties have a trifecta counting votes, the vote share of other parties decreases significantly by 0.21 pp. This decrease from traditional party trifectas is compounded in PR races with an additional significant decrease of 0.32 pp. We can also notice how two traditional party poll workers and one other party poll worker counting votes leads to a significant decrease of 0.18 pp. If instead the other parties have two poll workers and the traditional parties have one we still observe a significant decrease of 0.33 pp, yet this is compensated in PR races because we find that this allocation leads to a significant (at the 10 percent level) increase of 0.45 pp. Overall, these results are consistent with Prediction 3. Lastly, we find that “other” poll worker allocations lead to a significant decline of 0.22 pp in the vote share of other parties. This result is also plausible because now the “other” poll worker allocations include, for example, electoral tables with two traditional party poll workers and one unaffiliated poll worker (i.e., a poll worker registered as not supporting any political party according to the electoral authority). This sort of partisan poll worker allocations should ultimately benefit the traditional parties more. Hence, the results from this table are suggestive that collusion between traditional party poll workers and greater electoral irregularities (to the detriment of other parties) are more likely with PR races.

Table E10 examines the effect of poll workers on voting process outcomes (blank votes, null votes, and turnout) by pooling together observations from all elections and running equation (1) on its even columns. The dependent variables are blank vote shares, null (invalid) vote shares, and turnout recorded at each voting booth. Some of the results from Table E10 also appear on the results for the individual races, though because the number of observations is more limited, we might not have enough statistical power to observe as many statistically significant results. In Column (1) we can observe that poll workers do not have a significant effect on blank votes. In contrast, in Column (2) we find that electoral tables with just Colorado and Liberal poll workers significantly decrease the null vote share by around 0.08 pp. We can also notice that Colorado trifectas decrease the null vote share by approximately 0.19 pp. This result suggests that perhaps established party poll workers engage in electoral irregularities partly by awarding themselves null votes instead of discarding these. Still, these coefficients are an order of magnitude smaller than the vote share effects, so poll workers likely also reassign some other-party votes to their own party. Finally, on Column (3) we find two effects from partisan poll worker allocations. Electoral tables with two Liberal poll workers and one poll worker from neither the Colorado nor Liberal party lead to a significant increase of around 0.62 votes, and Colorado trifectas to a significant decrease of around 0.69 votes, compared to a mean of around 122.5 turnout per booth. These results are more ambiguous as an increase or decrease in turnout could potentially indicate real votes being counted, and there could be a mix of strategies with results being manipulated by either stuffing

ballot boxes or discarding opposition votes.

Table E10: Effect of partisan poll worker composition on election process results

	Blank vote share (1)	Null vote share (2)	Turnout (3)
Colorado trifecta	-0.0823 (0.0897)	-0.1886 (0.0708)	-0.6909 (0.3317)
Colorado advantage + Other	0.0231 (0.0423)	-0.0205 (0.0344)	-0.0543 (0.1436)
Colorado advantage + Liberal	0.0167 (0.0228)	-0.0753 (0.0190)	0.0263 (0.0786)
Liberal advantage + Colorado	-0.0051 (0.0244)	-0.0820 (0.0203)	0.1087 (0.0860)
Liberal advantage + Other	-0.1006 (0.0703)	-0.1013 (0.0662)	0.6179 (0.2535)
Liberal trifecta	0.0794 (0.2359)	-0.2517 (0.2199)	0.2832 (0.7629)
Other poll worker allocation	-0.0181 (0.0307)	0.0138 (0.0248)	-0.0452 (0.1046)
Mean of Dependent Variable	5.2896	3.2264	122.5388
Polling station-elect. race FE	✓	✓	✓
Colorado voter prop. control	✓	✓	✓
Liberal voter prop. control	✓	✓	✓
Other party voter prop. control	✓	✓	✓
Observations	117,229	117,229	111,493
R^2	0.6147	0.2120	0.6917

Note: A unit of observation is a voting booth-electoral race combination. Standard errors are robust. Vote shares are calculated by dividing by turnout (i.e., all votes including null and blank votes) and multiplying by 100. The number of observations on Column (3) is smaller because we restrict our sample to booths with exactly 200 registered voters. A trifecta indicates all three poll workers from the same party work at the voting booth. Advantage indicates that two out three poll workers from the same party work at the voting booth. Other refers to a poll worker that is not registered as supporting the Colorado or Liberal party. The excluded category corresponds to voting booths with “fair allocations” having one Colorado poll worker and one Liberal poll worker. The “other poll worker allocation” refers to any remaining poll worker allocation after accounting for “fair allocations” and the other allocations referenced with the independent variables. The sample is restricted to voting booths for which we have the ID numbers of all three poll workers (which is around 96% of all voting booths).

Table E11 reports the signature audit cited in Section I.A. We drew a random sample of 300 voting booths from those with a legible tally image and checked, for every race tally in the image, whether each of the three “FIRMA” (signature) lines and each of the three “C.I.N.” (ID number)

lines was filled in. A Python script measured the ink in each signature and ID box of every tally. The slots with the least ink, together with the tallies the script could not segment, were checked by hand from image crops (176 tally rows).

Table E11: Signatures and ID numbers on the race tallies of a random sample of voting booths

	Booths	Share
At least one race tally with fewer than three signatures	9	3.0%
At least one race tally with fewer than three ID numbers	54	18.0%
Booths audited	300	

Note: A unit of observation is a voting booth, and the sample is a random draw of 300 booths from the 20,523 booths with a legible tally image. The image of each booth holds one tally per race (1,133 tallies with a signature block in the sample, tallies transmitted by voice have none). A booth counts as having fewer than three signatures (ID numbers) if any of its race tallies lacks at least one of the three signatures (ID numbers). The 95% confidence intervals are 1.4% to 5.6% for signatures and 13.8% to 22.8% for ID numbers.

F. Establishment party support and partisan poll workers

Perhaps it is not a coincidence that most countries where explicitly partisan representatives can be poll workers or polling station administrators correspond to flawed democracies and hybrid regimes (Economist Intelligence Unit 2022), which are: Honduras, Armenia, Paraguay, Ukraine, Kosovo, El Salvador, and Colombia. All these countries have a history of established parties and their elites could have strategically designed these adversarial electoral institutions to help keep themselves in power. Albertus and Menaldo (2018) study how outgoing authoritarian elites constitutionalize protections for their interests and vote-counting institutions play a fundamental role in this aspect. Consistent with this, if we follow the definition of established parties from Lupu (2016) and use updated data on 111 Latin American presidential elections going from 1990 to 2018,⁵ we find a positive cross-country association between partisan poll workers and established party support: on average, established parties have received higher vote shares in Latin American countries that allow partisan poll workers.

Established parties according to Lupu (2016) are those “parties that in two consecutive elections either receive a plurality of the vote, winning the first round, or attract no less than one-third of the winning vote share.” Using the same definition of established parties for the 18 Latin American countries considered by Lupu (2016) for 1990 to 2018, yields the following established parties:

- **Argentina:** Partido Justicialista (PJ), Unión Cívica Radical (UCR)

⁵The Latin American elections data we updated come from Baker and Greene (2011a), originally compiled for Baker and Greene (2011b).

- **Bolivia:** Acción Democrática y Nacionalista (ADN), Conciencia de Patria (CONDEPA), Movimiento al Socialismo (MAS), Movimiento de Izquierda Revolucionario (MIR), Movimiento Nacionalista Revolucionario (MNR)
- **Brazil:** Partido da Social Democracia Brasileira (PSDB), Partido dos Trabalhadores (PT)
- **Chile:**⁶ Alianza por Chile, Concertación
- **Colombia:** Centro Democrático (CD), Partido Conservador Colombiano (PCC), Partido Liberal Colombiano (PLC), Partido de la U
- **Costa Rica:** Partido Liberación Nacional (PLN), Partido Unidad Social Cristiana (PUSC)
- **Dominican Republic:** Partido de la Liberación Dominicana (PLD), Partido Reformista Social Cristiano (PRSC), Partido Revolucionario Dominicano (PRD)
- **Ecuador:** Alianza Patria Altiva i Soberana (Alianza PAIS), Izquierda Democrática (ID), Partido Roldosista Ecuatoriano (PRE), Partido Social Cristiano (PSC)
- **El Salvador:** Alianza Republicana Nacional (ARENA), Frente Farabundo Martí para la Liberación Nacional (FMLN)
- **Guatemala:** Frente Republicano Guatemalteco (FRG), Partido de Avanzada Nacional (PAN)
- **Honduras:** Partido Liberal de Honduras (PLH), Partido Nacional de Honduras (PNH)
- **Mexico:** Partido Acción Nacional (PAN), Partido de la Revolución Democrática (PRD), Partido Revolucionario Institucional (PRI)
- **Nicaragua:** Frente Sandinista de Liberación Nacional (FSLN), Partido Liberal Constitucionalista (PLC)
- **Panama:** Partido Arnulfista (renamed Partido Panameñista), Partido Revolucionario Democrático (PRD)
- **Paraguay:** Asociación Nacional Republicana—Partido Colorado (ANR-PC), Partido Liberal Radical Auténtico (PLRA)
- **Peru:** Alianza Popular Revolucionaria Americana (APRA), Cambio 90, Partido Nacionalista Peruano, Perú Posible
- **Uruguay:** Frente Amplio, Partido Colorado, Partido Nacional
- **Venezuela:** Acción Democrática (AD), Movimiento V República (MVR) [which later became the Partido Socialista Unido de Venezuela (PSUV)], Partido Social Cristiano (COPEI)

SA Table F1 shows that established parties' vote shares are higher in the Latin American presidential elections held in countries with partisan poll workers: the mean is 76.8% against 66.5%, the minimum 34.6% against 2.8%, the median 83.7% against 67.2%, and the maximum 100% against 99.9%. SA Table F2 reports the same comparison as a regression: the vote share of established parties is about 10 percentage points higher in elections held under partisan poll workers, with or without election-year fixed effects. This is a descriptive cross-country correlation, not a causal estimate. Countries that allow partisan poll workers differ from the others in ways that also

⁶Following Lupu (2016), for Chile we code coalitions instead of individual parties given the “uniquely institutionalized and stable nature of electoral coalitions.”

bear on established parties' vote shares, among them the institutionalization of the party system, the age of the democracy, and, as noted above, the possibility that established parties designed these adversarial institutions to protect themselves. The correlation is consistent with the within-country, quasi-experimental evidence from Paraguay, but it cannot by itself establish that partisan poll workers cause established parties' electoral advantage.

Table F1: Established party vote share summary statistics

	Observations	Mean	Standard Deviation	Minimum	Median	Maximum
Without partisan poll workers:						
Established party vote share	73	66.47	26.16	2.76	67.23	99.87
With partisan poll workers:						
Established party vote share	38	76.79	20.00	34.63	83.69	100.00

Note: Each observation corresponds to the vote share for established parties in a Latin American presidential election from 1990 to 2018. All statistics are rounded up to the second digit after the decimal.

Table F2: Established party vote share and partisan poll workers: a cross-country correlation

	Established party vote share	
	(1)	(2)
Has partisan poll workers	10.324 (4.456)	10.765 (4.371)
Mean of Dependent Variable	70.000	69.899
Year FE		✓
Observations	111	109
R^2	0.0399	0.4838

Note: Each observation corresponds to the vote share for established parties in a Latin American presidential election from 1990 to 2018. Standard errors are robust. “Has partisan poll workers” refers to a country where explicitly partisan representatives can be polling station administrators or poll workers. The indicator is coded from the electoral rules in force in 2018 and treated as a country characteristic over the period. “Observations” refers to the number of non-singleton observations, so the number of observations slightly decreases when we include year fixed effects on the regression of Column (2).

G. Additional Robustness Checks

This section collects supporting tables and figures for the specifications discussed above. Tables G1 and G2 reproduce Tables 3 and 4 showing every covariate. Table G3 controls for the proportion of voters at each booth carrying one of the country's 100 most common surnames. Tables G4 and G5 re-estimate our main specification using the pre-registered allocation, first on its own and then alongside the *de facto* allocation, and Tables G11 and G12 control for the proportion of voters sharing a surname with a candidate in the presidential and governor races.

Table G6 restricts the sample to booths whose pre-registered allocation was fair. Tables G7 and G8 cluster standard errors by voting booth. Table G9 splits the fair reference category by the party of the third poll worker. Table G10 lets the voter-affiliation controls vary by race.

Table G1: Effect of partisan poll worker composition on election results—showing all covariates

	Colorado vote share (1)	Liberal/GANAR vote share (2)	Other party vote share (3)
Colorado trifecta	0.2820 (0.2029)	-0.5121 (0.1847)	0.2301 (0.1752)
Colorado advantage + Other	0.4318 (0.0954)	-0.4969 (0.0790)	0.0651 (0.0820)
Colorado advantage + Liberal	0.5125 (0.0493)	-0.3800 (0.0455)	-0.1325 (0.0433)
Liberal advantage + Colorado	-0.0328 (0.0525)	0.2063 (0.0505)	-0.1735 (0.0483)
Liberal advantage + Other	-0.2126 (0.1611)	-0.0035 (0.1406)	0.2161 (0.1442)
Liberal trifecta	-1.9481 (0.5951)	0.4756 (0.5947)	1.4724 (0.4649)
Other poll worker allocation	0.0246 (0.0646)	-0.1593 (0.0597)	0.1347 (0.0578)
Colorado voter prop.	0.3802 (0.0049)	-0.2898 (0.0043)	-0.0904 (0.0040)
Liberal voter prop.	-0.5370 (0.0052)	0.6334 (0.0049)	-0.0963 (0.0046)
Other party voter prop.	-0.1242 (0.0115)	-0.0220 (0.0094)	0.1461 (0.0090)
Mean of Dependent Variable	41.5808	34.4445	23.9746
Polling station-elect. race FE	✓	✓	✓
Observations	117,229	117,229	117,229
R^2	0.8425	0.8869	0.9308

Note: A unit of observation is a voting booth-electoral race combination. Standard errors are robust. All regressions control for the percent of Colorado, Liberal, and other party voters registered to vote at each booth. The Liberal party always ran separately or through the GANAR alliance (while mainly proposing its own candidates), hence we have an outcome variable for Liberal/GANAR vote shares. Vote shares are calculated by dividing by the number of valid votes (i.e., all votes excluding null and blank votes) and multiplying by 100. A trifecta indicates all three poll workers from the same party work at the voting booth. Advantage indicates that two out three poll workers from the same party work at the voting booth. Other refers to a poll worker that is not registered as supporting the Colorado or Liberal party. The excluded category corresponds to voting booths with “fair allocations” having one Colorado party poll worker, one Liberal poll worker, and one “other” poll worker. The “other poll worker allocation” refers to any remaining poll worker allocation after accounting for “fair allocations” and the other allocations referenced with the independent variables.

Table G2: Effect of partisan poll worker composition on election results by electoral system—showing all covariates

	Colorado vote share (1)	Liberal/GANAR vote share (2)	Other party vote share (3)
Colorado trifecta × PR race	-0.5625 (0.4166)	0.1364 (0.3948)	0.4261 (0.3124)
Colorado advantage + Other × PR race	0.0317 (0.1976)	0.0117 (0.1750)	-0.0434 (0.1440)
Colorado advantage + Liberal × PR race	0.1232 (0.1032)	0.1718 (0.0982)	-0.2949 (0.0754)
Liberal advantage + Colorado × PR race	0.0514 (0.1097)	0.0887 (0.1072)	-0.1401 (0.0848)
Liberal advantage + Other × PR race	0.1243 (0.3316)	-0.1741 (0.3116)	0.0498 (0.2596)
Liberal trifecta × PR race	0.0301 (1.1852)	-0.6823 (1.2005)	0.6522 (0.8066)
Other poll worker allocation × PR race	0.1203 (0.1356)	-0.1856 (0.1292)	0.0653 (0.1024)
Colorado trifecta	0.6614 (0.3292)	-0.6040 (0.3248)	-0.0574 (0.1983)
Colorado advantage + Other	0.4104 (0.1575)	-0.5047 (0.1485)	0.0943 (0.0877)
Colorado advantage + Liberal	0.4295 (0.0831)	-0.4957 (0.0816)	0.0663 (0.0449)
Liberal advantage + Colorado	-0.0673 (0.0883)	0.1469 (0.0874)	-0.0796 (0.0511)
Liberal advantage + Other	-0.2959 (0.2627)	0.1130 (0.2648)	0.1830 (0.1667)
Liberal trifecta	-1.9683 (0.9019)	0.9328 (0.9306)	1.0355 (0.4775)
Other poll worker allocation	-0.0564 (0.1096)	-0.0345 (0.1075)	0.0909 (0.0635)
Colorado voter prop.	0.3802 (0.0049)	-0.2898 (0.0043)	-0.0904 (0.0040)
Liberal voter prop.	-0.5370 (0.0052)	0.6333 (0.0049)	-0.0963 (0.0046)
Other party voter prop.	-0.1242 (0.0115)	-0.0220 (0.0094)	0.1461 (0.0090)
Mean of Dependent Variable	41.5808	34.4445	23.9746
Polling station-elect. race FE	✓	✓	✓
Observations	117,229	117,229	117,229
R ²	0.8425	0.8869	0.9308

Note: A unit of observation is a voting booth-electoral race combination. Standard errors are robust. All regressions control for the percent of Colorado, Liberal, and other party voters registered to vote at each booth. The Liberal party always ran separately or through the GANAR alliance (while mainly proposing its own candidates), hence we have an outcome variable for Liberal/GANAR vote shares. Vote shares are calculated by dividing by the number of valid votes (i.e., all votes excluding null and blank votes) and multiplying by 100. A trifecta indicates all three poll workers from the same party work at the voting booth. Advantage indicates that two out three poll workers from the same party work at the voting booth. Other refers to a poll worker that is not registered as supporting the Colorado or Liberal party. The excluded category corresponds to voting booths with “fair allocations” having one Colorado party poll worker and one Liberal poll worker. The “other poll worker allocation” refers to any remaining poll worker allocation after accounting for “fair allocations” and the other allocations referenced with the independent variables. “PR race” is an indicator variable for electoral races with a proportional representation (PR) system. The coefficient for PR race is omitted because it is collinear after controlling for polling station-electoral race fixed effects.

Table G3: Effect of partisan poll worker composition on election results controlling for the proportion of voters at voting booths with top 100 most common surnames in the country

	Colorado vote share (1)	Liberal/GANAR vote share (2)	Other party vote share (3)	Total votes (turnout) (4)
Colorado trifecta	0.3274 (0.2030)	-0.5213 (0.1849)	0.1939 (0.1766)	-0.6837 (0.3301)
Colorado advantage + Other	0.4540 (0.0952)	-0.4698 (0.0789)	0.0158 (0.0822)	-0.0434 (0.1427)
Colorado advantage + Liberal	0.5180 (0.0498)	-0.3578 (0.0458)	-0.1602 (0.0438)	0.0271 (0.0791)
Liberal advantage + Colorado	-0.0439 (0.0527)	0.2323 (0.0507)	-0.1884 (0.0485)	0.0778 (0.0865)
Liberal advantage + Other	-0.1831 (0.1604)	-0.0078 (0.1406)	0.1909 (0.1445)	0.5996 (0.2509)
Liberal trifecta	-1.9207 (0.5893)	0.4524 (0.5994)	1.4683 (0.4667)	0.5306 (0.7339)
Other poll worker allocation	0.0430 (0.0644)	-0.1655 (0.0596)	0.1225 (0.0579)	-0.0287 (0.1047)
Mean of Dependent Variable	41.5808	34.4445	23.9746	122.5388
Polling station-elect. race FE	✓	✓	✓	✓
Colorado voter prop. control	✓	✓	✓	✓
Liberal voter prop. control	✓	✓	✓	✓
Other party voter prop. control	✓	✓	✓	✓
Top 100 surnames prop.	✓	✓	✓	✓
Observations	117,229	117,229	117,229	111,493
R^2	0.8438	0.8878	0.9311	0.6942

Note: A unit of observation is a voting booth-electoral race combination. Standard errors are robust. The Liberal party always ran separately or through the GANAR alliance (while mainly proposing its own candidates), hence we have an outcome variable for Liberal/GANAR vote shares. Vote shares are calculated by dividing by the number of valid votes (i.e., all votes excluding null and blank votes) and multiplying by 100. A trifecta indicates all three poll workers from the same party work at the voting booth. Advantage indicates that two out three poll workers from the same party work at the voting booth. Other refers to a poll worker that is not registered as supporting the Colorado or Liberal party. The excluded category corresponds to voting booths with “fair allocations” having one Colorado party poll worker, one Liberal poll worker, and one “other” poll worker. The “other poll worker allocation” refers to any remaining poll worker allocation after accounting for “fair allocations” and the other allocations referenced with the independent variables. “Top 100 surnames prop.” denotes the inclusion of separate indicator variables for the proportion of voters having one of the 100 most common surnames in the country at each booth, with one indicator variable per surname.

Table G4: Correlation between pre-registered partisan poll worker composition on election results

	Colorado vote share (1)	Liberal/GANAR vote share (2)	Other party vote share (3)
Prereg. Colorado trifecta	-2.0554 (0.8382)	1.1300 (0.8333)	0.9253 (1.0322)
Prereg. Colorado advantage + Other	3.8046 (0.8920)	-1.5612 (0.6749)	-2.2434 (0.7105)
Prereg. Colorado advantage + Liberal	0.1294 (0.0838)	-0.2136 (0.0767)	0.0842 (0.0697)
Prereg. Liberal advantage + Colorado	-0.2510 (0.0904)	0.2553 (0.0834)	-0.0043 (0.0747)
Prereg. Liberal advantage + Other	-3.6482 (1.5619)	-1.2102 (2.0915)	4.8584 (1.5218)
Prereg. other poll worker allocation	2.9388 (0.9025)	-0.4990 (0.6488)	-2.4397 (0.8845)
Mean of Dependent Variable	41.5808	34.4445	23.9746
Polling station-elect. race FE	✓	✓	✓
Colorado voter prop. control	✓	✓	✓
Liberal voter prop. control	✓	✓	✓
Other party voter prop. control	✓	✓	✓
Observations	117,229	117,229	117,229
R^2	0.8424	0.8868	0.9308

Note: A unit of observation is a voting booth-electoral race combination. Standard errors are robust. The Liberal party always ran separately or through the GANAR alliance (while mainly proposing its own candidates), hence we have an outcome variable for Liberal/GANAR vote shares. Vote shares are calculated by dividing by the number of valid votes (i.e., all votes excluding null and blank votes) and multiplying by 100. The regressors correspond to pre-registered partisan poll worker allocations. A trifecta indicates all three poll workers from the same party work at the voting booth. There is no pre-registered Liberal trifecta in our data so we do not include it as a regressor. Three pre-registered categories are nearly empty: of the 20,365 voting booths in the sample, only 5 have a pre-registered Colorado trifecta, 8 a Colorado advantage with an “other” poll worker, and 2 a Liberal advantage with an “other” poll worker, against 1,541 with a Colorado advantage with a Liberal poll worker, 1,099 with a Liberal advantage with a Colorado poll worker, 195 with another allocation, and 17,515 with a fair allocation. The coefficients on the three sparse categories are estimated from a handful of booths and should be read accordingly. Advantage indicates that two out three poll workers from the same party work at the voting booth. Other refers to a poll worker that is not registered as supporting the Colorado or Liberal party. The excluded category corresponds to voting booths with “fair allocations” having one Colorado party poll worker, one Liberal poll worker, and one “other” poll worker. The “other poll worker allocation” refers to any remaining poll worker allocation after accounting for “fair allocations” and the other allocations referenced with the independent variables.

Table G5: Effect of *de facto* partisan poll worker composition on election results, controlling for the pre-registered allocation

	Colorado vote share (1)	Liberal/GANAR vote share (2)	Other party vote share (3)
<i>De facto</i> Colorado trifecta	0.2987 (0.2034)	-0.5102 (0.1850)	0.2114 (0.1757)
<i>De facto</i> Colorado advantage + Other	0.4280 (0.0957)	-0.4919 (0.0794)	0.0640 (0.0826)
<i>De facto</i> Colorado advantage + Liberal	0.5359 (0.0522)	-0.3759 (0.0479)	-0.1600 (0.0460)
<i>De facto</i> Liberal advantage + Colorado	-0.0026 (0.0550)	0.1824 (0.0529)	-0.1798 (0.0510)
<i>De facto</i> Liberal advantage + Other	-0.1987 (0.1616)	-0.0059 (0.1407)	0.2046 (0.1443)
<i>De facto</i> Liberal trifecta	-1.9537 (0.5947)	0.4712 (0.5957)	1.4824 (0.4636)
<i>De facto</i> other poll worker allocation	0.0232 (0.0646)	-0.1583 (0.0597)	0.1351 (0.0578)
Mean of Dependent Variable	41.5808	34.4445	23.9746
Polling station-elect. race FE	✓	✓	✓
Colorado voter prop. control	✓	✓	✓
Liberal voter prop. control	✓	✓	✓
Other party voter prop. control	✓	✓	✓
Pre-registered allocation controls	✓	✓	✓
Observations	117,229	117,229	117,229
R^2	0.8426	0.8869	0.9308

Note: A unit of observation is a voting booth-electoral race combination. Standard errors are robust. The Liberal party always ran separately or through the GANAR alliance (while mainly proposing its own candidates), hence we have an outcome variable for Liberal/GANAR vote shares. Vote shares are calculated by dividing by the number of valid votes (i.e., all votes excluding null and blank votes) and multiplying by 100. The reported regressors are the election-day (*de facto*) partisan poll worker allocations. The specification also includes a full set of indicators for the pre-registered allocation of the voting booth (coefficients not reported), so the *de facto* coefficients compare booths with different election-day compositions within each pre-registered category, so election-day changes provide the variation while booths whose composition did not change serve as comparison observations. A trifecta indicates all three poll workers from the same party work at the voting booth. There is no pre-registered Liberal trifecta in our data. Advantage indicates that two out three poll workers from the same party work at the voting booth. Other refers to a poll worker that is not registered as supporting the Colorado or Liberal party. The excluded category corresponds to voting booths with “fair allocations” having one Colorado party poll worker, one Liberal poll worker, and one “other” poll worker. The “other poll worker allocation” refers to any remaining poll worker allocation after accounting for “fair allocations” and the other allocations referenced with the independent variables.

Table G6: Effect of *de facto* partisan poll worker composition on election results, booths with a fair pre-registered allocation

	Colorado vote share (1)	Liberal/GANAR vote share (2)	Other party vote share (3)
Colorado trifecta	0.0009 (0.2325)	-0.2764 (0.2085)	0.2756 (0.2083)
Colorado advantage + Other	0.3844 (0.1063)	-0.4276 (0.0867)	0.0433 (0.0928)
Colorado advantage + Liberal	0.5070 (0.0568)	-0.3632 (0.0522)	-0.1438 (0.0506)
Liberal advantage + Colorado	0.0020 (0.0592)	0.1849 (0.0578)	-0.1869 (0.0564)
Liberal advantage + Other	-0.3512 (0.1667)	0.1089 (0.1437)	0.2423 (0.1536)
Liberal trifecta	-1.3044 (0.7157)	1.2711 (0.7293)	0.0333 (0.4120)
Other poll worker allocation	-0.0068 (0.0675)	-0.1802 (0.0619)	0.1870 (0.0603)
Mean of Dependent Variable	41.0947	34.2837	24.6216
Polling station-elect. race FE	✓	✓	✓
Colorado voter prop. control	✓	✓	✓
Liberal voter prop. control	✓	✓	✓
Other party voter prop. control	✓	✓	✓
Observations	101,155	101,155	101,155
R^2	0.8432	0.8874	0.9315

Note: A unit of observation is a voting booth-electoral race combination. Standard errors are robust. The sample keeps the 17,515 booths (86.0 percent of those with all three ID numbers) whose pre-registered allocation was fair, so the *de facto* allocation indicators capture election-day replacements only. Vote shares are calculated by dividing by the number of valid votes (i.e., all votes excluding null and blank votes) and multiplying by 100. A trifecta indicates all three poll workers from the same party work at the voting booth. Advantage indicates that two out of three poll workers from the same party work at the voting booth. Other refers to a poll worker that is not registered as supporting the Colorado or Liberal party. The excluded category corresponds to voting booths with fair allocations having one Colorado poll worker, one Liberal poll worker, and one other poll worker. The sample is also restricted to voting booths for which we have the ID numbers of all three poll workers (which is around 96% of all voting booths).

Table G7: Effect of partisan poll worker composition on election results, standard errors clustered by voting booth

	Colorado vote share (1)	Liberal/GANAR vote share (2)	Other party vote share (3)
Colorado trifecta	0.2820 (0.3920)	-0.5121 (0.3356)	0.2301 (0.2802)
Colorado advantage + Other	0.4318 (0.2005)	-0.4969 (0.1527)	0.0651 (0.1393)
Colorado advantage + Liberal	0.5125 (0.1033)	-0.3800 (0.0898)	-0.1325 (0.0739)
Liberal advantage + Colorado	-0.0328 (0.1110)	0.2063 (0.0993)	-0.1735 (0.0828)
Liberal advantage + Other	-0.2126 (0.3317)	-0.0035 (0.2647)	0.2161 (0.2436)
Liberal trifecta	-1.9481 (1.2953)	0.4756 (1.2266)	1.4724 (0.8450)
Other poll worker allocation	0.0246 (0.1348)	-0.1593 (0.1150)	0.1347 (0.0984)
Mean of Dependent Variable	41.5808	34.4445	23.9746
Polling station-elect. race FE	✓	✓	✓
Colorado voter prop. control	✓	✓	✓
Liberal voter prop. control	✓	✓	✓
Other party voter prop. control	✓	✓	✓
Observations	117,229	117,229	117,229
R^2	0.8425	0.8869	0.9308

Note: A unit of observation is a voting booth-electoral race combination. Standard errors are clustered by voting booth (one cluster per booth), which allows arbitrary correlation across the six races counted at the same booth. Coefficients are those of Table 3. Vote shares are calculated by dividing by the number of valid votes (i.e., all votes excluding null and blank votes) and multiplying by 100. A trifecta indicates all three poll workers from the same party work at the voting booth. Advantage indicates that two out of three poll workers from the same party work at the voting booth. Other refers to a poll worker that is not registered as supporting the Colorado or Liberal party. The excluded category corresponds to voting booths with fair allocations having one Colorado poll worker, one Liberal poll worker, and one other poll worker. The sample is restricted to voting booths for which we have the ID numbers of all three poll workers (which is around 96% of all voting booths).

Table G8: Effect of partisan poll worker composition on election results by electoral system, standard errors clustered by voting booth

	Colorado vote share (1)	Liberal/GANAR vote share (2)	Other party vote share (3)
Colorado trifecta $\times$ PR race	-0.5625 (0.2767)	0.1364 (0.3123)	0.4261 (0.3555)
Colorado advantage + Other $\times$ PR race	0.0317 (0.1332)	0.0117 (0.1379)	-0.0434 (0.1688)
Colorado advantage + Liberal $\times$ PR race	0.1232 (0.0694)	0.1718 (0.0764)	-0.2949 (0.0870)
Liberal advantage + Colorado $\times$ PR race	0.0514 (0.0719)	0.0887 (0.0879)	-0.1401 (0.0992)
Liberal advantage + Other $\times$ PR race	0.1243 (0.2025)	-0.1741 (0.2477)	0.0498 (0.2718)
Liberal trifecta $\times$ PR race	0.0301 (0.5733)	-0.6823 (0.8510)	0.6522 (0.7512)
Other poll worker allocation $\times$ PR race	0.1203 (0.0889)	-0.1856 (0.1012)	0.0653 (0.1131)
Colorado trifecta	0.6614 (0.4206)	-0.6040 (0.4012)	-0.0574 (0.2389)
Colorado advantage + Other	0.4104 (0.2051)	-0.5047 (0.1894)	0.0943 (0.1024)
Colorado advantage + Liberal	0.4295 (0.1082)	-0.4957 (0.1049)	0.0663 (0.0523)
Liberal advantage + Colorado	-0.0673 (0.1161)	0.1469 (0.1134)	-0.0796 (0.0591)
Liberal advantage + Other	-0.2959 (0.3391)	0.1130 (0.3288)	0.1830 (0.1964)
Liberal trifecta	-1.9683 (1.1452)	0.9328 (1.1063)	1.0355 (0.5459)
Other poll worker allocation	-0.0564 (0.1424)	-0.0345 (0.1365)	0.0909 (0.0728)
Mean of Dependent Variable	41.5808	34.4445	23.9746
Polling station-elect. race FE	✓	✓	✓
Colorado voter prop. control	✓	✓	✓
Liberal voter prop. control	✓	✓	✓
Other party voter prop. control	✓	✓	✓
Observations	117,229	117,229	117,229
R^2	0.8425	0.8869	0.9308

Note: A unit of observation is a voting booth-electoral race combination. Standard errors are clustered by voting booth (one cluster per booth). Coefficients are those of Table 4. PR race indicates the Senate, Chamber of Deputies, departmental council, and MERCOSUR parliament races. Vote shares are calculated by dividing by the number of valid votes (i.e., all votes excluding null and blank votes) and multiplying by 100. A trifecta indicates all three poll workers from the same party work at the voting booth. Advantage indicates that two out of three poll workers from the same party work at the voting booth. Other refers to a poll worker that is not registered as supporting the Colorado or Liberal party. The excluded category corresponds to voting booths with fair allocations having one Colorado poll worker, one Liberal poll worker, and one other poll worker. The sample is restricted to voting booths for which we have the ID numbers of all three poll workers (which is around 96% of all voting booths).

Table G9: Effect of partisan poll worker composition on election results, fair allocations split by the third poll worker's party

	Colorado vote share (1)	Liberal/GANAR vote share (2)	Other party vote share (3)
Colorado trifecta	0.2536 (0.2045)	-0.5080 (0.1862)	0.2545 (0.1766)
Colorado advantage + Other	0.4056 (0.0986)	-0.4960 (0.0823)	0.0904 (0.0847)
Colorado advantage + Liberal	0.4885 (0.0550)	-0.3853 (0.0511)	-0.1032 (0.0481)
Liberal advantage + Colorado	-0.0602 (0.0571)	0.2110 (0.0551)	-0.1508 (0.0522)
Liberal advantage + Other	-0.2376 (0.1628)	-0.0030 (0.1421)	0.2406 (0.1455)
Liberal trifecta	-1.9745 (0.5963)	0.4759 (0.5956)	1.4986 (0.4653)
Other poll worker allocation	-0.0000 (0.0692)	-0.1635 (0.0639)	0.1635 (0.0614)
Fair allocation, third worker other party	0.0996 (0.0725)	-0.3520 (0.0668)	0.2524 (0.0633)
Fair allocation, third worker multi-party	-0.2589 (0.0761)	0.1927 (0.0719)	0.0662 (0.0702)
Fair allocation, third worker unaffiliated	-0.0275 (0.0719)	0.1502 (0.0675)	-0.1227 (0.0633)
Mean of Dependent Variable	41.5808	34.4445	23.9746
Polling station-elect. race FE	✓	✓	✓
Colorado voter prop. control	✓	✓	✓
Liberal voter prop. control	✓	✓	✓
Other party voter prop. control	✓	✓	✓
Observations	117,229	117,229	117,229
R ²	0.8425	0.8869	0.9308

Note: A unit of observation is a voting booth-electoral race combination. Standard errors are robust. The excluded category is a fair allocation whose third poll worker is registered with FG, the party to which the Electoral Code assigns the third seat in 2018 (31.3 percent of booths). The three additional indicators mark fair allocations whose third worker is registered with another party (9.3 percent), with more than one party (5.9 percent), or with no party (7.5 percent). Vote shares are calculated by dividing by the number of valid votes (i.e., all votes excluding null and blank votes) and multiplying by 100. A trifecta indicates all three poll workers from the same party work at the voting booth. Advantage indicates that two out of three poll workers from the same party work at the voting booth. Other refers to a poll worker that is not registered as supporting the Colorado or Liberal party. The sample is restricted to voting booths for which we have the ID numbers of all three poll workers (which is around 96% of all voting booths).

Table G10: Effect of partisan poll worker composition on election results by electoral system, voter-affiliation controls interacted with race

	Colorado vote share (1)	Liberal/GANAR vote share (2)	Other party vote share (3)
Colorado trifecta $\times$ PR race	-0.5694 (0.4158)	0.1189 (0.3948)	0.4505 (0.3100)
Colorado advantage + Other $\times$ PR race	0.0368 (0.1973)	0.0266 (0.1749)	-0.0634 (0.1435)
Colorado advantage + Liberal $\times$ PR race	0.1193 (0.1031)	0.1778 (0.0979)	-0.2971 (0.0751)
Liberal advantage + Colorado $\times$ PR race	0.0262 (0.1096)	0.1112 (0.1068)	-0.1374 (0.0844)
Liberal advantage + Other $\times$ PR race	0.0422 (0.3319)	-0.1306 (0.3117)	0.0884 (0.2598)
Liberal trifecta $\times$ PR race	0.0070 (1.1921)	-0.9160 (1.2175)	0.9089 (0.8054)
Other poll worker allocation $\times$ PR race	0.1038 (0.1355)	-0.1675 (0.1292)	0.0637 (0.1021)
Colorado trifecta	0.6665 (0.3288)	-0.5947 (0.3260)	-0.0718 (0.1953)
Colorado advantage + Other	0.4071 (0.1572)	-0.5153 (0.1486)	0.1081 (0.0868)
Colorado advantage + Liberal	0.4323 (0.0830)	-0.5005 (0.0813)	0.0682 (0.0446)
Liberal advantage + Colorado	-0.0504 (0.0882)	0.1315 (0.0871)	-0.0812 (0.0505)
Liberal advantage + Other	-0.2410 (0.2630)	0.0845 (0.2654)	0.1565 (0.1674)
Liberal trifecta	-1.9513 (0.9118)	1.0844 (0.9529)	0.8669 (0.4690)
Other poll worker allocation	-0.0450 (0.1096)	-0.0468 (0.1077)	0.0919 (0.0632)
Mean of Dependent Variable	41.5808	34.4445	23.9746
Polling station-elect. race FE	✓	✓	✓
Colorado voter prop. $\times$ race	✓	✓	✓
Liberal voter prop. $\times$ race	✓	✓	✓
Other party voter prop. $\times$ race	✓	✓	✓
Observations	117,229	117,229	117,229
R^2	0.8429	0.8876	0.9310

Note: A unit of observation is a voting booth-electoral race combination. Standard errors are robust. The three voter-affiliation controls are interacted with indicators for each of the six electoral races, so their slopes vary by race. PR race indicates the Senate, Chamber of Deputies, departmental council, and MERCOSUR parliament races. Vote shares are calculated by dividing by the number of valid votes (i.e., all votes excluding null and blank votes) and multiplying by 100. A trifecta indicates all three poll workers from the same party work at the voting booth. Advantage indicates that two out of three poll workers from the same party work at the voting booth. Other refers to a poll worker that is not registered as supporting the Colorado or Liberal party. The excluded category corresponds to voting booths with fair allocations having one Colorado poll worker, one Liberal poll worker, and one other poll worker. The sample is restricted to voting booths for which we have the ID numbers of all three poll workers (which is around 96% of all voting booths).

Table G11: Effect of partisan poll worker composition on presidential election results, controlling for the proportion of voters sharing surnames with the candidates

	Colorado vote share (1)	Liberal/GANAR vote share (2)	Other vote share (3)
Colorado trifecta	0.6564 (0.4602)	-0.7651 (0.4660)	0.1087 (0.2548)
Colorado advantage + Other	0.4180 (0.2689)	-0.4828 (0.2609)	0.0648 (0.1373)
Colorado advantage + GANAR	0.4470 (0.1237)	-0.4922 (0.1224)	0.0452 (0.0607)
GANAR advantage + Colorado	-0.0836 (0.1055)	0.1425 (0.1053)	-0.0589 (0.0512)
GANAR advantage + Other	-0.3930 (0.2909)	0.4713 (0.2770)	-0.0783 (0.1439)
GANAR trifecta	0.3212 (0.5962)	-0.6507 (0.5784)	0.3295 (0.3169)
Other poll worker allocation	0.3188 (0.2256)	-0.1557 (0.2264)	-0.1632 (0.1098)
Mean of Dependent Variable	48.6944	45.3871	5.9185
Polling station FE	✓	✓	✓
Colorado voter prop. control	✓	✓	✓
GANAR voter prop. control	✓	✓	✓
Other party voter prop. control	✓	✓	✓
Prop. voters coincide w/ Colorado candidate surnames	✓	✓	✓
Prop. voters coincide w/ GANAR candidate surnames	✓	✓	✓
Prop. voters coincide w/ Other party candidate surnames	✓	✓	✓
Observations	20,287	20,287	20,287
R ²	0.8116	0.8205	0.7958

Note: A unit of observation is a voting booth. Standard errors are robust. Vote shares on Columns (1) to (3) are calculated by dividing by the number of valid votes (i.e., all votes excluding null and blank votes) and multiplying by 100. A trifecta indicates all three poll workers from the same party or alliance (in the case of GANAR) work at the voting booth. Advantage indicates that two out three poll workers from the same party work at the voting booth. Other refers to a poll worker that is not registered as supporting the Colorado or Liberal party. The excluded category corresponds to voting booths with “fair allocations” having one Colorado poll worker, one poll worker registered with a GANAR member party (the alliance of the Liberal party, FG, and other smaller left-wing parties), and one other poll worker. Because FG belongs to GANAR in the presidential race, the pre-registered team that the Electoral Code intends (one Colorado, one Liberal, and one FG poll worker) falls in the “GANAR advantage + Colorado” category in this table. The “other poll worker allocation” refers to any remaining poll worker allocation after accounting for “fair allocations” and the other allocations referenced with the independent variables. The regressions also control for the proportion of voters at each voting booth sharing surnames with the candidates for the Colorado party, the GANAR alliance, and the other party candidates. For example, to identify the proportion for the Colorado party, we calculated the proportion of voters at each voting booth that had either of the surnames of the candidates for the presidency and vicepresidency for that ticket: Abdo, Benitez, Velazquez, and Moreno. The sample is restricted to voting booths for which we have the ID numbers of all three poll workers (which is around 96% of all voting booths).

Table G12: Effect of partisan poll worker composition on governor election results, controlling for the proportion of voters sharing surnames with the candidates

	Colorado vote share (1)	Liberal/GANAR vote share (2)	Other vote share (3)
Colorado trifecta	0.7480 (0.4738)	-0.4299 (0.4603)	-0.3181 (0.3029)
Colorado advantage + Other	0.4169 (0.2281)	-0.4799 (0.2110)	0.0630 (0.1388)
Colorado advantage + Liberal	0.3907 (0.1220)	-0.4686 (0.1178)	0.0779 (0.0726)
Liberal advantage + Colorado	-0.0898 (0.1258)	0.1882 (0.1222)	-0.0984 (0.0803)
Liberal advantage + Other	-0.3737 (0.3618)	0.0437 (0.3957)	0.3300 (0.2688)
Liberal trifecta	-1.6285 (1.1044)	0.6524 (1.1095)	0.9761 (0.6793)
Other poll worker allocation	-0.1379 (0.1580)	-0.0954 (0.1533)	0.2333 (0.1052)
Mean of Dependent Variable	46.9013	42.3609	10.7379
Polling station FE	✓	✓	✓
Colorado voter prop. control	✓	✓	✓
Liberal voter prop. control	✓	✓	✓
Other party voter prop. control	✓	✓	✓
Prop. voters coincide w/ Colorado candidate surnames	✓	✓	✓
Prop. voters coincide w/ Liberal/GANAR candidate surnames	✓	✓	✓
Prop. voters coincide w/ Other party candidate surnames	✓	✓	✓
Observations	18,144	18,144	18,144
R^2	0.8142	0.8563	0.9084

Note: A unit of observation is a voting booth. The parties belonging to the GANAR alliance vary across Paraguayan departments for the governor elections, with FG running independently in some departments. However, the Liberal party always runs separately or through GANAR (and mainly proposing its own candidates), hence we have an outcome variable for Liberal/GANAR vote shares. Standard errors are robust. Vote shares on Columns (1) to (3) are calculated by dividing by the number of valid votes (i.e., all votes excluding null and blank votes) and multiplying by 100. A trifecta indicates all three poll workers from the same party work at the voting booth. Advantage indicates that two out of three poll workers from the same party work at the voting booth. Other refers to a poll worker that is not registered as supporting the Colorado or Liberal party. The excluded category corresponds to voting booths with “fair allocations” having one Colorado poll worker and one Liberal poll worker. The “other poll worker allocation” refers to any remaining poll worker allocation after accounting for “fair allocations” and the other allocations referenced with the independent variables. The regressions also control for the proportion of voters at each voting booth sharing surnames with the candidates for the Colorado party, the Liberal party or GANAR alliance, and the other party candidates. For example, to identify the proportion for the Colorado party we calculated the proportion of voters at each voting booth that had either of the surnames of the candidate for the governor in the voters’ relevant department (e.g., either Gonzalez or Vaesken for the Alto Parana department). The number of observations is lower for this electoral race as there is no governor election for Paraguayan voters abroad and those in the capital district. The sample is restricted to voting booths for which we have the ID numbers of all three poll workers (which is around 96% of all voting booths).

Table G13 provides summary statistics for all the variables used in the main tables of the paper, except the voting outcome variables, which are on Table G14. These variables are defined in detail on Section III. In contrast to the vote share variables, turnout is measured as the number of votes. Table G15 shows summary statistics for all the variables used in the SA tables and analyses.

Table G13: Main summary statistics

	Observations	Mean	Standard Deviation
Total pre-registered poll workers replaced at voting booth	21,211	1.1371	0.9316
Percent of Colorado voters at voting booth	21,211	34.2220	10.4397
Percent of Liberal voters at voting booth	21,211	17.3346	9.8823
Percent of other party voters (excluding the Colorado and Liberal parties) at voting booth	21,211	4.0607	2.4103
Number of registered voters at voting booth	21,211	199.9673	14.4187
Equation (1) indicator variables:			
Colorado trifecta	21,211	0.0074	0.0857
Colorado advantage + Other	21,211	0.0415	0.1995
Colorado advantage + Liberal	21,211	0.1693	0.3751
Liberal advantage + Colorado	21,211	0.1259	0.3318
Liberal advantage + Other	21,211	0.0130	0.1133
Liberal trifecta	21,211	0.0008	0.0283
Other poll worker allocation	21,211	0.1208	0.3259
Fair poll worker allocation	21,211	0.5211	0.4996

Note: All the means and standard deviations are rounded up to the fourth digit after the decimal.

To examine results for the presidential race, we use similar indicator variables to equation (1) except we consider GANAR alliance poll workers instead of just Liberal poll workers. GANAR alliance poll workers are those supporting either the Liberal party, the FG, or other smaller left-wing parties that form part of the alliance.⁷ This change also affects who we consider as “other” in the “GANAR advantage + Other” and “Colorado advantage + Other” variables, which now consists of those poll workers not registered as supporting either the Colorado party or the GANAR alliance. Our excluded category then corresponds to voting booths with fair allocations having one Colorado poll worker and one GANAR poll worker. Lastly, the indicator variable for “other poll worker allocations” here refers to any remaining poll worker allocation after accounting for fair allocations and the other allocations referenced above. These indicator variables are all mutually exclusive and together cover all voting booths. Our regression specification for presidential race

⁷As mentioned previously, the full list of parties that were part of GANAR are: Liberal party, PRF, PDP, PEN, Avanza País, PMAS, and the FG member parties. Virtually all GANAR poll workers support the Liberal party and FG (28,978 of the 29,294 GANAR poll workers). Only 316 poll workers support one of the other small left-wing parties that were part of GANAR.

Table G14: Vote share summary statistics

	Observations	Mean	Standard Deviation
Presidential election vote shares:			
Colorado vote share	21,139	48.625	12.311
GANAR vote share	21,139	45.439	12.564
Other party vote share	21,139	5.936	5.836
Blank vote share	21,139	2.435	2.333
Null vote share	21,139	2.751	1.984
Turnout	21,139	122.900	18.592
Senatorial election vote shares:			
Colorado vote share	21,107	32.765	12.558
Liberal vote share	21,107	24.484	12.643
Other party vote share	21,107	42.752	15.775
Blank vote share	21,107	5.366	4.069
Null vote share	21,107	3.621	2.660
Turnout	21,107	122.627	18.876
MERCOSUR parliament election vote shares:			
Colorado vote share	21,122	42.199	13.042
Liberal vote share	21,122	30.512	13.659
Other party vote share	21,122	27.289	14.068
Blank vote share	21,122	8.562	5.190
Null vote share	21,122	3.904	2.726
Turnout	21,122	122.630	18.744
Deputies election vote shares:			
Colorado vote share	20,932	39.295	14.101
Liberal/GANAR vote share	20,932	30.441	14.704
Other party vote share	20,932	30.264	19.012
Blank vote share	20,932	5.220	3.432
Null vote share	20,932	3.055	2.236
Turnout	20,932	123.369	17.111
Governor election vote shares:			
Colorado vote share	18,910	46.846	12.432
Liberal/GANAR vote share	18,910	42.390	13.713
Other party vote share	18,910	10.764	11.016
Blank vote share	18,910	3.727	2.641
Null vote share	18,910	2.585	1.895
Turnout	18,910	122.279	16.615
Departmental council election vote shares:			
Colorado vote share	18,893	40.423	12.872
Liberal/GANAR vote share	18,893	34.377	13.223
Other party vote share	18,893	25.200	13.246
Blank vote share	18,893	6.735	3.603
Null vote share	18,893	3.465	2.526
Turnout	18,893	122.079	16.661

Note: All means and standard deviations are rounded up to the fourth digit. Vote shares for parties are calculated by dividing by the number of valid votes (i.e., all votes excluding null and blank votes) and multiplying by 100. Vote shares for blank and null votes are calculated by dividing by turnout (total votes including null and blank votes) and multiplying by 100. The number of observations for the deputies' vote shares is lower as there is no deputies' election for Paraguayan voters abroad. The number of observations is lower for the governor and departmental council vote shares as there is no governor or departmental council election for Paraguayan voters abroad and those in the capital district. There are still slight differences in observations across races due to electoral contestations annulling the vote counts for certain races at some voting booths and missing vote tallies that were not received by the election authority for certain electoral races.

Table G15: SA analysis summary statistics

	Observations	Mean	Standard Deviation
Equation (3) indicator variables:			
Traditional party trifecta	21,211	0.3035	0.4598
Traditional party advantage + Other party	21,211	0.4130	0.4924
Other party advantage + Traditional party	21,211	0.0136	0.1157
Other party trifecta	21,211	0.0002	0.0154
Other poll worker allocation	21,211	0.2227	0.4160
Fair allocation (traditional vs. other coding)	21,211	0.0470	0.2117
Electoral contestation variables:			
Contested electoral result	122,595	0.0262	0.1598
Contested electoral result by a party that is not the Colorado party	122,595	0.0228	0.1493
Contested electoral result by a party that is not the Liberal party	122,595	0.0189	0.1360
Contestation of electoral result accepted	122,595	0.0060	0.0774
Electoral result annulled	122,595	0.0038	0.0615

Note: All the means and standard deviations are rounded up to the fourth digit after the decimal.

analyses is then as follows:

$$\begin{aligned}
 Y_{ab} = & \beta_0 + \beta_1 GANAR_Trif_{ab} + \beta_2 GANAR_Adv_Oth_{ab} + \beta_3 GANAR_Adv_Colorado_{ab} \\
 & + \beta_4 Colorado_Adv_GANAR_{ab} + \beta_5 Colorado_Adv_Oth_{ab} + \beta_6 Colorado_Trif_{ab} + \beta_7 Other_pw_alloc_{ab} \\
 & + \delta_1 voter_prop_Colorado_{ab} + \delta_2 voter_prop_GANAR_{ab} + \delta_3 voter_prop_Other_{ab} + \theta_a + \varepsilon_{ab}.
 \end{aligned}
 \tag{4}$$

Table G16 shows the effect of partisan poll workers on the presidential election by running equation (4). In Column (1) we can observe that the Colorado party vote share increases significantly by around 0.45 percentage points (pp) when a booth has two Colorado poll workers and one GANAR poll worker counting votes. This result is robust to controlling for the proportion of Colorado, GANAR, and other party voters at each voting booth. Similarly, in Column (2) we can notice that the GANAR vote share increases significantly at the 10 percent level by approximately 0.47 pp at booths with two GANAR poll workers and a poll worker not registered to the Colorado party or the GANAR parties. We can also notice how the GANAR vote share decreases by around 0.49 pp whenever the Colorado party has two poll workers counting votes. This result is robust to controlling for the proportion of partisan voters at each voting booth, and the direction varies symmetrically depending on which party/alliance has a poll worker majority at the electoral table. We do not observe any significant effects of these partisan poll worker allocations on the vote share of other parties. This might reflect that in SMDP races such as the presidency, biased poll

workers may divert votes from their main competition, while potentially ignoring other smaller parties. Next, in Column (4) the blank vote share decreases significantly at the 10 percent level when the allocation is that of two GANAR poll workers and a poll worker not registered to the Colorado party or the GANAR parties. Finally, we find no significant effects on the null vote share or turnout. We also do not find any significant coefficients for the “trifecta” dummies, which may be expected as we do not have many trifecta poll worker allocations for either party (just over 260 booths), leading to low statistical power.

Table G17 examines the effect of partisan poll workers on the senatorial election by running equation (1) (except we use polling station fixed effects instead of polling station-electoral race fixed effects), because for this electoral race the GANAR parties run under separate party lists. We observe a similar pattern for the Colorado and Liberal party vote shares as in Table G16. Whenever the Colorado party has two poll workers counting votes, its vote share increases significantly by around 0.52/0.53 pp. On the other hand, the Liberal party vote share increases significantly at the 10 percent level by around 0.21 pp at booths with two Liberal poll workers and one Colorado poll worker counting votes. In addition, when the Colorado party has two poll workers and there is one Liberal poll worker or one poll worker not registered to the Colorado or Liberal party, the Liberal party vote share decreases significantly by 0.25 pp and 0.34 pp respectively. Next, we observe that if the other parties have no poll worker present (particularly when a voting booth has 2 Colorado and 1 Liberal poll worker) the vote share for other parties goes down significantly by 0.27 pp. A Colorado party trifecta reduces the null vote share significantly (at the 10 percent level) by 0.31 pp. If the Liberal party has two poll workers and there is one Colorado poll worker or one poll worker not registered to the Colorado or Liberal party, the null vote share decreases by 0.10 pp and 0.42 pp respectively. Finally, we find no significant effects on the blank vote share or turnout.

Table G18 explores the effect of partisan poll workers on the MERCOSUR parliament election by running equation (1), because for this electoral race the GANAR parties also run under separate party lists. We observe the same pattern for the Colorado and Liberal party vote shares as in Table G17. If the Colorado party has two poll workers and the Liberal party has one counting votes, the Colorado party’s vote share increases significantly by 0.59 pp. If instead the Liberal party has two poll workers and the Colorado party has one counting votes, the Liberal party’s vote share rises significantly at the 10 percent level by 0.21 pp. Once again, when the Colorado party has two poll workers and there is one Liberal poll worker or one poll worker not registered to the Colorado or Liberal party, the Liberal party vote share decreases significantly by 0.3 pp and 0.52 pp respectively. Next, we continue finding that if the other parties have no poll worker representation (particularly when a voting booth has 2 Colorado and 1 Liberal poll worker) the vote share for other parties go down significantly by 0.29 pp. If there is a Liberal trifecta the blank vote share increases significantly by 1.23 pp. On the other hand, if there is Colorado trifecta the null vote

Table G16: Effect of partisan poll worker composition on presidential election results

	Colorado vote share (1)	Liberal/GANAR vote share (2)	Other vote share (3)	Blank vote share (4)	Null vote share (5)	Turnout (6)
Colorado trifecta	0.6563 (0.4599)	-0.7630 (0.4658)	0.1068 (0.2545)	-0.0080 (0.1301)	0.0560 (0.1519)	-0.8245 (0.7845)
Colorado advantage + Other	0.4175 (0.2689)	-0.4820 (0.2609)	0.0644 (0.1373)	-0.0679 (0.0708)	-0.0228 (0.0877)	0.1396 (0.4190)
Colorado advantage + GANAR	0.4488 (0.1236)	-0.4946 (0.1224)	0.0458 (0.0607)	0.0260 (0.0389)	-0.0571 (0.0408)	0.0933 (0.2020)
GANAR advantage + Colorado	-0.0829 (0.1055)	0.1409 (0.1052)	-0.0580 (0.0512)	0.0204 (0.0317)	0.0319 (0.0333)	0.0115 (0.1694)
GANAR advantage + Other	-0.3968 (0.2909)	0.4742 (0.2771)	-0.0774 (0.1438)	-0.1392 (0.0827)	-0.0379 (0.0932)	0.0740 (0.4439)
GANAR trifecta	0.3197 (0.5953)	-0.6510 (0.5773)	0.3312 (0.3167)	-0.2267 (0.1756)	0.0274 (0.2457)	0.4049 (0.8039)
Other poll worker allocation	0.3176 (0.2255)	-0.1554 (0.2264)	-0.1622 (0.1098)	-0.0507 (0.0619)	-0.0286 (0.0703)	0.2621 (0.3503)
Mean of Dependent Variable	48.6944	45.3871	5.9185	2.3829	2.7564	122.8357
Polling station FE	✓	✓	✓	✓	✓	✓
Colorado voter prop. control	✓	✓	✓	✓	✓	✓
GANAR voter prop. control	✓	✓	✓	✓	✓	✓
Other party voter prop. control	✓	✓	✓	✓	✓	✓
Observations	20,287	20,287	20,287	20,287	20,287	19,281
R^2	0.8116	0.8205	0.7958	0.4630	0.2462	0.7312

Note: A unit of observation is a voting booth. Standard errors are robust. Vote shares on Columns (1) to (3) are calculated by dividing by the number of valid votes (i.e., all votes excluding null and blank votes) and multiplying by 100. Vote shares on Columns (4) to (5) are calculated by dividing by turnout (i.e., all votes including null and blank votes) and multiplying by 100. The number of observations on Column (6) is smaller because we restrict our sample to booths with exactly 200 registered voters. A trifecta indicates all three poll workers from the same party or alliance (in the case of GANAR) work at the voting booth. Advantage indicates that two out three poll workers from the same party work at the voting booth. Other refers to a poll worker that is not registered as supporting the Colorado or Liberal party. The excluded category corresponds to voting booths with “fair allocations” having one Colorado poll worker, one poll worker registered with a GANAR member party (the alliance of the Liberal party, FG, and other smaller left-wing parties), and one other poll worker. Because FG belongs to GANAR in the presidential race, the pre-registered team that the Electoral Code intends (one Colorado, one Liberal, and one FG poll worker) falls in the “GANAR advantage + Colorado” category in this table. The “other poll worker allocation” refers to any remaining poll worker allocation after accounting for “fair allocations” and the other allocations referenced with the independent variables. The sample is restricted to voting booths for which we have the ID numbers of all three poll workers (which is around 96% of all voting booths).

Table G17: Effect of partisan poll worker composition on senatorial election results

	Colorado vote share (1)	Liberal vote share (2)	Other vote share (3)	Blank vote share (4)	Null vote share (5)	Turnout (6)
Colorado trifecta	-0.4535 (0.5754)	-0.2174 (0.4556)	0.6709 (0.5015)	-0.1953 (0.2276)	-0.3109 (0.1818)	-0.6024 (0.7970)
Colorado advantage + Other	0.5302 (0.2357)	-0.3406 (0.1720)	-0.1896 (0.2543)	0.1645 (0.1079)	-0.0565 (0.0891)	-0.0077 (0.3507)
Colorado advantage + Liberal	0.5185 (0.1194)	-0.2472 (0.1018)	-0.2713 (0.1326)	0.0099 (0.0545)	-0.0469 (0.0536)	-0.0449 (0.1940)
Liberal advantage + Colorado	-0.0618 (0.1272)	0.2154 (0.1167)	-0.1535 (0.1469)	0.0372 (0.0613)	-0.1044 (0.0564)	0.0854 (0.2146)
Liberal advantage + Other	-0.2183 (0.3735)	0.0088 (0.3259)	0.2095 (0.4385)	-0.0248 (0.1605)	-0.4170 (0.1552)	0.6086 (0.6438)
Liberal trifecta	-1.7440 (1.2990)	0.4261 (1.5652)	1.3179 (1.3759)	0.0910 (0.6409)	-0.6166 (0.5995)	-0.2237 (1.8600)
Other poll worker allocation	0.1012 (0.1507)	-0.1956 (0.1312)	0.0944 (0.1714)	0.0196 (0.0743)	0.0563 (0.0665)	-0.0627 (0.2653)
Mean of Dependent Variable	32.6369	24.4988	42.8642	5.2841	3.6022	122.5547
Polling station FE	✓	✓	✓	✓	✓	✓
Colorado voter prop. control	✓	✓	✓	✓	✓	✓
Liberal voter prop. control	✓	✓	✓	✓	✓	✓
Other party voter prop. control	✓	✓	✓	✓	✓	✓
Observations	20,257	20,257	20,257	20,257	20,257	19,249
R^2	0.8030	0.8597	0.8476	0.5612	0.1456	0.7075

Note: A unit of observation is a voting booth. Standard errors are robust. Vote shares on Columns (1) to (3) are calculated by dividing by the number of valid votes (i.e., all votes excluding null and blank votes) and multiplying by 100. Vote shares on Columns (4) to (5) are calculated by dividing by turnout (i.e., all votes including null and blank votes) and multiplying by 100. The number of observations on Column (6) is smaller because we restrict our sample to booths with exactly 200 registered voters. A trifecta indicates all three poll workers from the same party work at the voting booth. Advantage indicates that two out three poll workers from the same party work at the voting booth. Other refers to a poll worker that is not registered as supporting the Colorado or Liberal party. The excluded category corresponds to voting booths with “fair allocations” having one Colorado poll worker and one Liberal poll worker. The “other poll worker allocation” refers to any remaining poll worker allocation after accounting for “fair allocations” and the other allocations referenced with the independent variables. The sample is restricted to voting booths for which we have the ID numbers of all three poll workers (which is around 96% of all voting booths).

share declines significantly by 0.4 pp. When the Liberal party has two poll workers and there is one Colorado poll worker, the null vote share decreases significantly by 0.15 pp. In contrast, with other poll worker allocations the null vote share rises significantly by 0.15 pp. Finally, we find no significant effects on turnout.

With Table G19 we analyze the effect of partisan poll workers on the deputies election by running equation (1). For this electoral race, there are some departments where the GANAR alliance runs together and somewhere it does not. Also, the parties belonging to GANAR vary across departments for this race, with FG running independently in some departments. However, the Liberal party always runs separately or through GANAR (and mainly proposes its own candidates if it runs under GANAR), so we generate an outcome variable for Liberal/GANAR vote shares. We once again observe a similar pattern for the Colorado and Liberal/GANAR vote shares as in Tables G17 and G18. Whenever the Colorado party has two poll workers counting votes, its vote share grows significantly by 0.46 pp and 0.49 pp. If instead the Liberal party has two poll workers and the Colorado party has one counting votes, the Liberal/GANAR vote share increases significantly by 0.4 pp. Also, when the Colorado party has two poll workers and there is one Liberal poll worker or one poll worker not registered to the Colorado or Liberal party, the Liberal/GANAR vote share decreases significantly by 0.25 pp and 0.5 pp respectively. We again find that if the other parties have no poll workers counting votes the vote share for other parties decreases by 0.28 pp and 0.2 pp. If the Colorado party has two poll workers and there is one Liberal poll worker, the null vote share declines significantly (at the 10 percent level) by 0.08 pp. Finally, we find no significant effects on the blank vote share or turnout.

In Table G20 we analyze the effect of partisan poll workers on the departmental governor election by running equation (1). Similarly to Table G19, we use an outcome variable for Liberal/GANAR vote shares because the Liberal party either runs separately or through the GANAR alliance throughout the whole country for this electoral race. The results from this table are more similar to those of Table G16, as both explore the effects of partisan poll workers on elections defined by SMDP plurality voting. We find that whenever the Colorado party has two poll workers counting votes, its vote share grows by 0.39 pp and 0.42 pp. In addition, whenever the Colorado party has two poll workers the Liberal/GANAR vote share decreases significantly by 0.47 pp and 0.48 pp respectively. We also find that “other” poll worker allocations are associated with a significant increase in the vote share for other parties by 0.23 pp. This is plausible because these “other” poll worker allocations consist of any poll worker allocation excluding those that are denoted with the other independent variables and the fair allocations having one Colorado and one Liberal poll worker. Hence, these allocations include, for example, those electoral tables with 2 FG poll workers and those that have multiple unaffiliated poll workers who might be independent and more likely to support the non-traditional parties. If the Liberal party has two poll workers and

Table G18: Effect of partisan poll worker composition on MERCOSUR parliament election results

	Colorado vote share (1)	Liberal vote share (2)	Other vote share (3)	Blank vote share (4)	Null vote share (5)	Turnout (6)
Colorado trifecta	0.1034 (0.4875)	-0.7374 (0.4354)	0.6340 (0.4475)	-0.1638 (0.2928)	-0.3999 (0.1950)	-0.8075 (0.7886)
Colorado advantage + Other	0.3285 (0.2348)	-0.5169 (0.1849)	0.1885 (0.2172)	0.0882 (0.1306)	-0.0844 (0.0910)	-0.2450 (0.3533)
Colorado advantage + Liberal	0.5868 (0.1231)	-0.3011 (0.1078)	-0.2857 (0.1144)	0.0808 (0.0731)	-0.0700 (0.0517)	0.0551 (0.1898)
Liberal advantage + Colorado	-0.0009 (0.1334)	0.2127 (0.1222)	-0.2118 (0.1306)	0.0244 (0.0825)	-0.1448 (0.0561)	0.1304 (0.2181)
Liberal advantage + Other	-0.1213 (0.4570)	-0.0727 (0.3195)	0.1940 (0.4245)	0.0041 (0.2102)	0.0548 (0.1903)	0.4678 (0.7618)
Liberal trifecta	-2.4478 (1.7716)	0.6209 (1.5544)	1.8269 (1.3718)	1.2319 (0.6263)	-0.0940 (0.5312)	0.0525 (1.9021)
Other poll worker allocation	0.0748 (0.1665)	-0.1439 (0.1430)	0.0691 (0.1583)	0.0478 (0.0927)	0.1466 (0.0707)	0.0812 (0.2511)
Mean of Dependent Variable	42.1341	30.5360	27.3299	8.4776	3.9100	122.5657
Polling station FE	✓	✓	✓	✓	✓	✓
Colorado voter prop. control	✓	✓	✓	✓	✓	✓
Liberal voter prop. control	✓	✓	✓	✓	✓	✓
Other party voter prop. control	✓	✓	✓	✓	✓	✓
Observations	20,273	20,273	20,273	20,273	20,273	19,265
R^2	0.8022	0.8617	0.8485	0.5476	0.1869	0.7168

Note: A unit of observation is a voting booth. Standard errors are robust. Vote shares on Columns (1) to (3) are calculated by dividing by the number of valid votes (i.e., all votes excluding null and blank votes) and multiplying by 100. Vote shares on Columns (4) to (5) are calculated by dividing by turnout (i.e., all votes including null and blank votes) and multiplying by 100. The number of observations on Column (6) is smaller because we restrict our sample to booths with exactly 200 registered voters. A trifecta indicates all three poll workers from the same party work at the voting booth. Advantage indicates that two out three poll workers from the same party work at the voting booth. Other refers to a poll worker that is not registered as supporting the Colorado or Liberal party. The excluded category corresponds to voting booths with “fair allocations” having one Colorado poll worker and one Liberal poll worker. The “other poll worker allocation” refers to any remaining poll worker allocation after accounting for “fair allocations” and the other allocations referenced with the independent variables. The sample is restricted to voting booths for which we have the ID numbers of all three poll workers (which is around 96% of all voting booths).

Table G19: Effect of partisan poll worker composition on deputies' election results

	Colorado vote share (1)	Liberal vote share (2)	Other vote share (3)	Blank vote share (4)	Null vote share (5)	Turnout (6)
Colorado trifecta	0.2758 (0.4673)	0.0640 (0.3994)	-0.3398 (0.4646)	-0.1023 (0.1999)	-0.1779 (0.1607)	-0.4763 (0.7914)
Colorado advantage + Other	0.4876 (0.2322)	-0.4951 (0.1825)	0.0076 (0.2134)	-0.0292 (0.0972)	0.0553 (0.0870)	-0.0316 (0.3490)
Colorado advantage + Liberal	0.4551 (0.1159)	-0.2518 (0.1068)	-0.2033 (0.1126)	-0.0402 (0.0513)	-0.0782 (0.0436)	0.0085 (0.1860)
Liberal advantage + Colorado	-0.1125 (0.1274)	0.3970 (0.1266)	-0.2845 (0.1340)	-0.0377 (0.0562)	-0.0503 (0.0492)	0.0872 (0.2090)
Liberal advantage + Other	-0.3307 (0.3760)	0.0677 (0.3130)	0.2630 (0.3507)	-0.1604 (0.1585)	-0.1033 (0.1374)	0.6627 (0.5761)
Liberal trifecta	-2.2114 (1.5302)	0.7208 (1.3127)	1.4906 (1.0552)	-0.4972 (0.4973)	0.2142 (0.4985)	-0.1145 (2.0371)
Other poll worker allocation	0.0239 (0.1527)	-0.2210 (0.1398)	0.1970 (0.1478)	-0.0455 (0.0698)	-0.0351 (0.0565)	0.0058 (0.2508)
Mean of Dependent Variable	39.1551	30.2855	30.5593	5.1788	3.0362	123.1645
Polling station FE	✓	✓	✓	✓	✓	✓
Colorado voter prop. control	✓	✓	✓	✓	✓	✓
Liberal voter prop. control	✓	✓	✓	✓	✓	✓
Other party voter prop. control	✓	✓	✓	✓	✓	✓
Observations	20,138	20,138	20,138	20,138	20,138	19,134
R^2	0.8486	0.8812	0.9232	0.5051	0.1523	0.6657

Note: A unit of observation is a voting booth. The parties belonging to GANAR vary across Paraguayan departments for the deputies elections, with FG running independently in some departments. However, the Liberal party always runs separately or through GANAR (and mainly proposing its own candidates), hence we have an outcome variable for Liberal/GANAR vote shares. Standard errors are robust. Vote shares on Columns (1) to (3) are calculated by dividing by the number of valid votes (i.e., all votes excluding null and blank votes) and multiplying by 100. Vote shares on Columns (4) to (5) are calculated by dividing by turnout (i.e., all votes including null and blank votes) and multiplying by 100. The number of observations on Column (6) is smaller because we restrict our sample to booths with exactly 200 registered voters. A trifecta indicates all three poll workers from the same party work at the voting booth. Advantage indicates that two out of three poll workers from the same party work at the voting booth. Other refers to a poll worker that is not registered as supporting the Colorado or Liberal party. The excluded category corresponds to voting booths with “fair allocations” having one Colorado poll worker and one Liberal poll worker. The “other poll worker allocation” refers to any remaining poll worker allocation after accounting for “fair allocations” and the other allocations referenced with the independent variables. The number of observations is lower for this electoral race as there is no deputies' election for Paraguayan voters abroad. The sample is restricted to voting booths for which we have the ID numbers of all three poll workers (which is around 96% of all voting booths).

there is one Colorado poll worker, the null vote share declines significantly by 0.10 pp. Finally, we find no significant effects on the blank vote share or turnout.

Finally, Table G21 shows the effect of partisan poll workers on the departmental council election by running equation (1). Similarly to Tables G19 and G20, we use an outcome variable for Liberal/GANAR vote shares because the Liberal party either runs separately or through the GANAR alliance throughout the whole country for this electoral race. We continue finding a similar pattern for the Colorado and Liberal/GANAR vote shares as in Tables G17 to G19. Whenever the Colorado party has two poll workers counting votes, its vote share grows by 0.66 pp and 0.43 pp. In addition, whenever the Colorado party has two poll workers, the Liberal/GANAR vote share drops significantly by 0.51 pp and 0.62 pp respectively. We also find that if only the Colorado party is counting votes at an electoral table through a trifecta then the Liberal/GANAR vote share declines significantly by approximately 1.1 pp. In addition, other poll worker allocations lead to a decline of 0.31 pp in the Liberal/GANAR vote share.

Next, we again see that if the other parties have no poll worker present (particularly when a voting booth has 2 Liberal and 1 Colorado poll worker) the vote share for other parties declines by 0.22 pp. We also notice again that “other” poll worker allocations are associated with an increase in the vote share for other parties by 0.28 pp. Finally, we can observe that if only the Liberal party is counting votes at an electoral table through a trifecta the vote share of other parties rises by 2.5 pp. This estimate is relative to fair poll allocations with one Colorado and one Liberal poll worker. This coefficient suggests that Liberal poll workers may guard or favor other parties for these particular elections, plausibly because the Liberal party tends to be in the opposition with the other parties against the Colorado party. However, the Liberal party has trifectas for only 17 tables (compared to the Colorado party which has 158), so this result could be idiosyncratic and is driven by few observations. If the Colorado party has two poll workers and there is one Liberal poll worker, the null vote share declines significantly (at the 10 percent level) by 0.09 pp. Finally, we find no significant effects on the blank vote share or turnout.

Table G20: Effect of partisan poll worker composition on governor election results

	Colorado vote share (1)	Liberal/GANAR vote share (2)	Other vote share (3)	Blank vote share (4)	Null vote share (5)	Turnout (6)
Colorado trifecta	0.7428 (0.4735)	-0.4342 (0.4601)	-0.3086 (0.3025)	-0.1021 (0.1847)	-0.1286 (0.1459)	-0.9737 (0.8679)
Colorado advantage + Other	0.4158 (0.2281)	-0.4816 (0.2109)	0.0658 (0.1391)	-0.1097 (0.0815)	0.0792 (0.0704)	-0.0927 (0.3539)
Colorado advantage + Liberal	0.3881 (0.1220)	-0.4713 (0.1178)	0.0832 (0.0726)	-0.0041 (0.0444)	-0.0634 (0.0389)	0.0560 (0.1992)
Liberal advantage + Colorado	-0.0897 (0.1258)	0.1880 (0.1222)	-0.0983 (0.0803)	-0.0070 (0.0456)	-0.0969 (0.0395)	0.1161 (0.2070)
Liberal advantage + Other	-0.3790 (0.3616)	0.0420 (0.3956)	0.3369 (0.2692)	-0.1129 (0.1252)	0.0011 (0.1123)	0.5823 (0.5663)
Liberal trifecta	-1.6390 (1.0998)	0.6323 (1.1115)	1.0068 (0.6762)	0.2508 (0.6359)	-0.6177 (0.4783)	1.3708 (1.7457)
Other poll worker allocation	-0.1357 (0.1580)	-0.0948 (0.1532)	0.2305 (0.1053)	-0.0656 (0.0552)	0.0155 (0.0482)	-0.0833 (0.2598)
Mean of Dependent Variable	46.9013	42.3609	10.7379	3.6921	2.5681	122.1091
Polling station FE	✓	✓	✓	✓	✓	✓
Colorado voter prop. control	✓	✓	✓	✓	✓	✓
Liberal voter prop. control	✓	✓	✓	✓	✓	✓
Other party voter prop. control	✓	✓	✓	✓	✓	✓
Observations	18,144	18,144	18,144	18,144	18,144	17,289
R^2	0.8142	0.8563	0.9083	0.4340	0.2076	0.6474

Note: A unit of observation is a voting booth. The parties belonging to the GANAR alliance vary across Paraguayan departments for the governor elections, with FG running independently in some departments. However, the Liberal party always runs separately or through GANAR (and mainly proposing its own candidates), hence we have an outcome variable for Liberal/GANAR vote shares. Standard errors are robust. Vote shares on Columns (1) to (3) are calculated by dividing by the number of valid votes (i.e., all votes excluding null and blank votes) and multiplying by 100. Vote shares on Columns (4) to (5) are calculated by dividing by turnout (i.e., all votes including null and blank votes) and multiplying by 100. The number of observations on Column (6) is smaller because we restrict our sample to booths with exactly 200 registered voters. A trifecta indicates all three poll workers from the same party work at the voting booth. Advantage indicates that two out of three poll workers from the same party work at the voting booth. Other refers to a poll worker that is not registered as supporting the Colorado or Liberal party. The excluded category corresponds to voting booths with “fair allocations” having one Colorado poll worker and one Liberal poll worker. The “other poll worker allocation” refers to any remaining poll worker allocation after accounting for “fair allocations” and the other allocations referenced with the independent variables. The number of observations is lower for this electoral race as there is no governor election for Paraguayan voters abroad and those in the capital district. The sample is restricted to voting booths for which we have the ID numbers of all three poll workers (which is around 96% of all voting booths).

Table G21: Effect of partisan poll worker composition on departmental council election results

	Colorado vote share (1)	Liberal/GANAR vote share (2)	Other vote share (3)	Blank vote share (4)	Null vote share (5)	Turnout (6)
Colorado trifecta	0.5310 (0.4845)	-1.0895 (0.4901)	0.5585 (0.5097)	0.1194 (0.2401)	-0.1082 (0.1936)	-0.4681 (0.8651)
Colorado advantage + Other	0.4279 (0.2517)	-0.6167 (0.2004)	0.1889 (0.2249)	0.1299 (0.1260)	-0.0690 (0.0885)	-0.1489 (0.3679)
Colorado advantage + Liberal	0.6596 (0.1319)	-0.5145 (0.1215)	-0.1451 (0.1227)	0.0279 (0.0659)	-0.0872 (0.0503)	0.0176 (0.2008)
Liberal advantage + Colorado	0.0842 (0.1322)	0.1363 (0.1286)	-0.2204 (0.1285)	-0.0368 (0.0655)	-0.0487 (0.0537)	0.2002 (0.2101)
Liberal advantage + Other	-0.1209 (0.4059)	-0.2000 (0.3502)	0.3209 (0.3600)	-0.1678 (0.2431)	0.1194 (0.2214)	0.7217 (0.5918)
Liberal trifecta	-1.3125 (1.4742)	-1.2272 (1.5510)	2.5397 (1.3661)	-0.4390 (0.5731)	-0.3529 (0.5563)	1.2137 (1.6937)
Other poll worker allocation	0.0305 (0.1679)	-0.3077 (0.1588)	0.2772 (0.1629)	0.0005 (0.1020)	-0.0168 (0.0701)	-0.1811 (0.2689)
Mean of Dependent Variable	40.3654	34.3802	25.2544	6.7051	3.4382	121.8964
Polling station FE	✓	✓	✓	✓	✓	✓
Colorado voter prop. control	✓	✓	✓	✓	✓	✓
Liberal voter prop. control	✓	✓	✓	✓	✓	✓
Other party voter prop. control	✓	✓	✓	✓	✓	✓
Observations	18,130	18,130	18,130	18,130	18,130	17,275
R^2	0.8084	0.8354	0.8333	0.3534	0.1755	0.6485

Note: A unit of observation is a voting booth. The parties belonging to the GANAR alliance vary across Paraguayan departments for the departmental council elections, with FG running independently in some departments. However, the Liberal party always runs separately or through GANAR (and mainly proposing its own candidates), hence we have an outcome variable for Liberal/GANAR vote shares. Standard errors are robust. Vote shares on Columns (1) to (3) are calculated by dividing by the number of valid votes (i.e., all votes excluding null and blank votes) and multiplying by 100. Vote shares on Columns (4) to (5) are calculated by dividing by turnout (i.e., all votes including null and blank votes) and multiplying by 100. The number of observations on Column (6) is smaller because we restrict our sample to booths with exactly 200 registered voters. A trifecta indicates all three poll workers from the same party work at the voting booth. Advantage indicates that two out of three poll workers from the same party work at the voting booth. Other refers to a poll worker that is not registered as supporting the Colorado or Liberal party. The excluded category corresponds to voting booths with “fair allocations” having one Colorado poll worker and one Liberal poll worker. The “other poll worker allocation” refers to any remaining poll worker allocation after accounting for “fair allocations” and the other allocations referenced with the independent variables. The number of observations is lower for this electoral race as there is no departmental council election for Paraguayan voters abroad and those in the capital district. The sample is restricted to voting booths for which we have the ID numbers of all three poll workers (which is around 96% of all voting booths).

Next, we examine whether the presence of any partisan poll workers (regardless of the number) can explain the electoral results. To do so, we generate indicator variables for whether there is at least one of the following types of poll workers in an electoral table: Colorado supporter, Liberal supporter, GANAR supporter (for presidential races), other party supporter, multi-party supporter, and unaffiliated poll worker. Hence, this yields the following regression specification format for all races except the presidential one:

$$\begin{aligned}
Y_{ab} = & \beta_0 + \beta_1 \text{Any_Colorado}_{ab} + \beta_2 \text{Any_Liberal}_{ab} + \beta_3 \text{Any_Othpart}_{ab} + \beta_4 \text{Any_Multipart}_{ab} \\
& + \beta_5 \text{Any_Unaffiliated}_{ab} + \delta_1 \text{voter_prop_Colorado}_{ab} + \delta_2 \text{voter_prop_Liberal}_{ab} \\
& + \delta_3 \text{voter_prop_Other}_{ab} + \theta_a + \varepsilon_{ab}.
\end{aligned} \tag{5}$$

For the presidential race we have this analogous specification:

$$\begin{aligned}
Y_{ab} = & \beta_0 + \beta_1 \text{Any_Colorado}_{ab} + \beta_2 \text{Any_GANAR}_{ab} + \beta_3 \text{Any_Othpart}_{ab} + \beta_4 \text{Any_Multipart}_{ab} \\
& + \beta_5 \text{Any_Unaffiliated}_{ab} + \delta_1 \text{voter_prop_Colorado}_{ab} + \delta_2 \text{voter_prop_GANAR}_{ab} \\
& + \delta_3 \text{voter_prop_Other}_{ab} + \theta_a + \varepsilon_{ab}.
\end{aligned} \tag{6}$$

We analyze whether the presence (regardless of the number) of partisan poll workers can explain the electoral results by running equations (5) and (6). Although our previous regression specifications are more informative because the majority of poll workers at an electoral table decide the official results for each voting booth, we include these specifications for completeness. As they omit the number of workers from each party, identical presence indicators can describe opposite majorities (two Colorado and one Liberal worker versus one Colorado and two Liberal workers), so the coefficients are descriptive projections across heterogeneous compositions. Overall, the most consistent patterns within Tables G22 to G27 are as follows. Having a Liberal poll worker (or GANAR poll worker for the presidential race) counting votes tends to decrease the Colorado party's vote share and increase the Liberal/GANAR vote share. In addition, having a poll worker from the other parties count votes tends to significantly increase the vote share of these other parties, but only on PR races.

Table G22: Effect of partisan poll worker composition on presidential election process results

	Colorado vote share (1)	Liberal/GANAR vote share (2)	Other vote share (3)	Blank vote share (4)	Null vote share (5)	Turnout (6)
Any Colorado poll worker	0.0825 (0.2171)	-0.0740 (0.2117)	-0.0085 (0.1086)	0.1221 (0.0626)	0.0550 (0.0734)	-0.2232 (0.3268)
Any GANAR poll worker	-0.5063 (0.1952)	0.4793 (0.1935)	0.0270 (0.0970)	0.0597 (0.0523)	-0.0107 (0.0613)	-0.1596 (0.3073)
Any other party poll worker	0.1526 (0.1438)	-0.2116 (0.1417)	0.0590 (0.0732)	-0.0424 (0.0444)	0.0188 (0.0473)	0.2039 (0.2384)
Any multi-party poll worker	-0.2396 (0.1350)	0.2815 (0.1348)	-0.0418 (0.0682)	-0.0078 (0.0396)	0.0373 (0.0453)	-0.3034 (0.2155)
Any unaffiliated poll worker	-0.1877 (0.1329)	0.1883 (0.1321)	-0.0006 (0.0630)	-0.0025 (0.0363)	-0.0603 (0.0406)	0.0400 (0.2038)
Mean of Dependent Variable	48.6944	45.3871	5.9185	2.3829	2.7564	122.8357
Polling station FE	✓	✓	✓	✓	✓	✓
Colorado voter prop. control	✓	✓	✓	✓	✓	✓
GANAR voter prop. control	✓	✓	✓	✓	✓	✓
Other party voter prop. control	✓	✓	✓	✓	✓	✓
Observations	20,287	20,287	20,287	20,287	20,287	19,281
R^2	0.8114	0.8202	0.7957	0.4631	0.2462	0.7312

Note: A unit of observation is a voting booth. Standard errors are robust. Vote shares for the first three columns are calculated by dividing by the number of valid votes (i.e., all votes excluding null and blank votes) and multiplying by 100. Vote shares for columns (4) and (5) are calculated by dividing by turnout (total votes including null and blank votes) and multiplying by 100. The number of observations on Column (6) is smaller because we restrict our sample to booths with exactly 200 registered voters. “Any Colorado poll worker” is an indicator variable for whether any of the three poll workers at the voting booth support the Colorado party. “Any GANAR poll worker” is an indicator variable for whether any of the three poll workers at the voting booth support the political parties that are part of the GANAR alliance (the Liberal party, FG, among other smaller left-wing parties). “Any other party poll worker” is an indicator variable for whether any of the three poll workers at the voting booth support an individual political party that is not the Colorado party or a party from the GANAR alliance. “Any multi-party poll worker” is an indicator variable for whether any of the three poll workers at the voting booth support multiple political parties. “Any unaffiliated poll worker” is an indicator variable for whether any of the three poll workers at the voting booth is not registered to any party and does not support any political party as an electoral agent according to the election authority’s data. The sample is restricted to voting booths for which we have the ID numbers of all three poll workers (which is around 96% of all voting booths).

Table G23: Effect of partisan poll worker composition on senatorial election process results

	Colorado vote share (1)	Liberal vote share (2)	Other vote share (3)	Blank vote share (4)	Null vote share (5)	Turnout (6)
Any Colorado poll worker	-0.0971 (0.2176)	0.0304 (0.1980)	0.0667 (0.2528)	0.0949 (0.1086)	0.1730 (0.0962)	-0.2752 (0.3679)
Any Liberal poll worker	-0.2293 (0.1571)	0.4160 (0.1253)	-0.1866 (0.1691)	-0.0937 (0.0732)	0.0329 (0.0624)	-0.0218 (0.2500)
Any other party poll worker	-0.1603 (0.0981)	-0.0954 (0.0856)	0.2557 (0.1094)	-0.0731 (0.0469)	0.0938 (0.0426)	0.0208 (0.1633)
Any multi-party poll worker	-0.3398 (0.1440)	0.2473 (0.1256)	0.0925 (0.1650)	0.0653 (0.0701)	0.1035 (0.0638)	-0.4149 (0.2411)
Any unaffiliated poll worker	-0.1207 (0.1414)	0.3198 (0.1206)	-0.1991 (0.1567)	-0.0216 (0.0655)	0.1149 (0.0610)	0.0827 (0.2312)
Mean of Dependent Variable	32.6369	24.4988	42.8642	5.2841	3.6022	122.5547
Polling station FE	✓	✓	✓	✓	✓	✓
Colorado voter prop. control	✓	✓	✓	✓	✓	✓
Liberal voter prop. control	✓	✓	✓	✓	✓	✓
Other party voter prop. control	✓	✓	✓	✓	✓	✓
Observations	20,257	20,257	20,257	20,257	20,257	19,249
R^2	0.8027	0.8597	0.8476	0.5613	0.1454	0.7076

Note: A unit of observation is a voting booth. Standard errors are robust. Vote shares for the first three columns are calculated by dividing by the number of valid votes (i.e., all votes excluding null and blank votes) and multiplying by 100. Vote shares for columns (4) and (5) are calculated by dividing by turnout (total votes including null and blank votes) and multiplying by 100. The number of observations on Column (6) is smaller because we restrict our sample to booths with exactly 200 registered voters. “Any Colorado poll worker” is an indicator variable for whether any of the three poll workers at the voting booth support the Colorado party. “Any Liberal poll worker” is an indicator variable for whether any of the three poll workers at the voting booth support the Liberal party. “Any other party poll worker” is an indicator variable for whether any of the three poll workers at the voting booth support some party other than the Colorado or Liberal party. “Any multi-party poll worker” is an indicator variable for whether any of the three poll workers at the voting booth support multiple political parties. “Any unaffiliated poll worker” is an indicator variable for whether any of the three poll workers at the voting booth is not registered to any party and does not support any political party as an electoral agent according to the election authority’s data. The sample is restricted to voting booths for which we have the ID numbers of all three poll workers (which is around 96% of all voting booths).

Table G24: Effect of partisan poll worker composition on MERCOSUR parliament election process results

	Colorado vote share (1)	Liberal vote share (2)	Other vote share (3)	Blank vote share (4)	Null vote share (5)	Turnout (6)
Any Colorado poll worker	-0.0596 (0.2459)	0.0856 (0.2043)	-0.0261 (0.2293)	0.0135 (0.1328)	-0.1159 (0.1082)	-0.4668 (0.3873)
Any Liberal poll worker	-0.2722 (0.1616)	0.4712 (0.1359)	-0.1990 (0.1530)	-0.1350 (0.0922)	0.0349 (0.0663)	0.0774 (0.2461)
Any other party poll worker	-0.2528 (0.1017)	0.0262 (0.0906)	0.2267 (0.0969)	-0.0427 (0.0602)	0.1349 (0.0433)	-0.0457 (0.1612)
Any multi-party poll worker	-0.3861 (0.1526)	0.2697 (0.1364)	0.1164 (0.1512)	-0.0338 (0.0894)	0.1035 (0.0668)	-0.3551 (0.2389)
Any unaffiliated poll worker	-0.2775 (0.1481)	0.2400 (0.1289)	0.0375 (0.1386)	-0.1672 (0.0866)	0.1655 (0.0632)	0.0534 (0.2323)
Mean of Dependent Variable	42.1341	30.5360	27.3299	8.4776	3.9100	122.5657
Polling station FE	✓	✓	✓	✓	✓	✓
Colorado voter prop. control	✓	✓	✓	✓	✓	✓
Liberal voter prop. control	✓	✓	✓	✓	✓	✓
Other party voter prop. control	✓	✓	✓	✓	✓	✓
Observations	20,273	20,273	20,273	20,273	20,273	19,265
R^2	0.8021	0.8617	0.8485	0.5476	0.1869	0.7169

Note: A unit of observation is a voting booth. Standard errors are robust. Vote shares for the first three columns are calculated by dividing by the number of valid votes (i.e., all votes excluding null and blank votes) and multiplying by 100. Vote shares for columns (4) and (5) are calculated by dividing by turnout (total votes including null and blank votes) and multiplying by 100. The number of observations on Column (6) is smaller because we restrict our sample to booths with exactly 200 registered voters. “Any Colorado poll worker” is an indicator variable for whether any of the three poll workers at the voting booth support the Colorado party. “Any Liberal poll worker” is an indicator variable for whether any of the three poll workers at the voting booth support the Liberal party. “Any other party poll worker” is an indicator variable for whether any of the three poll workers at the voting booth support some party other than the Colorado or Liberal party. “Any multi-party poll worker” is an indicator variable for whether any of the three poll workers at the voting booth support multiple political parties. “Any unaffiliated poll worker” is an indicator variable for whether any of the three poll workers at the voting booth is not registered to any party and does not support any political party as an electoral agent according to the election authority’s data. The sample is restricted to voting booths for which we have the ID numbers of all three poll workers (which is around 96% of all voting booths).

Table G25: Effect of partisan poll worker composition on deputies' election process results

	Colorado vote share (1)	Liberal/GANAR vote share (2)	Other vote share (3)	Blank vote share (4)	Null vote share (5)	Turnout (6)
Any Colorado poll worker	-0.0197 (0.2241)	0.2296 (0.1992)	-0.2099 (0.2125)	0.2235 (0.0963)	0.0168 (0.0839)	-0.3832 (0.3522)
Any Liberal poll worker	-0.2440 (0.1539)	0.2385 (0.1330)	0.0055 (0.1465)	0.0183 (0.0669)	0.0503 (0.0558)	-0.0048 (0.2442)
Any other party poll worker	-0.1106 (0.0975)	-0.0869 (0.0919)	0.1975 (0.0968)	0.0666 (0.0428)	0.0585 (0.0373)	-0.0463 (0.1566)
Any multi-party poll worker	-0.2949 (0.1431)	0.0061 (0.1333)	0.2889 (0.1433)	0.0790 (0.0648)	0.0400 (0.0542)	-0.3397 (0.2342)
Any unaffiliated poll worker	-0.1864 (0.1400)	0.1189 (0.1285)	0.0675 (0.1392)	-0.0077 (0.0621)	0.1151 (0.0507)	0.0642 (0.2241)
Mean of Dependent Variable	39.1551	30.2855	30.5593	5.1788	3.0362	123.1645
Polling station FE	✓	✓	✓	✓	✓	✓
Colorado voter prop. control	✓	✓	✓	✓	✓	✓
Liberal voter prop. control	✓	✓	✓	✓	✓	✓
Other party voter prop. control	✓	✓	✓	✓	✓	✓
Observations	20,138	20,138	20,138	20,138	20,138	19,134
R^2	0.8485	0.8810	0.9232	0.5053	0.1523	0.6657

Note: A unit of observation is a voting booth. The parties belonging to GANAR vary across Paraguayan departments for the deputies' elections, with FG running independently in some departments. However, the Liberal party always runs separately or through GANAR (and mainly proposing its own candidates), hence we have an outcome variable for Liberal/GANAR vote shares. Standard errors are robust. Vote shares for the first three columns are calculated by dividing by the number of valid votes (i.e., all votes excluding null and blank votes) and multiplying by 100. Vote shares for columns (4) and (5) are calculated by dividing by turnout (total votes including null and blank votes) and multiplying by 100. The number of observations on Column (6) is smaller because we restrict our sample to booths with exactly 200 registered voters. "Any Colorado poll worker" is an indicator variable for whether any of the three poll workers at the voting booth support the Colorado party. "Any Liberal poll worker" is an indicator variable for whether any of the three poll workers at the voting booth support the Liberal party. "Any other party poll worker" is an indicator variable for whether any of the three poll workers at the voting booth support some party other than the Colorado or Liberal party. "Any multi-party poll worker" is an indicator variable for whether any of the three poll workers at the voting booth support multiple political parties. "Any unaffiliated poll worker" is an indicator variable for whether any of the three poll workers at the voting booth is not registered to any party and does not support any political party as an electoral agent according to the election authority's data. The number of observations is lower for this electoral race as there is no deputies' election for Paraguayan voters abroad. The sample is restricted to voting booths for which we have the ID numbers of all three poll workers (which is around 96% of all voting booths).

Table G26: Effect of partisan poll worker composition on governor election process results

	Colorado vote share (1)	Liberal/GANAR vote share (2)	Other vote share (3)	Blank vote share (4)	Null vote share (5)	Turnout (6)
Any Colorado poll worker	0.1762 (0.2245)	0.2022 (0.2257)	-0.3784 (0.1475)	0.1219 (0.0786)	-0.0175 (0.0700)	-0.2750 (0.3582)
Any Liberal poll worker	-0.2485 (0.1543)	0.3342 (0.1488)	-0.0857 (0.0996)	0.0538 (0.0554)	0.0247 (0.0473)	0.0109 (0.2495)
Any other party poll worker	-0.1025 (0.1004)	0.0370 (0.0978)	0.0655 (0.0623)	0.0171 (0.0368)	0.0577 (0.0318)	-0.0118 (0.1641)
Any multi-party poll worker	-0.3299 (0.1493)	0.3469 (0.1474)	-0.0170 (0.0997)	0.0350 (0.0538)	0.1039 (0.0467)	-0.4842 (0.2447)
Any unaffiliated poll worker	-0.2156 (0.1412)	0.1759 (0.1380)	0.0397 (0.0913)	-0.0539 (0.0509)	0.1118 (0.0443)	-0.0291 (0.2276)
Mean of Dependent Variable	46.9013	42.3609	10.7379	3.6921	2.5681	122.1091
Polling station FE	✓	✓	✓	✓	✓	✓
Colorado voter prop. control	✓	✓	✓	✓	✓	✓
Liberal voter prop. control	✓	✓	✓	✓	✓	✓
Other party voter prop. control	✓	✓	✓	✓	✓	✓
Observations	18,144	18,144	18,144	18,144	18,144	17,289
R^2	0.8141	0.8562	0.9083	0.4341	0.2076	0.6474

Note: A unit of observation is a voting booth. The parties belonging to GANAR vary across Paraguayan departments for the governor elections, with FG running independently in some departments. However, the Liberal party always runs separately or through GANAR (and mainly proposing its own candidates), hence we have an outcome variable for Liberal/GANAR vote shares. Standard errors are robust. Vote shares for the first three columns are calculated by dividing by the number of valid votes (i.e., all votes excluding null and blank votes) and multiplying by 100. Vote shares for columns (4) and (5) are calculated by dividing by turnout (total votes including null and blank votes) and multiplying by 100. The number of observations on Column (6) is smaller because we restrict our sample to booths with exactly 200 registered voters. “Any Colorado poll worker” is an indicator variable for whether any of the three poll workers at the voting booth support the Colorado party. “Any Liberal poll worker” is an indicator variable for whether any of the three poll workers at the voting booth support the Liberal party. “Any other party poll worker” is an indicator variable for whether any of the three poll workers at the voting booth support some party other than the Colorado or Liberal party. “Any multi-party poll worker” is an indicator variable for whether any of the three poll workers at the voting booth support multiple political parties. “Any unaffiliated poll worker” is an indicator variable for whether any of the three poll workers at the voting booth is not registered to any party and does not support any political party as an electoral agent according to the election authority’s data. The number of observations is lower for this electoral race as there is no governor election for Paraguayan voters abroad and those in the capital district. The sample is restricted to voting booths for which we have the ID numbers of all three poll workers (which is around 96% of all voting booths).

Table G27: Effect of partisan poll worker composition on departmental council election process results

	Colorado vote share (1)	Liberal/GANAR vote share (2)	Other vote share (3)	Blank vote share (4)	Null vote share (5)	Turnout (6)
Any Colorado poll worker	0.0016 (0.2377)	0.3187 (0.2172)	-0.3203 (0.2196)	0.2684 (0.1348)	-0.0894 (0.1099)	-0.5116 (0.3655)
Any Liberal poll worker	-0.3905 (0.1672)	0.6400 (0.1501)	-0.2495 (0.1566)	-0.1041 (0.0865)	0.0685 (0.0651)	0.1697 (0.2606)
Any other party poll worker	-0.3702 (0.1066)	0.0915 (0.1014)	0.2787 (0.1026)	0.0121 (0.0540)	0.0891 (0.0425)	-0.0860 (0.1655)
Any multi-party poll worker	-0.5191 (0.1526)	0.4955 (0.1430)	0.0236 (0.1509)	0.1301 (0.0925)	0.0345 (0.0635)	-0.3711 (0.2474)
Any unaffiliated poll worker	-0.3012 (0.1505)	0.3181 (0.1373)	-0.0170 (0.1412)	-0.0711 (0.0817)	0.0650 (0.0592)	-0.1341 (0.2308)
Mean of Dependent Variable	40.3654	34.3802	25.2544	6.7051	3.4382	121.8964
Polling station FE	✓	✓	✓	✓	✓	✓
Colorado voter prop. control	✓	✓	✓	✓	✓	✓
Liberal voter prop. control	✓	✓	✓	✓	✓	✓
Other party voter prop. control	✓	✓	✓	✓	✓	✓
Observations	18,130	18,130	18,130	18,130	18,130	17,275
R^2	0.8083	0.8353	0.8333	0.3537	0.1756	0.6485

Note: A unit of observation is a voting booth. The parties belonging to GANAR vary across Paraguayan departments for the departmental council elections, with FG running independently in some departments. However, the Liberal party always runs separately or through GANAR (and mainly proposing its own candidates), hence we have an outcome variable for Liberal/GANAR vote shares. Standard errors are robust. Vote shares for the first three columns are calculated by dividing by the number of valid votes (i.e., all votes excluding null and blank votes) and multiplying by 100. Vote shares for columns (4) and (5) are calculated by dividing by turnout (total votes including null and blank votes) and multiplying by 100. The number of observations on Column (6) is smaller because we restrict our sample to booths with exactly 200 registered voters. “Any Colorado poll worker” is an indicator variable for whether any of the three poll workers at the voting booth support the Colorado party. “Any Liberal poll worker” is an indicator variable for whether any of the three poll workers at the voting booth support the Liberal party. “Any other party poll worker” is an indicator variable for whether any of the three poll workers at the voting booth support some party other than the Colorado or Liberal party. “Any multi-party poll worker” is an indicator variable for whether any of the three poll workers at the voting booth support multiple political parties. “Any unaffiliated poll worker” is an indicator variable for whether any of the three poll workers at the voting booth is not registered to any party and does not support any political party as an electoral agent according to the election authority’s data. The number of observations is lower for this electoral race as there is no departmental council election for Paraguayan voters abroad and those in the capital district. The sample is restricted to voting booths for which we have the ID numbers of all three poll workers (which is around 96% of all voting booths).

We obtained legal documents from the election authority indicating all electoral tables where electoral results were contested (*impugnaciones electorales*) at the electoral race level, and we codified these into a dataset (TSJE 2018b). For each electoral contestation, we have information on the voting booth and race contested, the parties that contested the results, and the contestation's outcome. Among the contestation outcomes possible is annulling the tally of the electoral table for the particular race being contested, which entails that zero votes are counted from that electoral table for that particular race. To determine whether a contestation is accepted, the election authority checks the tallies submitted by each of the poll workers. If a majority (two or three) of the poll workers' tallies agree with the electoral contestation, then the results are amended. Yet given how this system works and that some parties have an outright majority or collude at a booth, contestations may not be successful despite being legitimate. Overall, there are not many electoral contestations: 3,213 out of 122,595 possible table-race combinations. Out of 3,213 contestations in this election, only 738 were accepted. The low number of contestations may be partly because parties need to have either poll workers, poll watchers, or empowered party officers present to observe irregularities, and only the established parties are well-resourced enough to cover booths throughout the country. In addition, given that contestations are decided by what the majority of poll workers wrote down, some parties may not even submit a contestation which they expect to be ultimately rejected. Party activists from smaller parties confirmed this to us.

Table G28 explores whether certain allocations of partisan poll workers on voting booths are associated with more electoral contestations where results are challenged. Overall, there are not many contestations in our sample: 3,116 out of 117,742 (2.65%) possible voting booth-electoral race combinations.⁸ This is not fully surprising because most often parties need to have either poll workers, poll watchers, or empowered party officers present to file a contestation. Only the Colorado party and Liberal party (to a somewhat lesser degree) are well-resourced enough to cover most of the country. The election authority adjudicates contestations by checking the tallies filled out individually by the three poll workers and confirming that at least two of the three poll workers have tallies that coincide. If at least two tallies coincide, then the results from those tallies are confirmed as the official results. This conspicuously weakens the electoral contestation process given our evidence that individual political parties tend to have a majority in more than a third of all voting booths, and that some parties may collude at the tally-writing process. If all three tallies differ, then the tally for the relevant electoral race in that voting booth is annulled, meaning that zero votes are counted for that race in that voting booth.

Table G28 shows that contestations are much less common when a single party controls a

⁸The sample of 117,742 comes from the voting booths for which we have the ID numbers of all three poll workers (which is around 96% of all voting booths). If we consider all voting booths in the elections, we have contestations in 3,213 out of 122,595 (2.62%) possible voting booth-electoral race combinations, as shown on SA Table G15.

voting booth through a trifecta of poll workers. This is unsurprising because if no poll worker from another party is present, any irregularity is less likely to be reported. Due to the low number of contestations, even fewer voting booths get their counts amended due to contestations being accepted or that the tally for a particular race at a particular voting booth is annulled. Hence, although contestations are a recourse that political parties use in other contexts to demand greater electoral integrity, this recourse appears to be quite ineffective in the Paraguayan context. Electoral courts are meant to resolve electoral disputes and address allegations of electoral irregularities by either recognizing their legitimacy or discarding them if they do not prove legitimate. Given our findings that partisan poll workers tend to favor their own parties, particularly when they have a majority at a voting booth, that we observe fewer contestations under single-party control is itself likely problematic.

Tables G29 to G31 additionally test whether the alphabetic ranking of voters' surnames (which determines the booth where they vote) is uncorrelated with electoral results for this election. To do so, we first calculate the alphabetic ranking of voters' last names within their polling station. This ranking is normalized by the number of voters in each polling station such that all within-polling station rankings go from 0 to 1. That is, voters with the earliest names in the alphabet in their polling station have a ranking near 0, whereas the voter with the latest name in the alphabet has a ranking of 1. We then calculate the average voter alphabetic ranking that each voting booth has. One table shows regressions of the Colorado party vote share for each electoral race on the alphabetic ranking of the voting booths. We find no significant correlation between voters' surname order and their preference towards the Colorado party. The other two tables run similar tests but instead use the Liberal party and GANAR vote share respectively. Both of these tables show similar non-significant results. Hence, these placebo tests also indicate that voters' first letters of their last names do not correlate with their political preferences.

Our analyses using data on voters' surnames and their political preferences support our assumption that voters are assigned to voting booths quasi-randomly. Hence, any strong deviance in results recorded across booths within a polling station are likely due to the partisan representation of poll workers in the booth after we control for polling station fixed effects.

Table G28: Effect of partisan poll worker composition on electoral contestations

	Contested (1)	Contested by non-Colorado (2)	Contested by non-Liberal (3)	Contestation accepted (4)	Tally annulled (5)
Colorado trifecta	-0.0218 (0.0036)	-0.0168 (0.0035)	-0.0115 (0.0029)	-0.0074 (0.0009)	-0.0051 (0.0007)
Colorado advantage + Other	0.0005 (0.0026)	-0.0003 (0.0024)	-0.0011 (0.0021)	-0.0009 (0.0012)	-0.0007 (0.0010)
Colorado advantage + Liberal	0.0002 (0.0013)	0.0007 (0.0012)	0.0006 (0.0011)	0.0004 (0.0007)	0.0007 (0.0005)
Liberal advantage + Colorado	0.0002 (0.0014)	0.0004 (0.0013)	0.0001 (0.0012)	-0.0010 (0.0007)	-0.0011 (0.0005)
Liberal advantage + Other	-0.0041 (0.0036)	-0.0037 (0.0034)	-0.0017 (0.0032)	-0.0019 (0.0018)	-0.0001 (0.0017)
Liberal trifecta	-0.0154 (0.0040)	-0.0148 (0.0039)	-0.0119 (0.0038)	-0.0050 (0.0018)	-0.0038 (0.0014)
Other poll worker allocation	-0.0002 (0.0018)	-0.0003 (0.0017)	0.0001 (0.0016)	0.0005 (0.0010)	0.0006 (0.0008)
Mean of Dependent Variable	0.0265	0.0230	0.0189	0.0059	0.0036
Polling station-elect. race FE	✓	✓	✓	✓	✓
Colorado voter prop. control	✓	✓	✓	✓	✓
Liberal voter prop. control	✓	✓	✓	✓	✓
Other party voter prop. control	✓	✓	✓	✓	✓
Observations	117,680	117,680	117,680	117,680	117,680
R^2	0.2046	0.2225	0.1624	0.0591	0.0603

Note: A unit of observation is a voting booth-electoral race combination. Standard errors are robust. The specification includes polling station by electoral race fixed effects and the three voter-affiliation controls of equation (1). “Contested” is an indicator variable for whether the electoral results for this electoral race and voting booth were contested. “Contested by non-Colorado (Liberal)” is an indicator variable for the voting booth having its results contested by a political party that is not the Colorado party (Liberal party). “Contestation accepted” is an indicator variable for whether the contestation towards the electoral results in this voting booth were accepted by the election authority, thus amending the count. “Tally annulled” is an indicator variable for whether the electoral results from this voting booth were annulled (i.e., 0 votes are counted).

Table G29: Relationship between voters' within-polling station alphabetic ranking and Colorado party votes

	President vote share (1)	Senator vote share (2)	MERCOSUR parliament vote share (3)	Deputies vote share (4)	Governor vote share (5)	Departmental council vote share (6)
Avg. surname alphabetical ranking	0.3681 (0.2935)	-0.0366 (0.3007)	0.1144 (0.3123)	-0.0515 (0.3384)	0.0583 (0.3157)	-0.0391 (0.3269)
Mean of Dependent Variable	48.6974	32.6445	42.1395	39.1594	46.9023	40.3697
Observations	20,298	20,268	20,284	20,149	18,153	18,139
R^2	0.0001	0.0000	0.0000	0.0000	0.0000	0.0000

Note: A unit of observation is a voting booth. Standard errors are robust. Vote shares for each electoral race are calculated by dividing by the number of valid votes (i.e., all votes excluding null and blank votes) and multiplying by 100. The “Avg. surname alphabetic ranking” is calculated by ranking all voters within each polling station by the alphabetic order of their surname and then calculating the average ranking per voting booth. Each column is a separate regression for one electoral race, so the number of observations varies across columns: there is no governor or departmental council election for voters abroad or in the capital district, and no polling station fixed effects are used, so no singleton booths are dropped.

Table G30: Relationship between voters' within-polling station alphabetic ranking and Liberal party votes

	Senator vote share (1)	MERCOSUR parliament vote share (2)	Deputies vote share (3)	Governor vote share (4)	Departmental council vote share (5)
Avg. surname alphabetical ranking	-0.2408 (0.3071)	-0.3436 (0.3305)	-0.4191 (0.4649)	-0.0390 (0.4473)	-0.6417 (0.4118)
Mean of Dependent Variable	24.5030	30.5376	28.7323	44.8315	36.1613
Observations	20,268	20,285	12,489	8,812	11,032
R^2	0.0000	0.0001	0.0001	0.0000	0.0002

Note: A unit of observation is a voting booth. Standard errors are robust. Vote shares for each electoral race are calculated by dividing by the number of valid votes (i.e., all votes excluding null and blank votes) and multiplying by 100. The “Avg. surname alphabetic ranking” is calculated by ranking all voters within each polling station by the alphabetic order of their surname and then calculating the average ranking per voting booth. Each column is a separate regression for one electoral race, so the number of observations varies across columns: there is no governor or departmental council election for voters abroad or in the capital district, and no polling station fixed effects are used, so no singleton booths are dropped, and the Liberal vote share is defined only in the races and departments where the Liberal party ran on its own list rather than through the GANAR alliance.

Table G31: Relationship between voters' within-polling station alphabetic ranking and GANAR votes

	President vote share (1)	Deputies vote share (2)	Governor vote share (3)	Departmental council vote share (4)
Avg. surname alphabetical ranking	-0.3711 (0.3005)	0.1472 (0.5413)	0.3387 (0.5169)	0.2284 (0.5635)
Mean of Dependent Variable	45.3860	32.8310	40.0363	31.6273
Observations	20,298	7,660	9,341	7,107
R^2	0.0001	0.0000	0.0000	0.0000

Note: A unit of observation is a voting booth. Standard errors are robust. Vote shares for each electoral race are calculated by dividing by the number of valid votes (i.e., all votes excluding null and blank votes) and multiplying by 100. The “Avg. surname alphabetic ranking” is calculated by ranking all voters within each polling station by the alphabetic order of their surname and then calculating the average ranking per voting booth. Each column is a separate regression for one electoral race, so the number of observations varies across columns: there is no governor or departmental council election for voters abroad or in the capital district, and no polling station fixed effects are used, so no singleton booths are dropped, and the GANAR vote share is defined only in the races and departments where the alliance ran.